STOCK CARBON MAZUAY

Project title	Stock Carbon Mazuay
Project ID	5343
Credit period	January 1, 2025 to December 31, 2044
Original issuance date	For registration, DD-Month-YYYY is the date the project description was completed following the completion of the audit
Most recent issuance date	March 20, 2025
Version	Version 1
VCS standard version	VCS 4.7 — Updated version on April 16, 2024.
Prepared by	Stock Capital Holding LTDA.

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Stock Carbon Mazuay Project of Gleba Mazuay, located in Silves, Amazonas, is focused on the conservation and sustainable use of native vegetation in an ecologically rich region of the Amazon. The Gleba is situated within a context of Dense Ombrophilous Forest of the Lowlands, in addition to other ecosystems such as floodplain forests, savannas, and wetlands.

According to IBGE 2024 data, the State of Amazonas has a territorial area of approximately 1,559,256.365 km^2 and its capital is the city of Manaus. With a population density of 2.53 inhabitants/km^2, Amazonas is home to an estimated population of 4,281,209 inhabitants. The Human Development Index (HDI) is 0.7, considered high, but with significant inequalities among its 62 municipalities. The monthly per capita household income is R$1,172.00.

The region faces significant environmental challenges, such as the expansion of road infrastructure and mining activities, which threaten the integrity of ecosystems and the environmental services provided by native vegetation. To mitigate these risks, the Stock Carbon Mazuay Project aims to preserve native vegetation, manage forests sustainably, protect water resources, and quantify carbon flux.

The project's actions include environmental monitoring, control of deforestation and forest degradation, restoration of degraded areas, and engagement with local communities. The use of remote sensing techniques, such as the analysis of vegetation indices (NDVI, EVI, LAI, and FPAR), will enable the estimation of Gross Primary Productivity (GPP), Autotrophic Respiration (RA), and Net Primary Production (NPP) in the region, which are essential for the quantification of carbon credits under the VCS Program.

Stock Capital is responsible for the proposal, implementation, monitoring, and certification of the project. The environmental analysis conducted within a 50 km radius from Gleba Mazuay highlights the region's importance for biodiversity conservation, water resource maintenance, and climate change mitigation. The project aims to conserve 988.66 hectares of dense ombrophilous forest, generating 718,455.73 VCUs over 20 years, with an annual average of 35,922.79 VCUs throughout the project's lifespan.

This initiative seeks to maximize the climate impact of carbon projects, providing benefits for local communities and biodiversity. The preservation of native vegetation not only contributes to climate change mitigation but also promotes ecosystem sustainability and the well-being of the populations that depend on them.

1.2 Audit History

Table 1 – Audit Schedule of the Stock Carbon Mazuay Project.

Audit type	**Period**	**Program**	**Validation/verification body name**	**Number of years**
Validation/ verification				

1.3 Sectoral Scope and Project Type

The table below presents relevant information about the AFOLU projects (category), such as sector scope and project activity type.

Sectoral scope	14
AFOLU project category1	Emissions Reduction from Deforestation and Forest Degradation (REDD+)
Project activity type	Avoiding Planned Deforestation (APD)

1.4 Project Eligibility

1.4.1 General eligibility

The Stock Carbon Mazuay Project is included within the scope of the VCS Program as an Emissions Reduction from Deforestation and Forest Degradation (REDD+) activity. According to the VM0015 methodology, the project aims to reduce greenhouse gas (GHG) emissions through the preservation and sustainable management of native vegetation, combating deforestation and forest degradation. The VM0049 methodology provides a globally applicable framework for carbon capture and storage activities, which can be adapted for REDD+ projects.

The Stock Carbon Mazuay Project meets the requirements of the VCS Program, including the listing timeline in the pipeline, the holding of the opening meeting with the validation/verification body, and the validation timeline. All necessary steps for approval have been followed according to the program's requirements.

The methodology applied in the Stock Carbon Mazuay Project is eligible for the VCS Program, adhering to best practices for REDD+ activities according to the VM0015 methodology. The project is not a fragmented part of a larger activity that would exceed capacity limits. The activities are carried out independently and comply with the scale and capacity limits set.

According to VCS v4.5, REDD+ projects in AFOLU are eligible to reduce GHG emissions by decreasing deforestation and/or forest degradation. Deforestation is the conversion of forested land into non-forested land induced by human activity. Degradation is the reduction of canopy cover or carbon stocks without converting the forest into non-forested land. Projects must meet internationally accepted definitions of forest and qualify as such for at least 10 years prior.

Stock Capital uses advanced remote sensing techniques to monitor and quantify carbon credits. The project aims to conserve 988.66 hectares of forest, generating climate and social benefits over 20 years, with the production of 718,455.86 VCUs. Additionally, the project follows the IPCC guidelines for REDD+ activities, ensuring the integrity and effectiveness of climate change mitigation measures.

1.4.2 AFOLU Project Eligibility

In the area in question, a detailed forest inventory was carried out for the purpose of Sustainable Forest Management, with the objective of implementing, in the future, management practices that promote forest conservation and carbon sequestration.

A continuous monitoring system will be implemented in accordance with VCS methodologies VM0012 and VM0006, and periodic environmental impact reports will be produced. The data on stored and sequestered carbon were obtained using NDVI.

1.4.3 Transfer project eligibility

Not Applicable

1.5 Project Design

Indicate if the project has been designed as:

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

1.5.1 Grouped project design

The design of the project in question, characterized by being in a single location or installation, occurs in a single geographic location or facility. Therefore, it does not apply to the item specified.

1.6 Project Proponent

Organization name	Stock Capital Holding LTDA.
Contact person	Mr. Jonatan Augusto Ville Lubian
Title	Founder and CEO
Address	Alameda Dom Pedro II, 155 - Batel - Curitiba - Paraná - Brazil - Postal Code: 80420060
Telephone	+55 (41) 9 8788-8230
Email	augusto@stockcapital.com.br

1.7 Other Entities Involved in the Project

Organization name	Cipó Engenharia e Tecnologia de Biomassa LTDA.
Role in the project	Project Developer
Contact person	Mr. Dennis Souza Russelakis de Oliveira
Title	Founder and CEO
Address	Rua Paulo Fortes, 6595 - Aponiã. Porto Velho – RO – Brazil – Postal Code: 76824082
Telephone	+55 (69) 9 9241-4040
Email	cipoengenharia@gmail.com

Organization name	Martinelli & Guimarães Advocacia.
Role in the project	Legal Advisory
Contact person	Mr. Gustavo Martinelli Tanganelli Gazotto
Title	Managing Partner
Address	Alameda Princesa Izabel, 1975 - Bigorrilho - Curitiba - Paraná - Brazil - Postal Code: 80730-080
Telephone	+55 (41) 3077-7758
Email	gustavo@martinelliguimaraes.com.br

1.8 Ownership

According to the provided information, the Stock Carbon Mazuay Project complies with the requirements of the VCS Program regarding project ownership.

Gleba Mazuay, owned by Mr. Josué Rogério de Souza, as evidenced by property records (registration), has entered into a formal contract with Stock Capital Holding LTDA for the implementation and management of the Stock Carbon Mazuay Forest Conservation and Management Project.

Mr. Josué Rogério de Souza has expressly authorized Stock Capital Holding LTDA to develop and implement the project. All records, documents, and relevant information related to the ownership of Gleba and the rights for project implementation are properly organized and filed.

Thus, it is possible to verify the ownership of the project by Mr. Josué Rogério de Souza and the appropriate consent and authorization for its implementation by Stock Capital Holding LTDA, in accordance with the requirements of the VCS Program, demonstrating the commitment and essence of the forest conservation project.

1.9 Project Start Date

Project start date	12-november-2024
Justification	The agreement between Stock Capital Holding LTDA and Landowner was celebrated November 12, 2024.

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☐ Ten years, fixed ☒ Other – Twenty years (The Stock Carbon Mazuay project has a credit generation period of 20 years, starting on January 1, 2025, and ending on December 31, 2044).
Start and end date of first or fixed crediting period	January 1, 2025 to December 31, 2044

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

Indicate the estimated annual GHG emission reductions/removals (ERRs) of the project:

☒ < 300,000 tCO2e/year (project)

☐ ≥ 300,000 tCO2e/year (large project)

Complete the table below for the first (if renewable) or fixed crediting period:

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO2e)
January 1, 2025 to December 31, 2025	29,569.28
January 1, 2026 to December 31, 2026	30,160.67
January 1, 2027 to December 31, 2027	30,763.88
January 1, 2028 to December 31, 2028	31,379.16
January 1, 2029 to December 31, 2029	32,006.74
January 1, 2030 to December 31, 2030	32,646.87
January 1, 2031 to December 31, 2031	33,299.81
January 1, 2032 to December 31, 2032	33,965.81
January 1, 2033 to December 31, 2033	34,645.12
January 1, 2034 to December 31, 2034	35.338,03
January 1, 2035 to December 31, 2035	36.044,79
January 1, 2036 to December 31, 2036	36.765,68
January 1, 2037 to December 31, 2037	37.501,00
January 1, 2038 to December 31, 2038	38.251,02
January 1, 2039 to December 31, 2039	39.016,04
January 1, 2040 to December 31, 2040	39.796,36
January 1, 2041 to December 31, 2041	40.592,28
January 1, 2042 to December 31, 2042	41.404,13
January 1, 2043 to December 31, 2043	42.232,21

January 1, 2044 to December 31, 2044	43.076,86
Total estimated ERRs during the first or fixed crediting period	**718,455.73**
Total number of years	**20**
Average annual ERRs	**35,922.79**

1.12 Description of the Project Activity

The Stock Carbon Mazuay Forest Conservation and Management Project aims to preserve and sustainably use native vegetation in an area of significant environmental relevance in the Amazon region. The main activities and technologies/measures to be implemented in the project include:

- Establish long-term agreements with landowners to conserve native vegetation cover in areas subject to legal deforestation in the Amazon. The complementary measures include mitigating leakage, promoting net positive benefits for communities, conserving biodiversity, and monitoring the project.

- Conduct environmental monitoring using remote sensing techniques, such as vegetation index analysis (NDVI, EVI, IAF/LAI, and FPAR), to track vegetation dynamics and estimate Gross Primary Productivity (GPP), Autotrophic Respiration (RA), and Net Primary Carbon Flux (NPP).

- Conduct a forest biomass inventory to estimate aboveground biomass in trees using permanent plots within the project area and LiDAR data collected with drones. The plots will be organized into clusters of 8 hectares, with subplots of 2 hectares each. Drone-LiDAR techniques will be applied in a modeling approach. Non-tree biomass and belowground biomass will be estimated indirectly using secondary data and standard conversion factor values. Deadwood and litter will be directly measured in the same permanent plots. Biomass estimates will be fixed during the first baseline period and will be remeasured every 10 years. Changes in carbon stocks will be estimated using the stock difference method, with a 90% confidence interval. New estimates will prevail over previous ones if they fall outside the 90% confidence interval of the prior estimates. It is important to note that the area in question has a 100% forest inventory, as evidenced by the attached results.

- To control and prevent illegal deforestation and degradation of native vegetation, surveillance and enforcement actions will be implemented. These actions will be supported by monitoring technologies, such as land-use change maps and satellite imagery, collected from platforms like PRODES2 and DETER3/INPE4, GLOBAL FOREST WATCH, and FIRMS/NASA. The burned area will be measured using the ΔNBR6 spectral index and verified with high-resolution satellite images from Sentinel-2, along with ground truth verification on-site.

- To involve local communities in promoting the sustainable use of forest resources and biodiversity conservation. Initially, detailed socioeconomic surveys are conducted to understand the communities' conditions, including their sources of income, access to education, health, and infrastructure. The communities are involved in the process, ensuring that their needs and knowledge are considered. Based on this data, environmental education programs are developed, addressing the importance of conservation and the sustainable use of resources. The activities include practical workshops, training on sustainable techniques, awareness campaigns, and partnerships with local schools and NGOs. The progress of the programs is continuously monitored, and participatory reports are produced to ensure transparency. The benefits include empowering communities, generating sustainable income, and improving quality of life. Community engagement is essential for the success of forest conservation and sustainable resource use efforts.

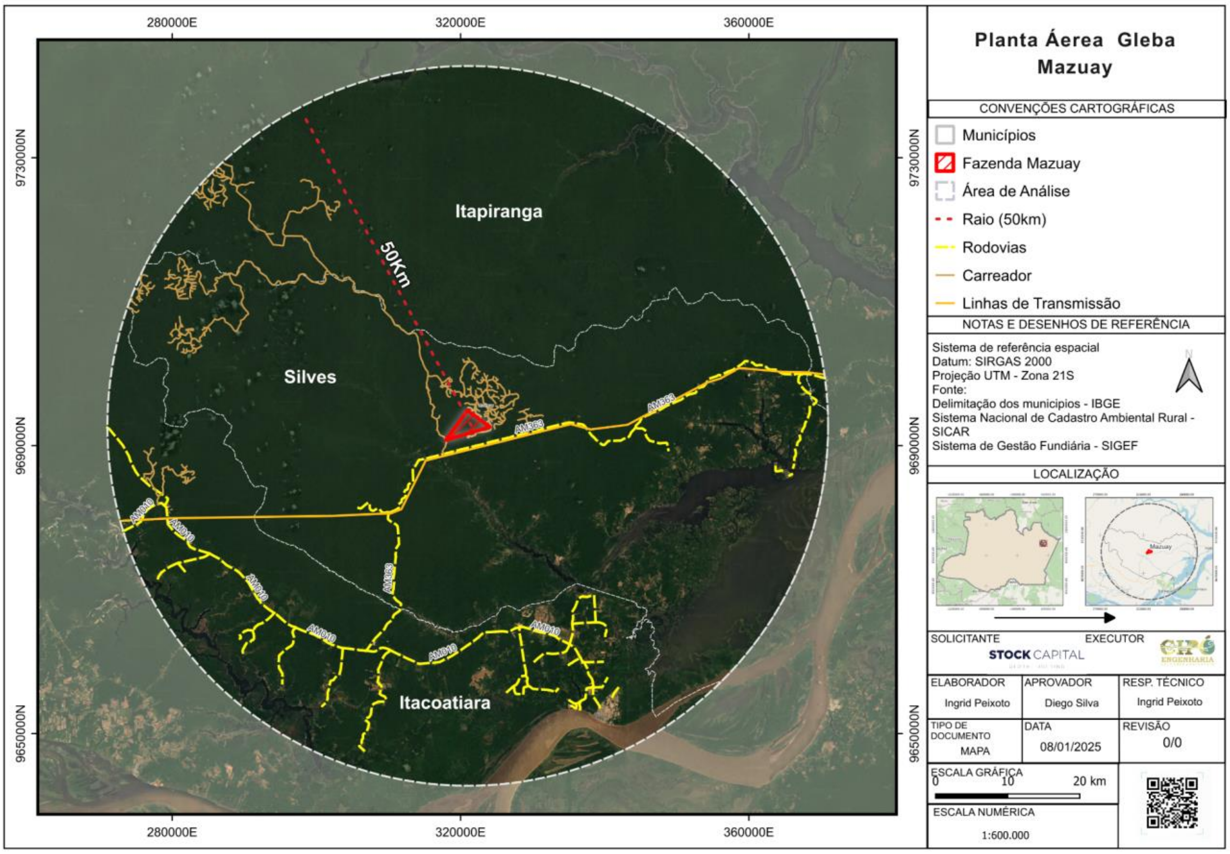
- The recovery of degraded areas will be carried out through reforestation activities and sustainable forest management in areas that have suffered degradation, aiming at the restoration of vegetation cover and the improvement of ecosystem services. The project implementation schedule foresees the start of activities according to the needs identified during the continuous monitoring and verification program throughout the entire duration of the project. The Forest Conservation and Management Project of Gleba Mazuay is located in a region of the Amazon that is not covered by a jurisdictional REDD+ program.

1.13 Project Location

Leaving from Manaus International Airport - Eduardo Gomes, head east for approximately 450 meters and turn right onto Avenida Santos Dumont. Continue for 2.5 km until you turn left onto Avenida Torquato Tapajós, traveling for 2.4 km. At the roundabout, stay on the main route and continue on Avenida Torquato Tapajós for another 11.7 km. Then, continue on Rodovia Deputado Vital de Mendonça for about 205 km. Turn left onto AM-363 and drive for 35.5 km until you reach the access road to Gleba Mazuay.

The aerial mapping of Gleba Mazuay shows a significant increase in access roads in the eastern portion, heading northwest, covering the municipalities of Silves and Itapiranga. These access roads are primary routes used for agricultural activities, natural resource extraction, and local transportation. However, their uncontrolled growth may lead to severe environmental consequences.

The presence of access roads in Permanent Preservation Areas (APPs) poses a risk to the integrity of these regions, increasing the vulnerability of local vegetation and fauna. The fragmentation of natural ecological corridors compromises the connectivity between forest fragments, affecting the dynamics of various species (NASCIMENTO et al., 2021). This risk is exacerbated by illegal logging, mining, and other illegal activities facilitated by the access roads.



1.13.1 Characterization of the Study Area — Physical Environment

1.13.1.1 Climate

Understanding the climate is fundamental for the sustainable management of agricultural and environmental activities in a given region. This chapter presents a detailed climatological analysis of the region where Gleba Mazuay is located, following the guidelines established by the World Meteorological Organization (WMO) and the Verification and Registration of Emission Reductions and Removals of Greenhouse Gases (VERRAS).

The climatological analysis of the Gleba Mazuay region reveals a humid tropical climate with a dry season (Aw), with specific characteristics of temperature, precipitation, and extreme events. This understanding of the local climate is fundamental for the planning and sustainable management of the Gleba, enabling the adoption of practices adapted to climatic conditions and the mitigation of potential impacts of climate change. The implementation of the recommendations presented will contribute to the long-term resilience and sustainability of Gleba Mazuay.

Climate Classification - Climate Analysis of Gleba Mazuay

According to the Köppen climate classification, the Gleba Mazuay region has an Aw climate, characterized as a humid tropical climate with a dry season. The main characteristics of this climate are:

- **Elevated temperatures:** Temperatures in the Gleba Mazuay are relatively high throughout the year, with annual averages around 26°C. The hottest months occur between December and March, with maximum temperatures often exceeding 35°C. Even during winter, minimum temperatures rarely fall below 10°C.

- **Seasonal Precipitation:** Precipitation in the Gleba Mazuay exhibits clear seasonality, with a rainy season lasting from October to April and a dry season from May to September. During the rainy season, monthly precipitation values exceed 200 mm, reaching peaks above 300 mm. In contrast, during the dry season, precipitation stays below 100 mm, with some months recording less than 50 mm.

- **Interannual Variability:** Throughout the analyzed period, from 2011 to 2023, significant variability in annual precipitation is observed, with values ranging from 1,500 mm to 2,500 mm. This interannual variability may be related to larger-scale climatic phenomena, such as the El Niño-Southern Oscillation (ENSO).

In summary, Gleba Mazuay is located in a region with a humid tropical climate with a dry season, characterized by elevated temperatures and a marked seasonality in rainfall distribution. This information is essential for the planning and management of agricultural activities on the property, aiming at adaptation to local climatic conditions.

1.13.1.2 Temperature

The map presents the annual average temperature of Gleba Mazuay in 2024, with values interpolated using the IDW (Inverse Distance Weighting) interpolation method, based on measurements taken from automatic and conventional stations provided by INMET. A spatial variation in the thermal distribution of the region is observed, highlighting the influence of land use and land cover on the recorded temperatures.

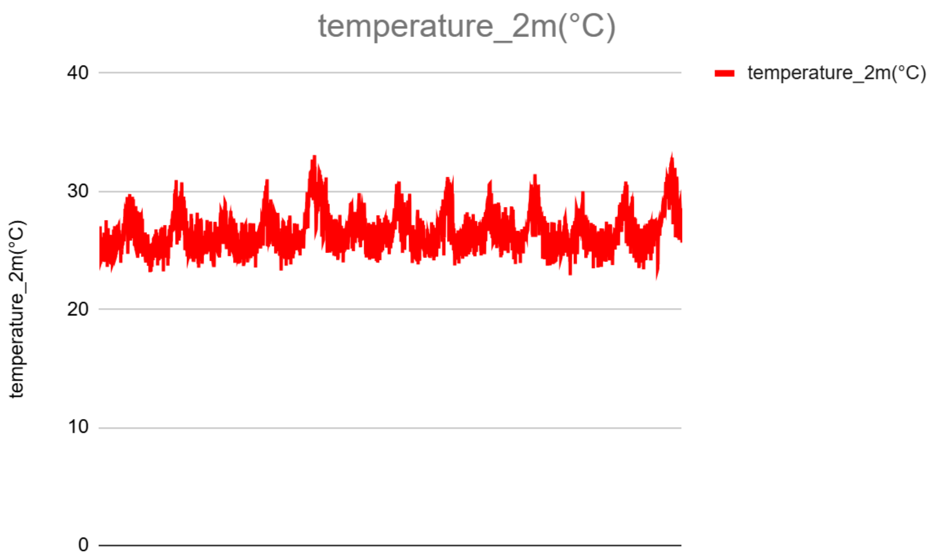
Areas with exposed soil exhibit the highest temperatures, reinforcing the direct relationship between vegetation cover and thermal modulation. Dense vegetation intercepts part of the incoming solar radiation, reducing surface heating and contributing to lower average temperatures. This dynamic highlights the importance of forest cover in local climate regulation, impacting CO_2 absorption and the efficiency of the photosynthetic process, as previously discussed by Albert et al. (2011) and Gobel et al. (2019).

In the equatorial climate context, the region exhibits high average temperatures, ranging between 26°C and 35°C. This thermal behavior is characteristic of equatorial climates, which have low annual and daily thermal amplitude. The interaction between solar radiation, vegetation cover, and the thermal properties of the soil reinforces the need for conservation strategies and sustainable vegetation management to mitigate thermal extremes and promote the stability of the regional microclimate.

Temperature Analysis at Gleba Mazuay

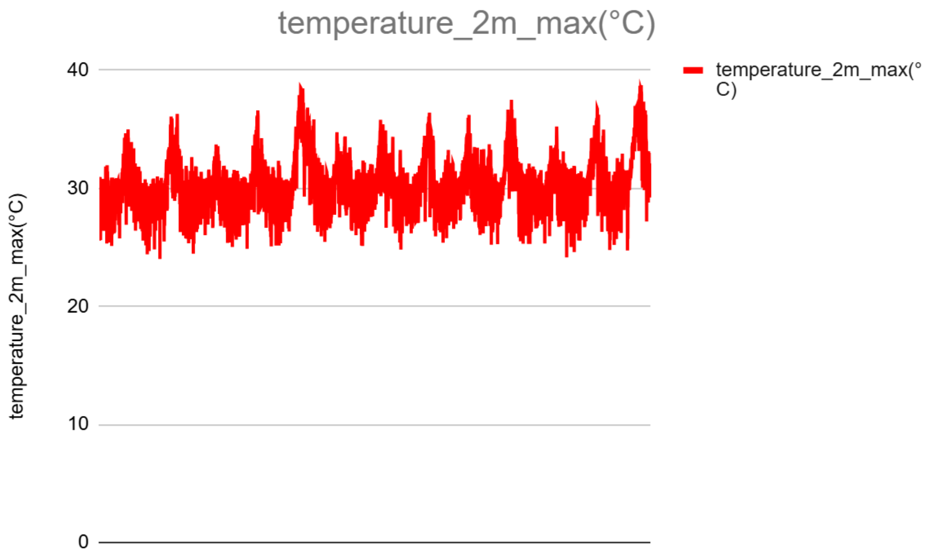
Temperature 2m (°C)

The 2m temperature (°C) graph shows the variation in average temperature throughout the year. Values range between 22°C and 30°C, with an annual average of around 26°C. Once again, seasonality is observed, with higher temperatures in the summer months and milder temperatures in winter. This information is essential for planning agricultural activities and managing crops in Gleba Mazuay.



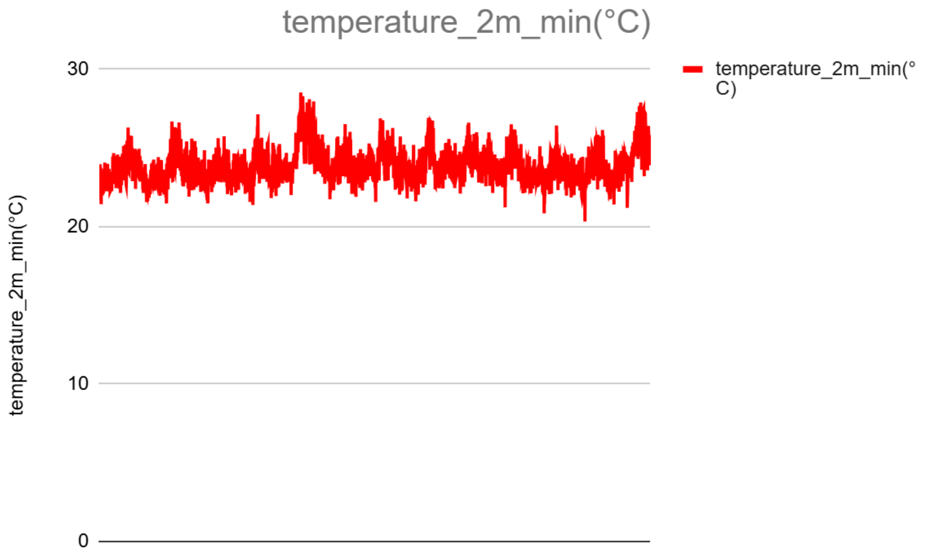
Temperature 2m max (°C)

The 2m temperature maximum (°C) graph shows that values range between 30°C and 40°C throughout the year in Gleba Mazuay. Peaks in maximum temperature occur during the summer months, from December to March, often exceeding 35°C. In contrast, during the winter months, from June to August, maximum values remain around 30°C. This seasonal variation is characteristic of the region's humid tropical climate.



Temperature 2m min (°C)

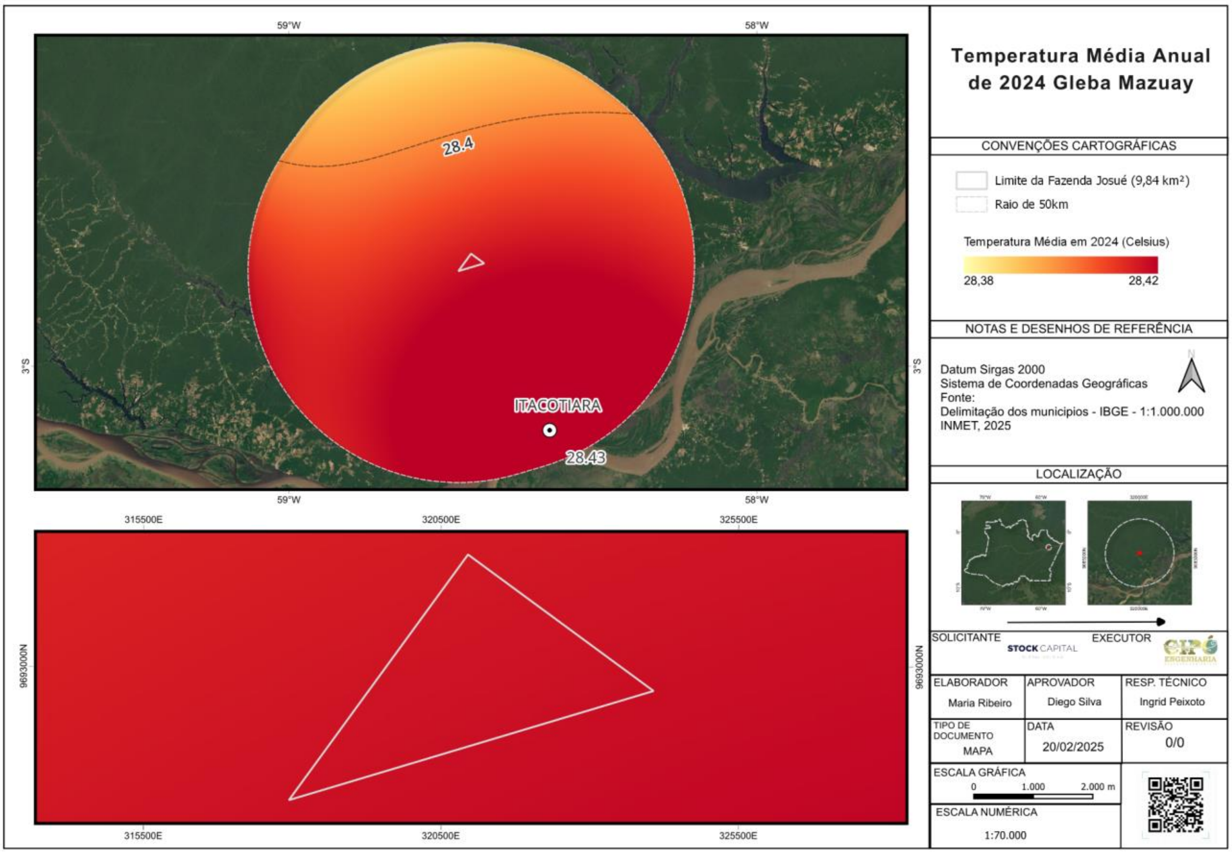
The 2m minimum temperature (°C) graph shows that values range between 10°C and 20°C throughout the year. Minimum temperatures occur during the winter months, from June to August, reaching as low as 10°C. In the summer months, minimum temperatures range between 18°C and 20°C. This seasonal variation in minimum temperatures is a key factor for selecting adapted crops and managing agricultural practices in Gleba Mazuay.



In summary, the analysis of the three temperature graphs reveals a climate pattern typical of humid tropical regions, with clear seasonality and significant variations between maximum, average, and minimum values. This information is essential for the planning and management of agricultural activities in Gleba Mazuay, aiming at adaptation to local climatic conditions.

Temperature Analysis for the Year 2024 of Gleba Mazuay

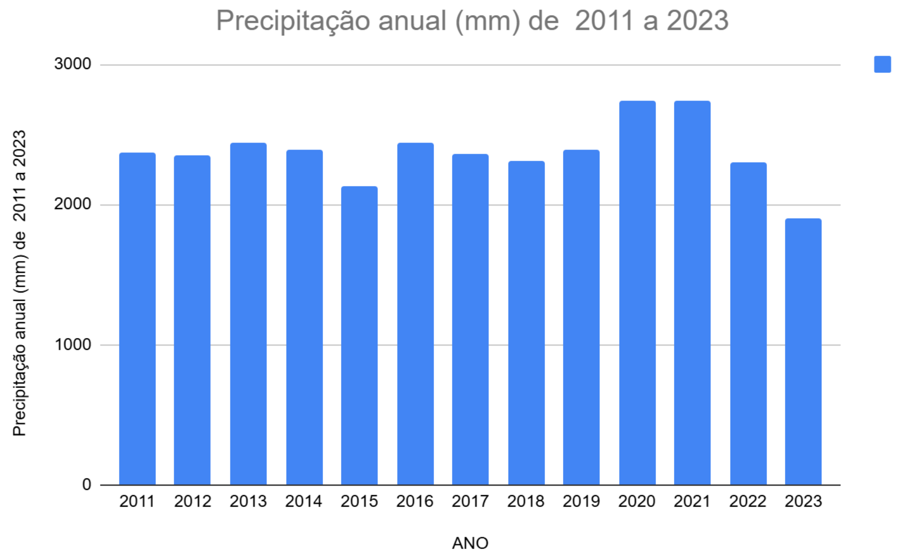
The annual average temperature recorded in Gleba Mazuay in 2024 was 28.38°C, typical of tropical climate regions. The seasonal temperature variation is moderate, with the highest temperatures occurring during the summer months and the lowest during the winter.



1.13.1.3 Precipitation

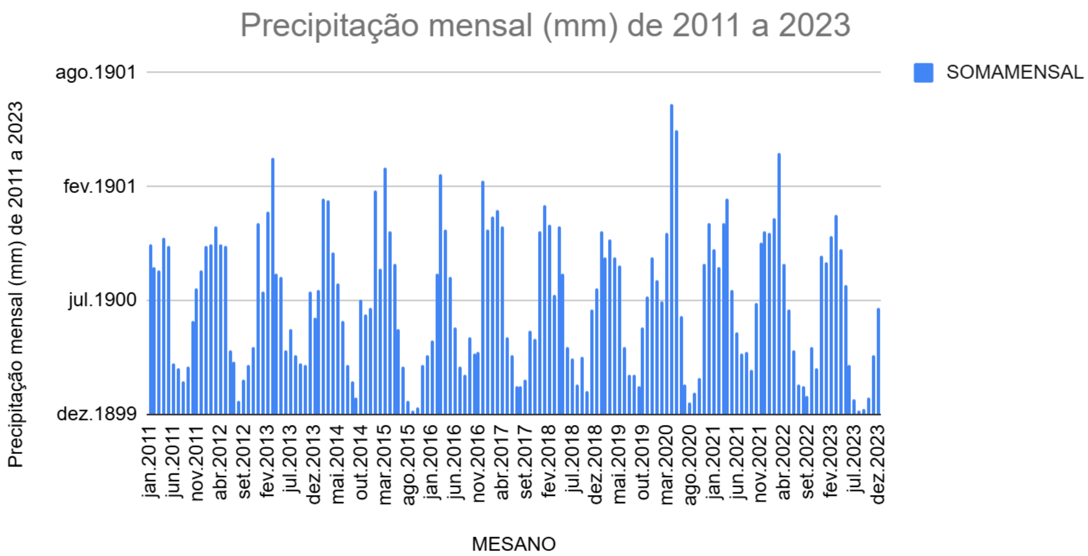
Annual Precipitation (mm) from 2011 to 2023

The annual precipitation graph in Gleba Mazuay shows significant variation over the years analyzed, from 2011 to 2023. The annual values range between 1,500 mm and 2,500 mm, with an average of around 2,000 mm. It is observed that precipitation shows a gradual increasing trend over this period, which may indicate possible changes in the region's climate pattern.



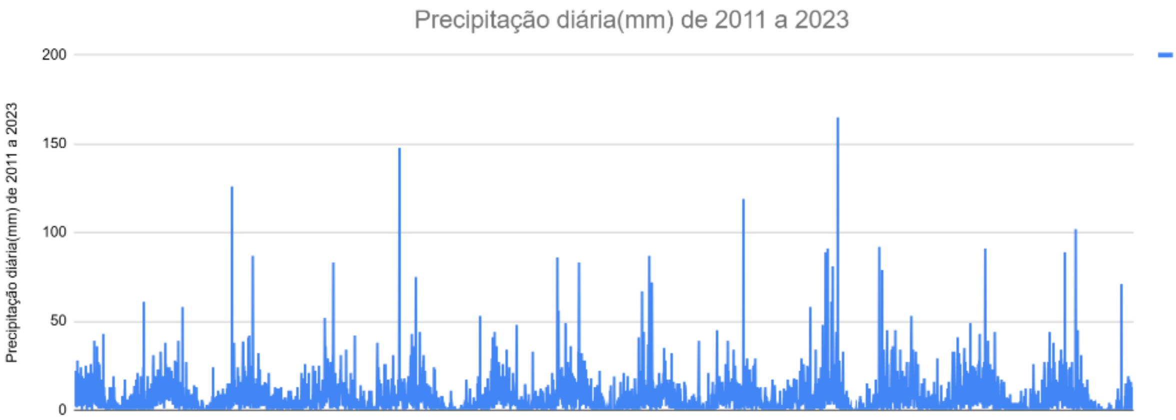
Monthly Precipitation (mm) from 2011 to 2023

The monthly precipitation graph reveals the seasonal distribution of rainfall in Gleba Mazuay. A clear distinction can be identified between the rainy season, which extends from October to April, and the dry season, from May to September. During the rainy season, monthly values exceed 200 mm, reaching peaks above 300 mm. In contrast, during the dry season, precipitation stays below 100 mm, with some months recording less than 50 mm. This information is crucial for the planning and management of agricultural activities on the property.



Daily Precipitation (mm) from 2011 to 2023

The daily precipitation graph shows the variability and occurrence of extreme rainfall events in Gleba Mazuay. Throughout the analyzed period, several peaks in daily precipitation above 50 mm can be observed, indicating the possibility of intense rainfall events. This information is relevant for the design of drainage systems, erosion prevention, and the management of crops sensitive to flooding.



In summary, the analysis of the precipitation graphs reveals a humid tropical climate pattern, with distinct seasonality and interannual variability. This information is essential for the planning and efficient management of water resources, selection of adapted crops, and implementation of sustainable agricultural practices in Gleba Mazuay.

Daily Precipitation Analysis in Gleba Mazuay

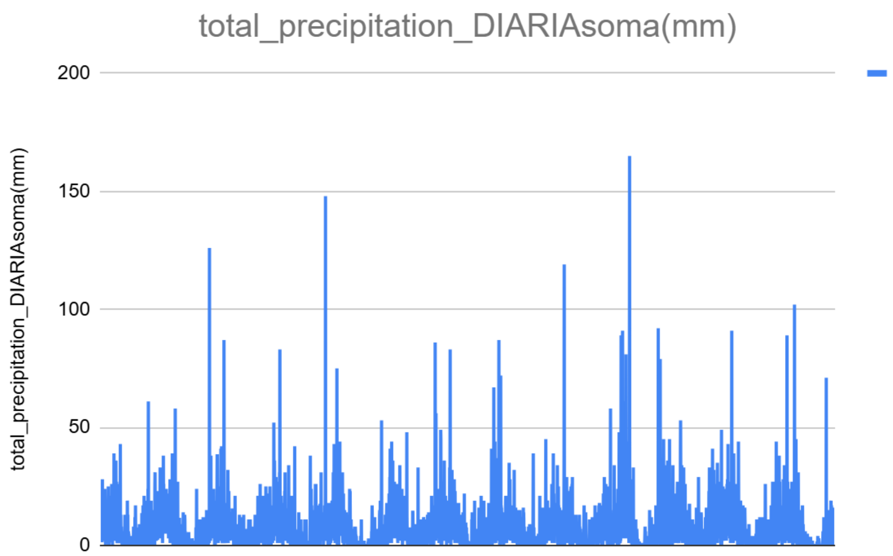
The graph presents the accumulated daily precipitation (mm) in Gleba Mazuay over the analyzed period. It is observed that daily precipitation shows significant variability, with peaks reaching over 150 mm on some days. These intense rainfall events may be associated with extreme climate phenomena such as storms, cold fronts, or convective systems.

Some days recorded precipitation exceeding 100 mm, which may represent a risk of flooding, soil erosion, and damage to crops and property infrastructure. Esses eventos extremos de precipitação exigem atenção especial no planejamento e gestão das atividades agrícolas.

On the other hand, it is also possible to identify periods with lower daily precipitation, below 50 mm. These drier conditions can affect water availability for crops and agricultural productivity, requiring the adoption of efficient water management practices, such as irrigation.

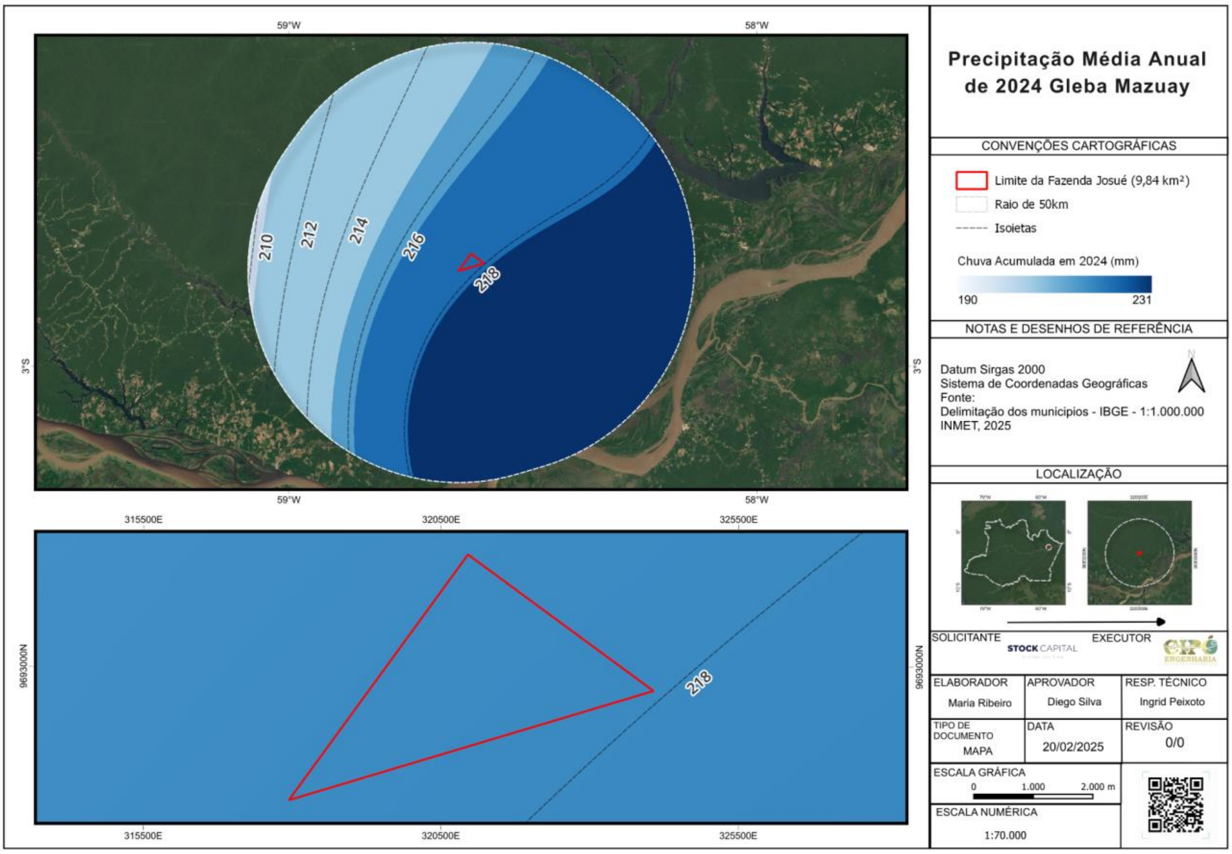
This analysis of daily precipitation variability is crucial for the development of adaptation and resilience strategies in Gleba Mazuay. Continuous monitoring of these extreme events, the implementation of adequate drainage systems, and the planning of resilient crops and agricultural practices are some of the challenges to be addressed.

In summary, the daily precipitation graph highlights the need for careful water resource management in Gleba Mazuay, aiming to balance periods of intense rainfall and drought, ensuring the sustainability and productivity of agricultural activities.



Precipitation Analysis for the Year 2024 in Gleba Mazuay

The average annual precipitation in Gleba Mazuay in 2024 was 2,212 mm. The seasonal distribution of rainfall is well-defined, with a rainy season (summer) and a dry season (winter). The months from January to April and October to December recorded the highest rainfall values, while the months from June to September tend to be drier.



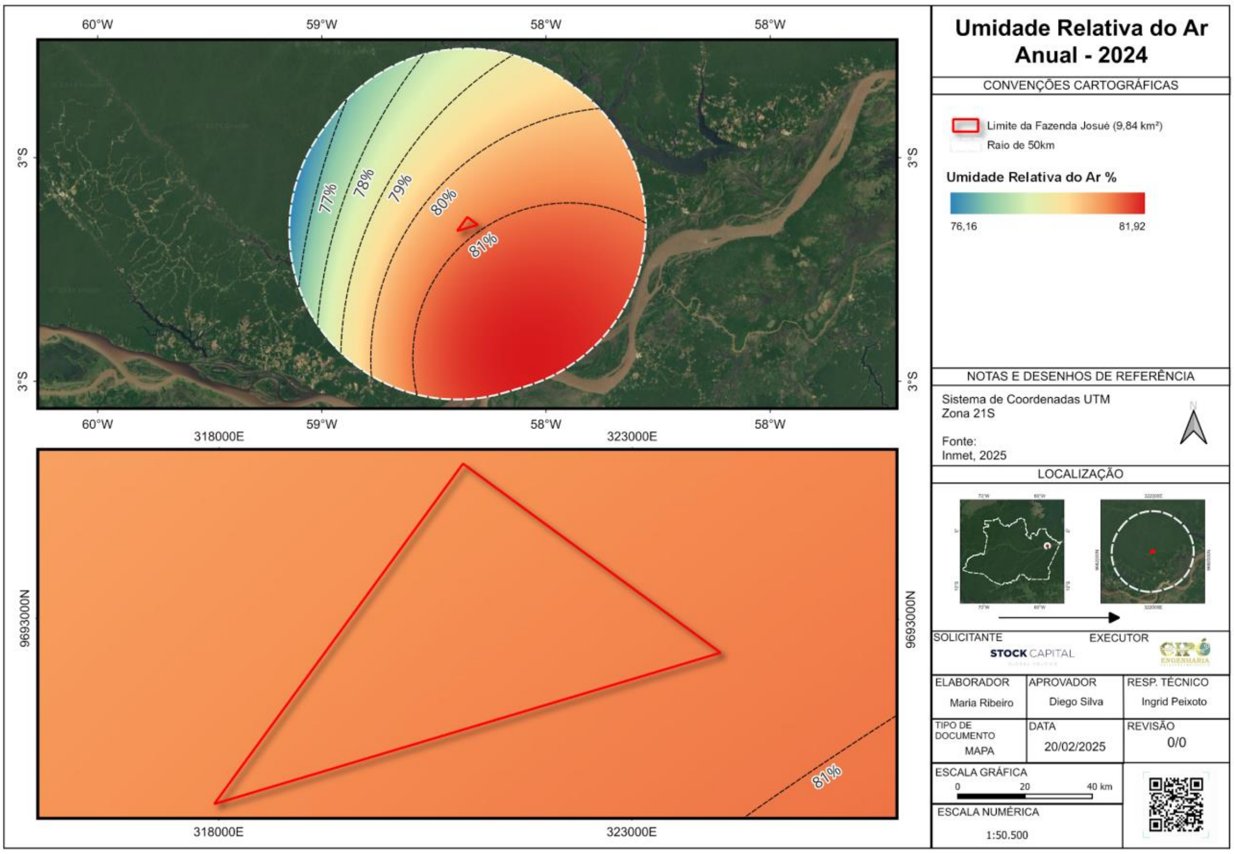
Extreme Events

Some years stand out due to the occurrence of extreme precipitation events, such as 2020 and 2021, which recorded the highest annual values during the analyzed period (2011-2023). These events may be related to large-scale climatic phenomena, such as the El Niño-Southern Oscillation (ENSO).

1.13.1.4 Relative Humidity

The map presents the annual distribution of relative humidity for the year 2024 in the Gleba Mazuay area. The data indicates a variation in relative humidity, with values ranging between 76.16% and 81.92%, as represented by the color bands on the map. The southern regions of the study area show the highest humidity levels, while the northern areas record slightly lower values.

Relative humidity is an essential factor in the region's climatic dynamics, directly influencing temperature, evapotranspiration, and precipitation patterns. In regions with a tropical humid climate, such as the Amazon, humidity tends to remain high throughout the year, with small seasonal fluctuations. However, variations can occur due to the influence of air masses and regional climatic phenomena.

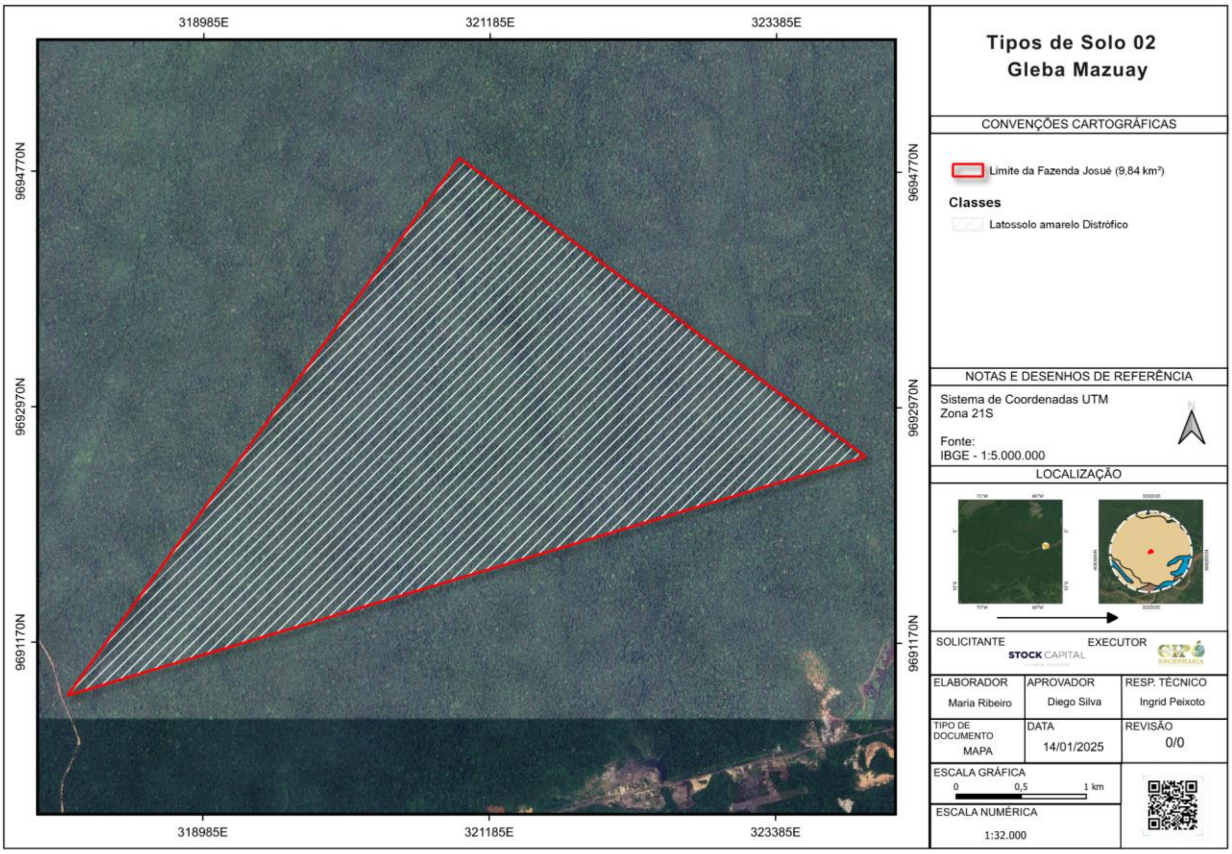


1.13.1.5 Soils

Three main classes were identified: Haplic Gleisol, Clay with High Eutrophic Activity, Dystrophic Yellow Oxisol and Continental Water Mass.

- **Gleysols:** These soils are typically associated with areas of permanent or periodic water saturation, which is reflected in their chemical and physical characteristics. These soils are often found in poorly drained areas or places subject to flooding. The constant presence of water can lead to soil compaction if management is inadequate, directly affecting its structure and the circulation of air and water. From an environmental perspective, Gleysols play a fundamental role as water recharge areas, regulating the flow of water to adjacent ecosystems.

- **Yellow Oxisol:** It occupies the largest portion of the area and is characterized as a deep, well-drained soil with low natural fertility. This soil class is formed under intense leaching in humid tropical climates, which explains its yellowish color resulting from the accumulation of iron and aluminum oxides. From an ecological perspective, Oxisols play an important role in supporting tropical forests and are also suitable for agroforestry systems and sustainable management practices that respect their regenerative capacity.



Land Use and Cover

The use and soil cover map shows a predominance of forest formations, but with a significant diversity of other forms of use that reflect different ecological and anthropogenic dynamics in the region.

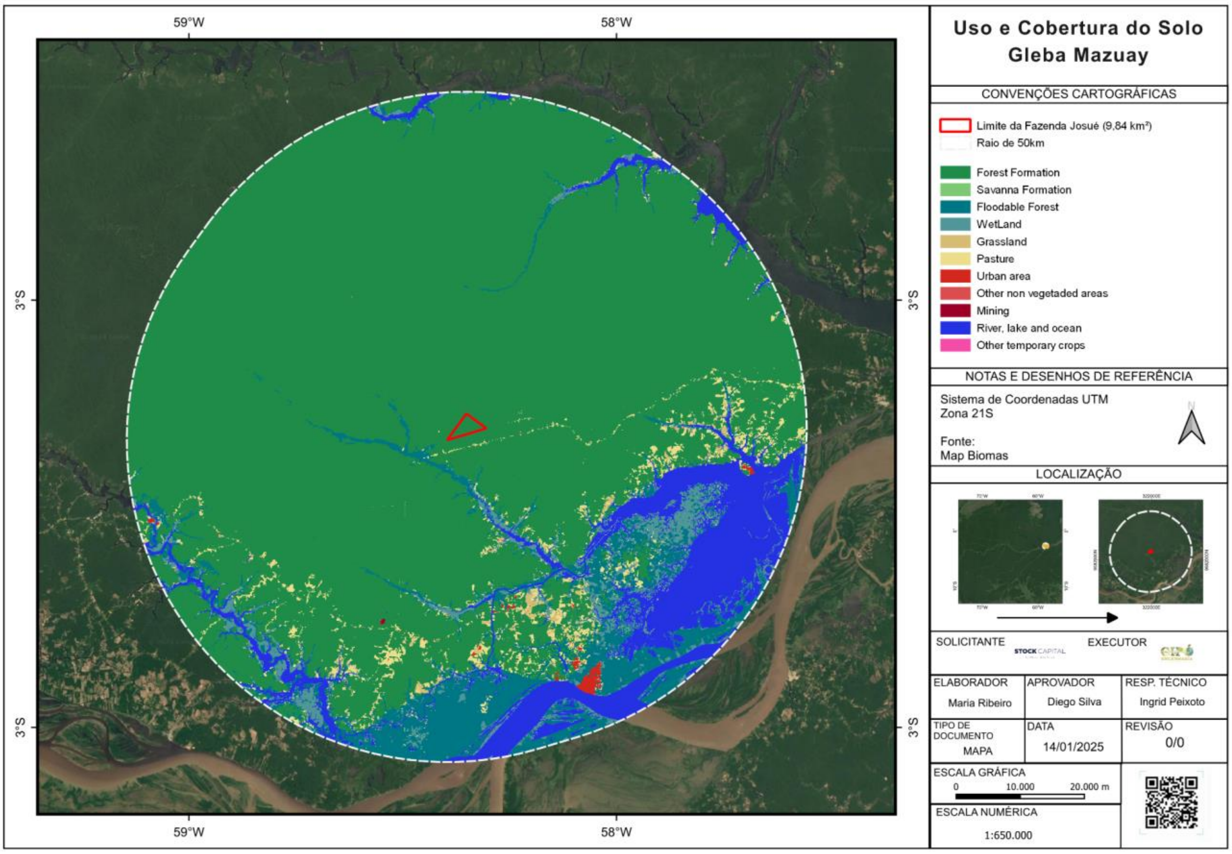
- **Forest Formation:** It occupies the largest part of the area within the 50 km radius and within the Gleba, representing the Amazonian Forest ecosystem. This formation is crucial for the conservation of local biodiversity and the balance of hydrological and climatic cycles. Forests act as carbon sinks, essential for mitigating the effects of climate change.

- **Pasture:** areas of pasture are present around the Gleba, suggesting that the region is also used for agriculture and livestock activities, such as cattle ranching. The conversion of forest areas into pastures is a common practice in the Amazon, but it can result in biodiversity loss, soil degradation, and greenhouse gas emissions if not managed sustainably.

- **Grassland Formation:** Appears in smaller areas, characterized by open and herbaceous vegetation. These areas may be used for grazing or less intensive agricultural activities, but they may also represent important natural ecosystems, such as fields and cerrado.

- **Flooded and Wetland Areas:** are observed near bodies of water, such as the Urubu and Amazonas rivers. These ecosystems are vital for maintaining both aquatic and terrestrial biodiversity, acting as refuge zones for various species and regulators of the local hydrological regime.

- **Rivers, Lakes, and Ocean:** Water bodies, including the Anebá, Caru, Caribê, Sanabani, Urubu rivers, and the Amazonas River, play an essential role in the landscape dynamics. They not only provide water resources and support biodiversity, but are also fundamental for human activities such as transportation, fishing, and potentially irrigation. Furthermore, it is evident that a large portion of the urban area shown on the map is located near the Amazonas River.

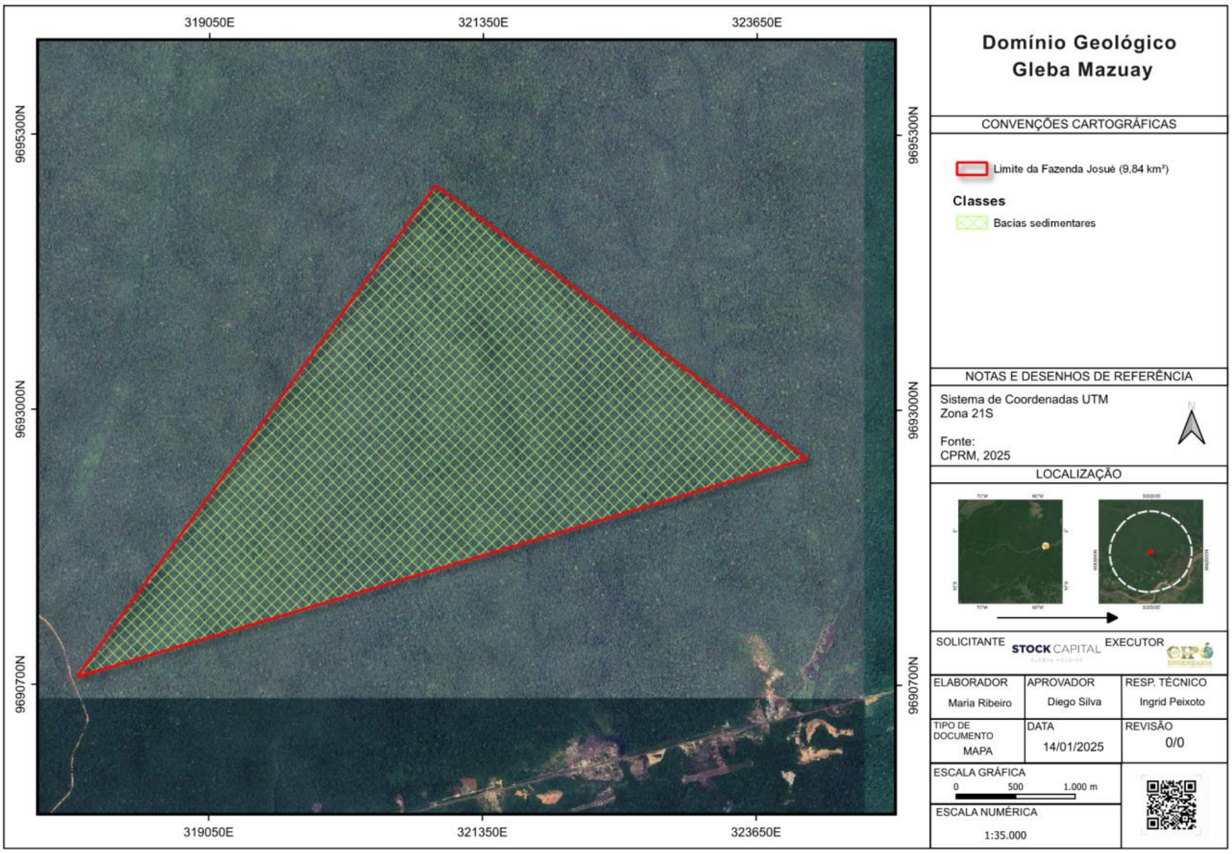


1.13.1.6 Geological Domains

To conduct an environmental analysis linking geological and hydrogeological domains, it is essential to understand how geological formations influence the hydrogeological characteristics of a region.

In the study area, the predominance of geological classes from sedimentary basins and Cenozoic formations is observed, along with alluvial deposits and the Cenozoic Alter do Chão Formation for hydrogeological domains.

The alluvial deposit is composed of sediments such as sand, silt, and clay, which are typical of fluvial areas that accumulate material transported by watercourses.

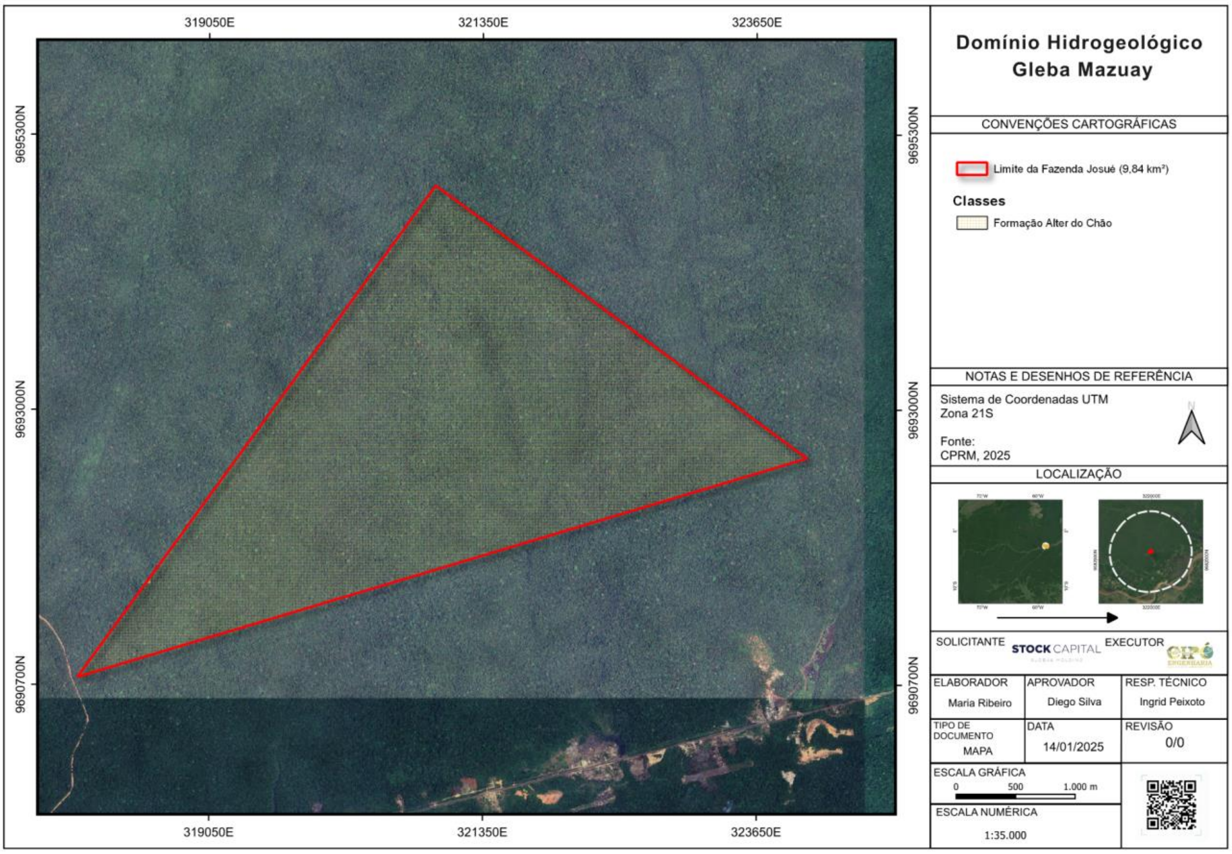


These Cenozoic formations are relatively recent and result in soils with varied characteristics. The presence of sand, with its high permeability, facilitates water infiltration and the recharge of local aquifers, contributing to a water productivity that can range from low to moderate. However, silt and clay, with their lower permeability, limit this infiltration, leading to greater water retention on the surface and increasing the risk of waterlogging and erosion during periods of intense rainfall.

On the other hand, the sedimentary basin with the Alter do Chão Formation is characterized by layers of medium to coarse sandstone and mudstone.

Sandstone, with its high porosity and permeability, functions as an excellent aquifer, providing high groundwater productivity. This storage and water flow capacity is a significant advantage for water supply, making the area valuable for human and agricultural use. However, this same characteristic can be a vulnerability, as it facilitates the rapid spread of contaminants, requiring strict pollution protection measures. Claystone, with its lower permeability, can act as a confining layer, helping to limit the vertical flow of water and partially protecting the aquifer from surface contamination.

In the alluvial deposit, attention should be focused on drainage and flood prevention, while in the sedimentary basin, the priority should be the sustainable management of the aquifer, ensuring water quality and availability in the long term. This analysis allows for better planning of land use and water resources, promoting practices that minimize environmental impacts and ensure the ecological balance of the region.



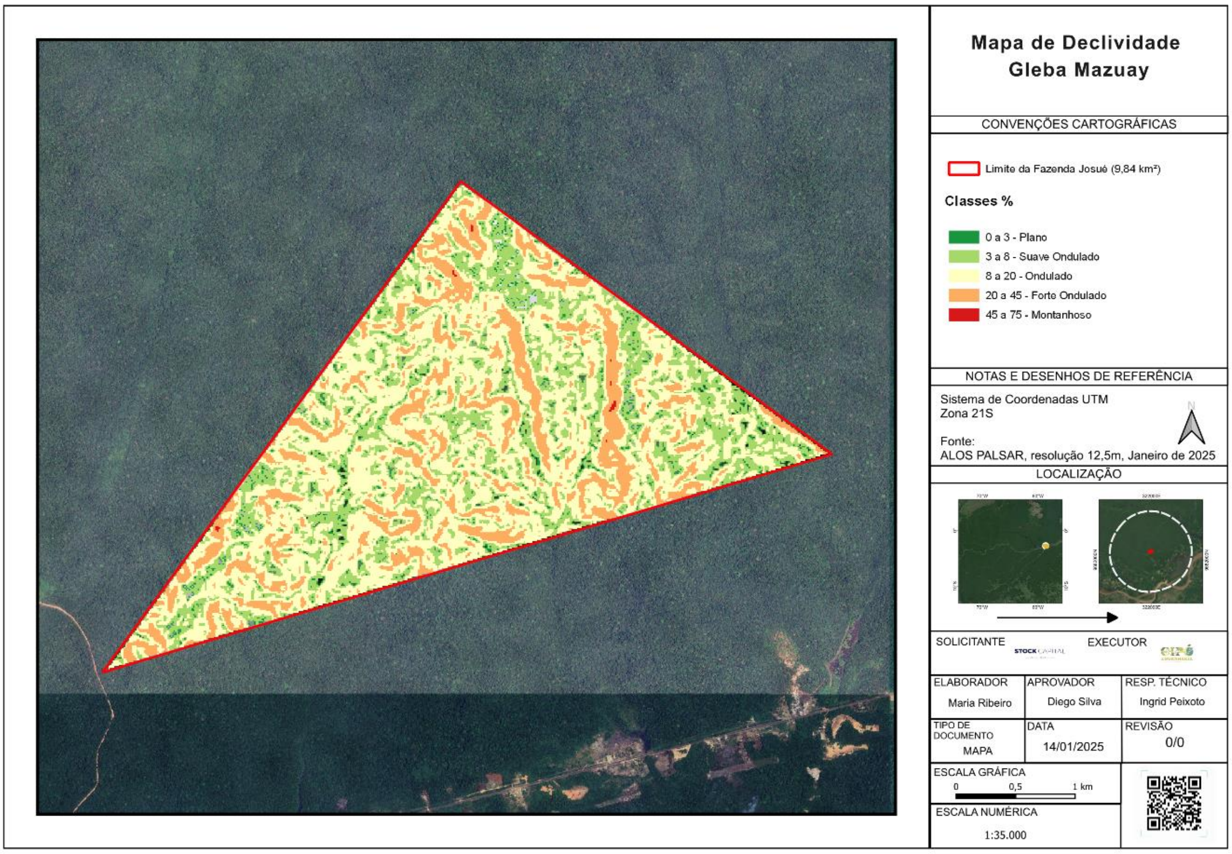
Slope Map and Hypsometric Map with Contour Lines

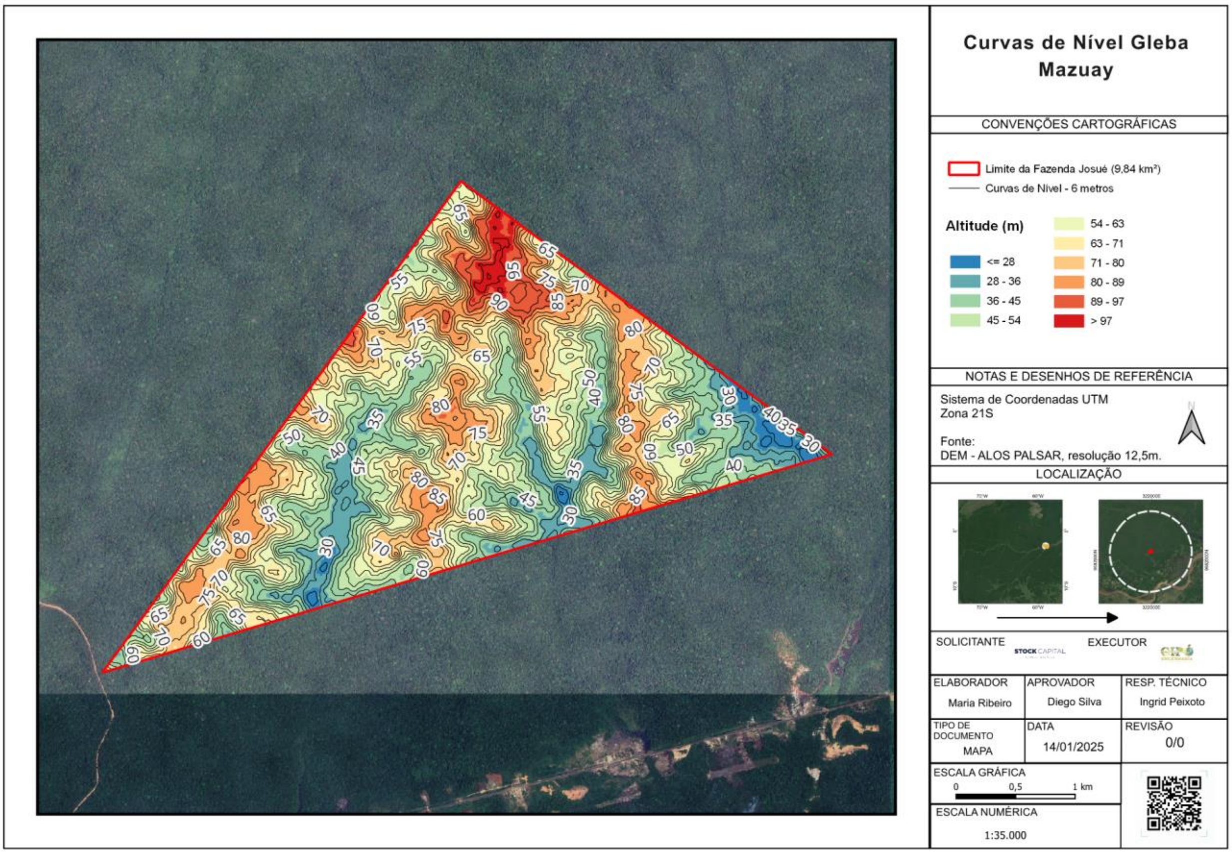
According to the slope map, we can identify a variation in the slope classes, as per the map legend. The slope is presented as a percentage, with the following classification according to EMBRAPA:

- 0 to 3%: Flat
- 3 to 8%: Gently Undulating
- 8 to 20%: Undulating o
- 20 to 45%: Strongly Undulating
- 45 to 75%: Mountainous

The analysis of the relief, based on these classes, allows for a better understanding of the spatial distribution of the land inclinations.

The hypsometry, presented on the level curve map, illustrates the altitudes through colors and contour lines, representing altitudes between 30 and 80 meters. This graphic method is crucial for understanding the relief and visualizing the altimetric distribution of the terrain. The contour lines were obtained using the ALOS PALSAR digital elevation model, with a resolution of 12.5 meters, providing a detailed representation of the topography.





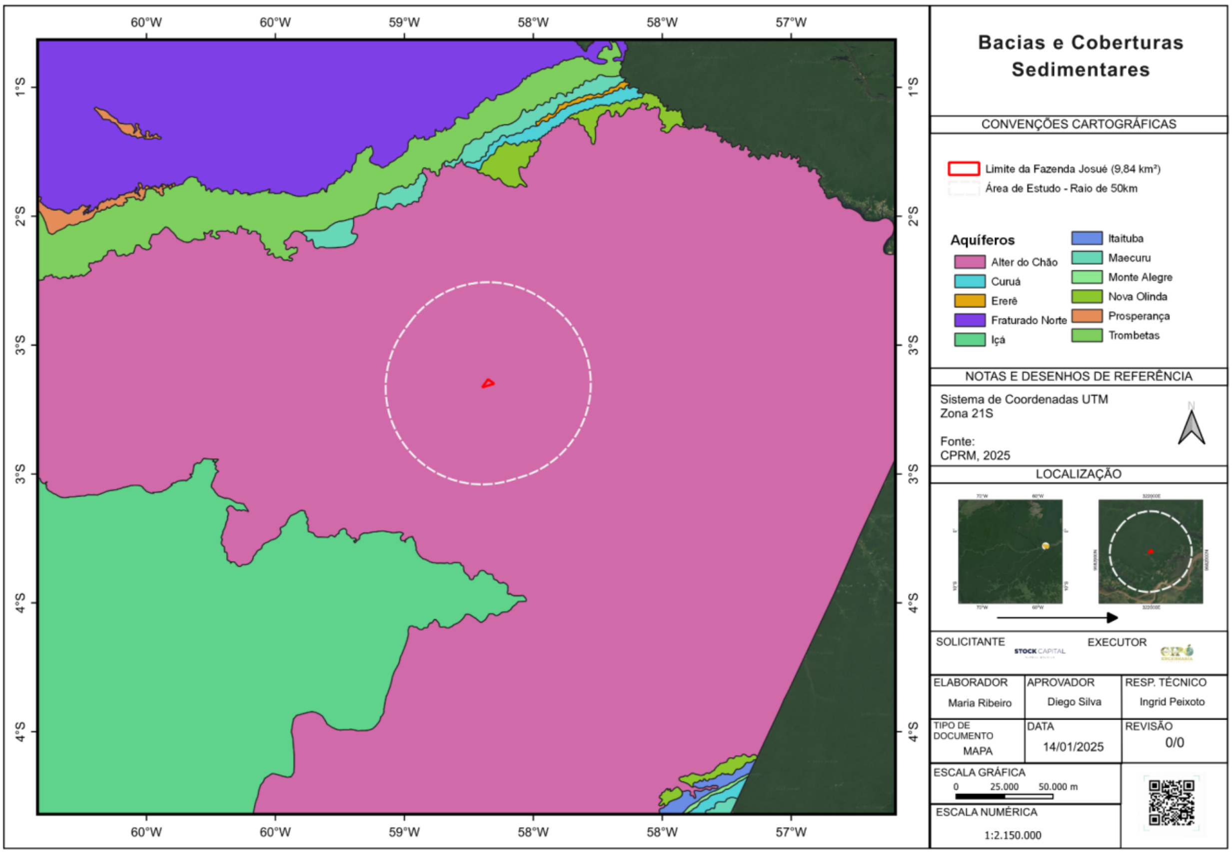
1.13.1.7 Sedimentary Basins

On the Sedimentary Basins Map, we have the main representations of the aquifers surrounding the study area. Regarding the study area, it is situated in the Alter do Chão aquifer, a widely recognized nomenclature.

According to previous studies conducted by CPRM (2015), the Alter do Chão Aquifer is located in the Amazon region, covering a vast area of the Amazon Basin. It has an approximately irregular shape with an area of about 437,500 km^2, being considered one of the largest aquifers in the world. Due to the presence of medium to coarse sandstone layers with thicknesses that can range from 100 to 300 meters, this unit is extensively exploited as a source of groundwater, mainly for urban and rural water supply. The presence of clay layers intercalating the sandstone layers gives the aquifer a semi-confined character, though it can be considered unconfined in some specific regions. The waters from this system are suitable for human consumption.

According to CPRM (2015), based on the correlation of geological profiles from various wells in the area, it was possible to identify three main horizons with a predominance of sandy layers within the system: the first is within the first 100 meters of depth, the second between 100 and 200 meters, and the third below 300 meters. The water quality in this aquifer system is notably good, with low levels of cations and anions; in most wells in this region, the water is classified as sodium-bicarbonate, with Na^+ and HCO_3^- concentrations generally below 10 and 50 mg/L, respectively. In some wells, however, K+ concentrations are significant, with a maximum of 6 mg/L. The low ionic concentrations are reflected in the electrical conductivity values, with a minimum near 15 μS/cm and a maximum around 120 μS/cm.

The Alter do Chão Aquifer System is represented by the sediments at the top of the sedimentary sequence of the Amazon Basin, having a significant recharge area that covers various regions of the Amazon. The water quality is considered excellent, although there are concerns related to the natural vulnerability to pollution, especially in urban areas due to the shallow water table and the lack of basic sanitation, as well as contamination risks associated with poorly constructed wells. Proper management and protection of this water resource are crucial to ensure its long-term sustainability.



1.13.2 Characterization of the Surrounding Area

1.13.2.1 State Highway Section

The Mazuay Gleba, located in the municipality of Silves, Amazonas, is situated in a complex environmental context, characterized by extensive areas that are relevant to the environmental, economic, and social context of the region. The environmental analysis was conducted based on a 50 km radius buffer, covering the neighboring municipalities of Itapiranga to the north and Itacoatiara to the south. This spatial delimitation allowed for a detailed assessment of land use and cover, vegetation remnants, and the environmental restrictions present in the region.

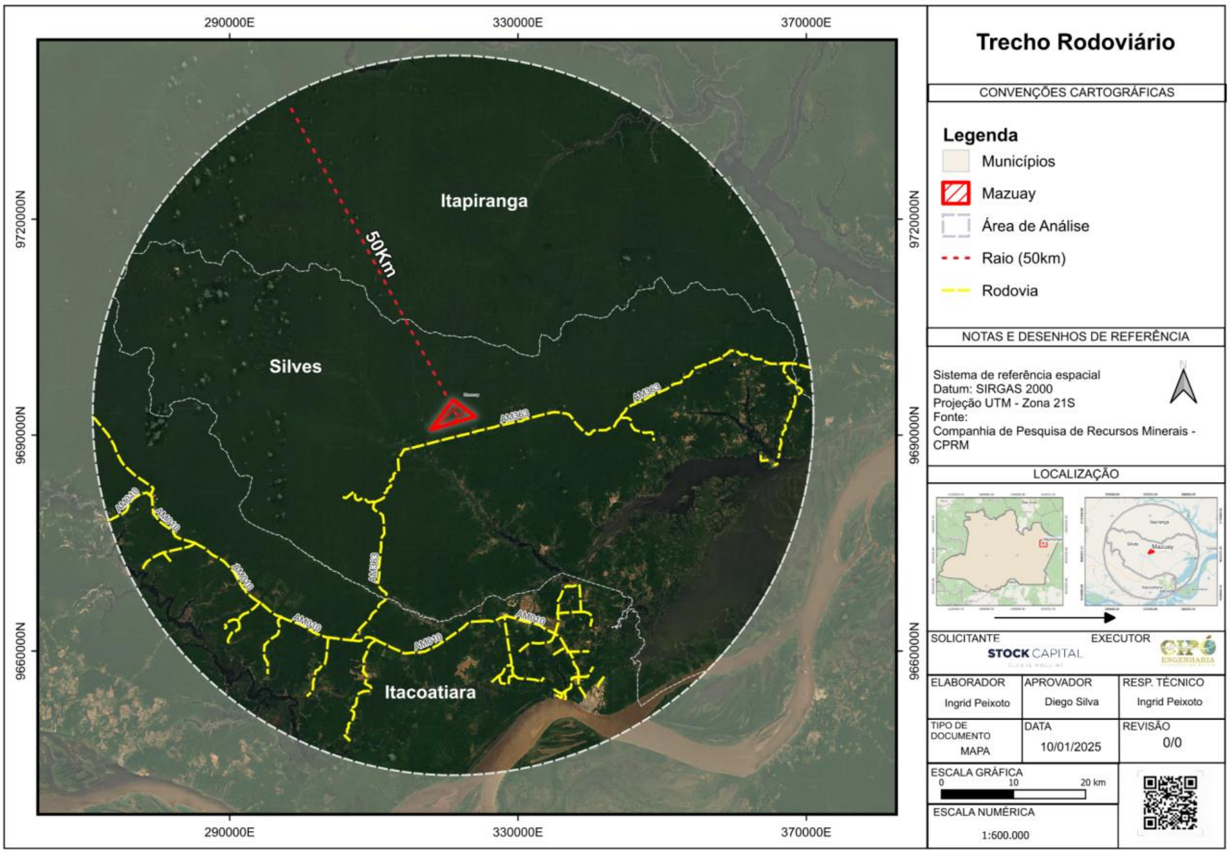
The analysis of the road map near the Mazuay Gleba highlights the importance of the AM-363 highway in regional connectivity and the environmental impacts associated with its presence. This road plays a crucial role in the territorial and economic integration of the municipality of Silves and its neighboring areas, facilitating the flow of goods, supplies, and people. However, the presence of road infrastructure in environmentally sensitive regions can pose ecological challenges that must be properly managed.

The paving and maintenance of the AM-363 highway have direct environmental implications, including habitat fragmentation, increased surface runoff, and a higher risk of soil and water body contamination due to waste and effluents from traffic. Along the margins of the AM-363, there are consolidated areas, communities, industries, and even health centers, reinforcing the need for integrated environmental planning to mitigate adverse impacts and promote sustainable land use practices.

Unregulated occupation along the highway margins can exacerbate environmental degradation, intensifying erosive processes and altering natural hydrological patterns. Road construction facilitates access to previously isolated areas, which may increase pressure on natural resources, leading to deforestation and unsustainable exploitation of timber and minerals (FERREIRA et al., 2019).

Given this scenario, it is essential that the management of AM-363 is guided by principles of sustainability and impact mitigation. Strategies such as the creation of ecological corridors, the implementation of effective drainage systems, and the monitoring of environmental quality along the highway are fundamental to ensuring a balance between economic development and environmental preservation (OLIVEIRA et al., 2018).

Finally, the analysis of the presence of AM-363 near Gleba Mazuay presents both challenges and opportunities. The connectivity provided by the road can be a positive factor for the region’s logistics; however, it is essential to ensure that its environmental impact is minimized through strategic actions that take into account the ecological vulnerability of the Amazon and the needs of local populations.

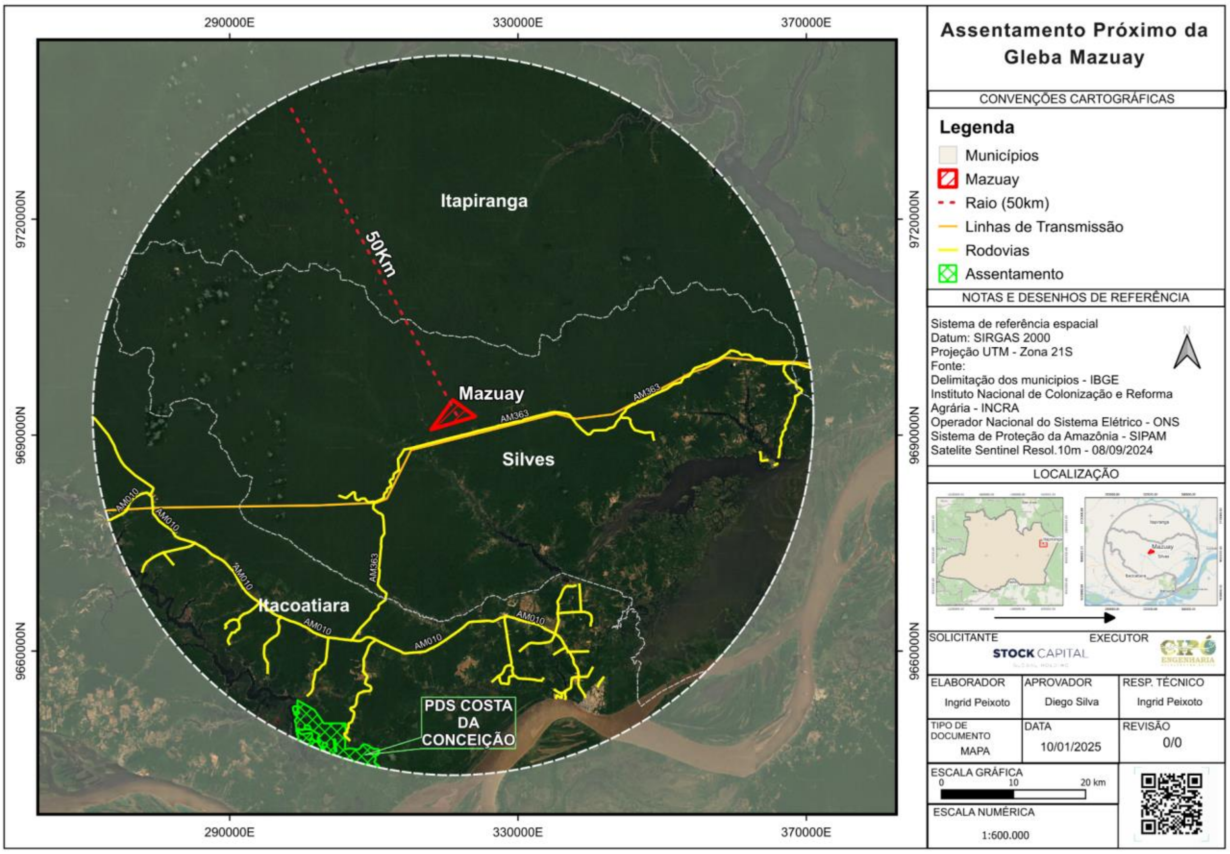


1.13.2.2 Settlements

The first map reveals that the nearest settlement to Gleba Mazuay is located in Itacoatiara, approximately 47 km away. This dispersion indicates low population density and may be associated with restrictions imposed by conservation areas and the presence of natural resources (IBGE, 2023).

Additionally, the proximity of conservation areas may influence the spatial organization of settlements, as environmental policies restrict occupation in certain areas. According to the Embrapa Amazônia Ocidental report (2024), regions like this require territorial planning initiatives that reconcile the sustainable use of natural resources with the encouragement of population settlement in strategic areas.

On the other hand, the dispersion of human settlements may also be related to the search for fertile and accessible land in areas outside conservation units. Studies by Almeida and Santos (2021) indicate that local populations often rely on land near water bodies for subsistence, highlighting the importance of protecting these areas to prevent land conflicts and adverse environmental impacts.



1.13.2.3 Fire Events

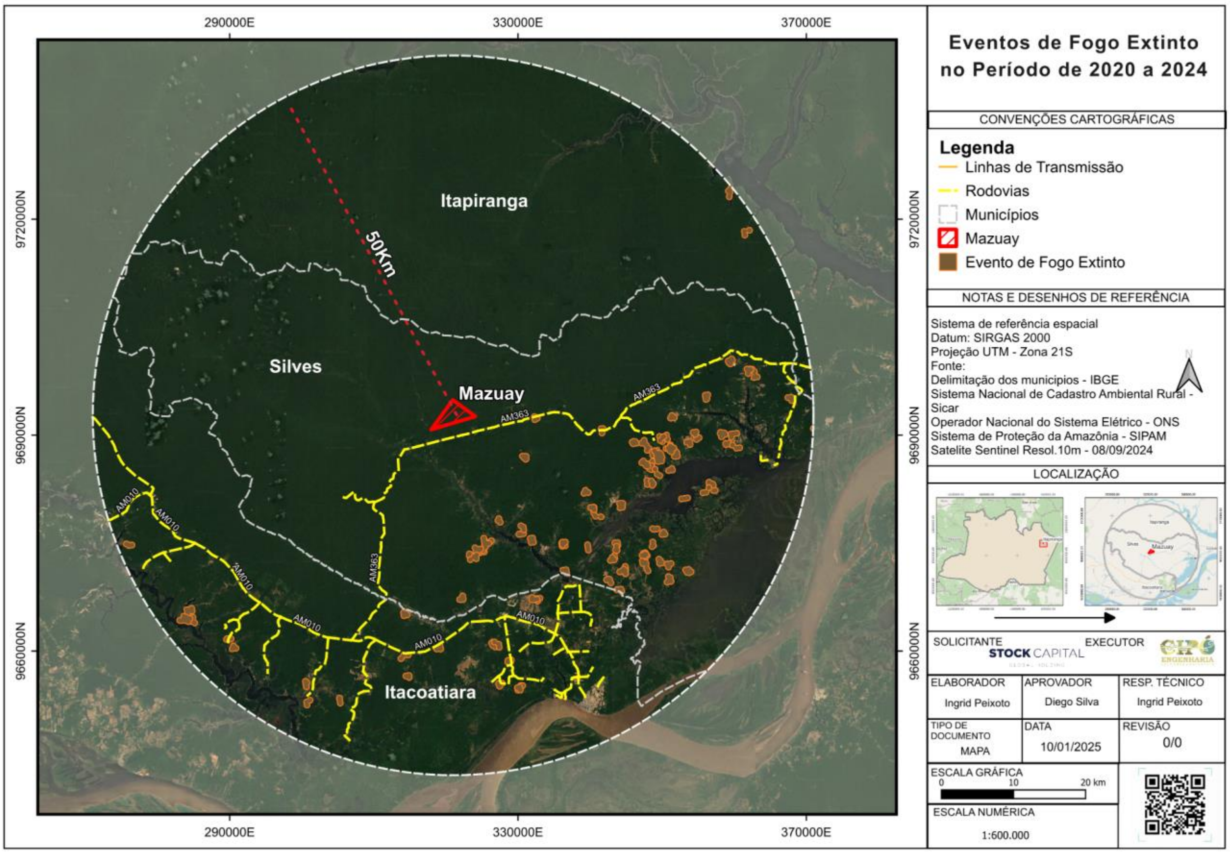
Data from the Management and Operational Center of the Amazon Protection System (CENSIPAM) show that, in 2023, fire events occurred primarily around the Urubu, Caru, and Anabá rivers, as well as near the AM-363 highway. These events are often associated with deforestation and controlled burns for agricultural expansion. The use of fire as a land management tool is a recurring practice in the Amazon region, but its implications are severe for environmental balance.

Burning events directly affect soil quality, reducing its fertility due to the loss of organic matter and essential nutrients (Fearnside, 2021). Moreover, they contribute to the release of large amounts of greenhouse gases, such as carbon dioxide (CO_2) and methane (CH_4), exacerbating global climate change (INPE, 2023).

Another significant impact is on local biodiversity. Areas affected by fire, especially those near bodies of water such as the rivers mentioned, suffer habitat loss for numerous aquatic and terrestrial species. Studies by Aragão et al. (2020) highlight that recurrent fire events lead to ecosystem degradation, promoting the replacement of native forests with secondary vegetation or pasture areas.

The proximity of the fires to the AM 363 highway also suggests the possibility of accidental or intentional fires associated with land clearing for agriculture and livestock.

This calls for monitoring and prevention strategies, such as the implementation of early warning systems and awareness-raising actions for local communities about the environmental and socio-economic risks of burning.

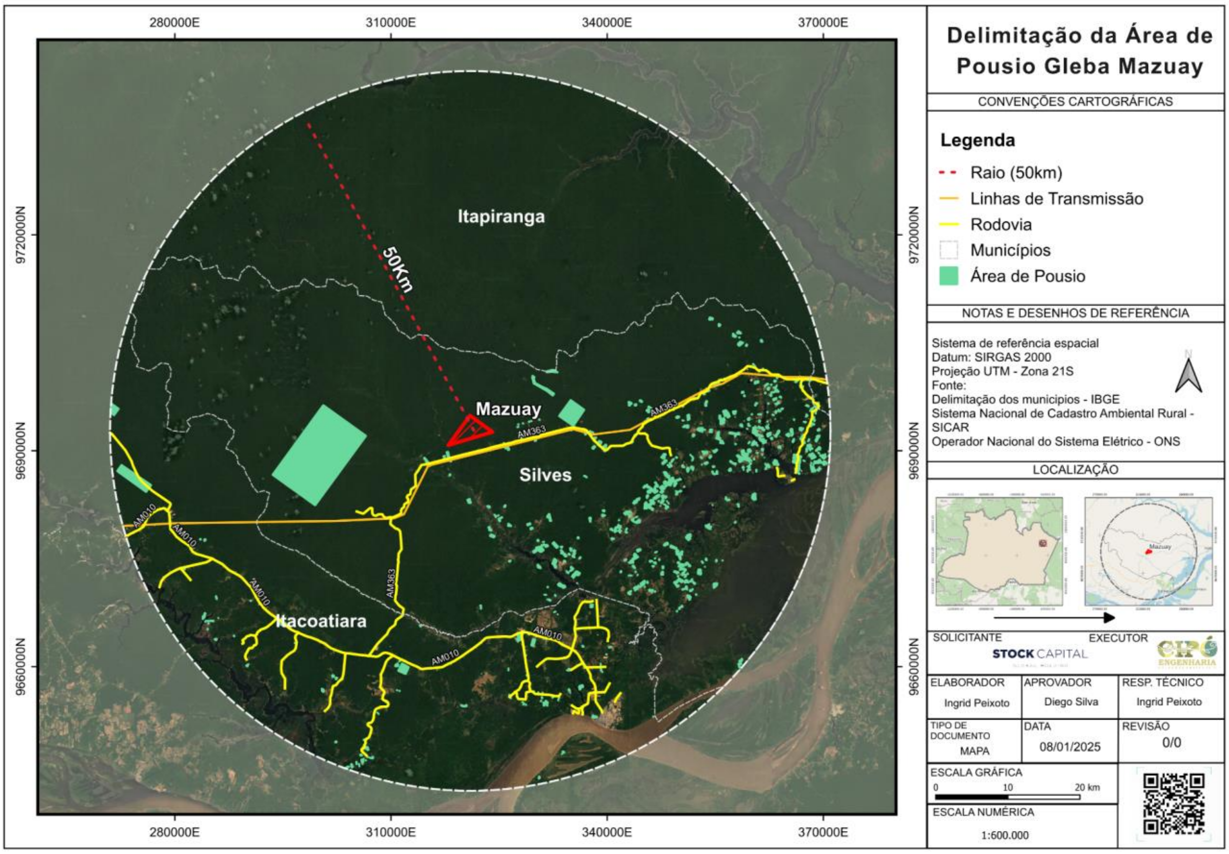


1.13.2.4 Fallow Area

The map shows the delineation of fallow areas in Gleba Mazuay, highlighting that both to the east and west there are significant extensions of this land use category. Fallow refers to the period when land is left in rest for the natural recovery of soil fertility and ecological structure (Vieira et al., 2021).

This practice is common in sustainable agricultural systems, especially in the Amazon, where continuous land use can lead to soil degradation and nutrient loss (Sá et al., 2020). Fallow helps in the recovery of organic matter and the restoration of essential ecological processes, such as nutrient cycling and the regeneration of native vegetation.

However, the effectiveness of this strategy depends on the duration of the fallow period and the absence of degradation factors, such as wildfires or soil compaction (Cerri et al., 2019). The monitoring of these areas can be carried out through satellite imagery and remote sensing analysis, ensuring that the fallow is fulfilling its environmental and productive function.



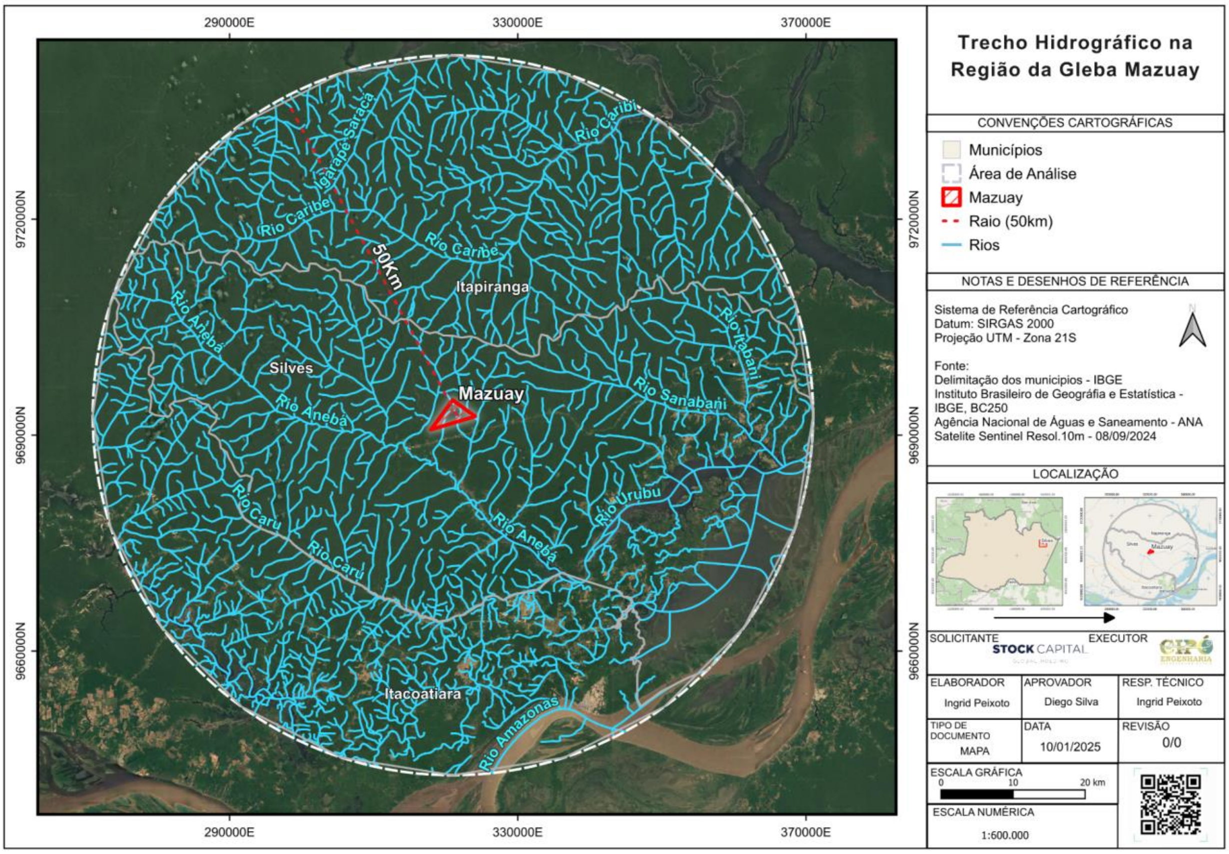
1.13.2.5 Water Resources

The rivers that make up the area include the Caribé, Sambani, Igarapé Iguaçu, and others. These watercourses play a crucial role in supporting local biodiversity, supplying communities, and maintaining the hydrological cycle. The presence of springs within the Gleba Mazuay, such as those feeding the Anabá River, highlights the need to preserve these areas to ensure the continuity of water flow and water quality.

The region's rivers are also essential for the subsistence of riparian communities, who rely on these resources for fishing, transportation, and domestic consumption. Studies by Nascimento et al. (2020) highlight those riparian areas, such as those found in the Gleba Mazuay, are important habitats for aquatic species and serve as ecological corridors that connect different ecosystems.

Additionally, the proximity to major rivers such as the Amazon increases the potential for connectivity of Gleba Mazuay with river transportation routes, which are essential for logistics in the Amazon region. However, the conservation of these areas faces challenges, such as sedimentation caused by deforestation and improper land use. According to Silva and Santos (2021), sustainable management practices, including the maintenance of riparian vegetation, are essential to mitigate environmental impacts and ensure the integrity of these water bodies.

Another important concern is the impact of mining and fires on local water resources. Mining activities, common in nearby areas, can lead to contamination by heavy metals, while fires increase the load of sediments and nutrients in rivers, altering their quality and biodiversity. Integrated management strategies, such as continuous water quality monitoring and the establishment of protection zones, are recommended to preserve these vital resources (ANA, 2023).



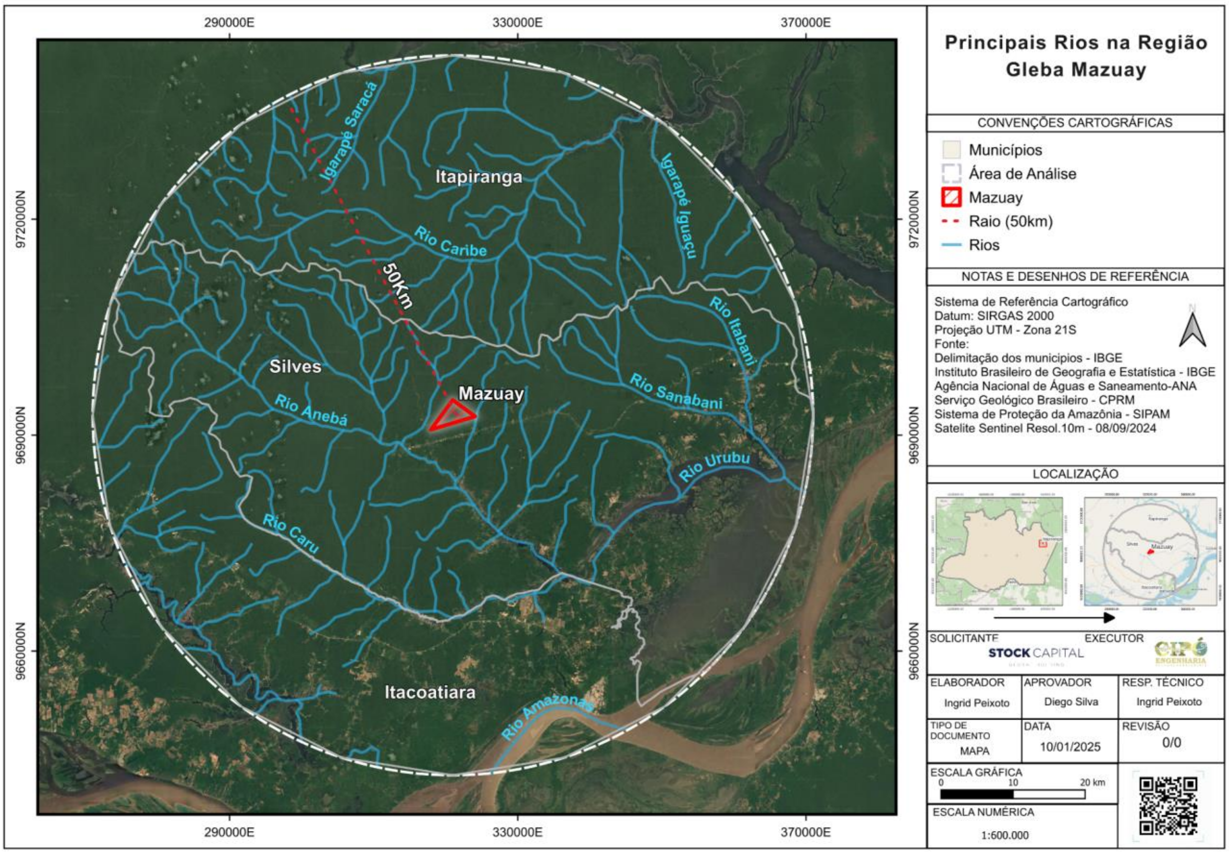
Main Hydrographic Sections Influencing Gleba Mazuay

The hydrographic analysis of the Gleba Mazuay region highlights the presence of significant water bodies within a 50 km radius. Among the most important rivers are the Anebá River, which has tributaries flowing through the area of Gleba, the Urubu River, the Sanabani River, the Caru River, the Caribe River, and the Amazon River. In addition to these larger rivers, two igarapés were identified to the north of the region: the Saracá Igarapé and the Iguaçu Igarapé.

The presence of these water resources plays a fundamental role in the environmental dynamics of Gleba Mazuay. The Amazon River, being one of the largest in the world, has a significant influence on the region's humidity, rainfall regime, and groundwater recharge. Smaller rivers, such as the Anebá and its tributaries, have essential functions in draining the area and supplying water for agricultural activities.

The protection of these watercourses is essential to ensure the sustainability of productive activities and the conservation of local biodiversity. This requires proper environmental management practices, such as maintaining riparian forests, preventing sedimentation, and monitoring water quality. Furthermore, the influence of the streams in the hydrographic network reinforces the need to preserve springs and prevent contamination from improper land use practices.

The interaction between conservation units, mining, and the region's hydrographic network highlights the importance of integrated environmental planning for Gleba Mazuay. The adoption of sustainable strategies can minimize negative impacts and enhance ecological benefits for the property and its surroundings.

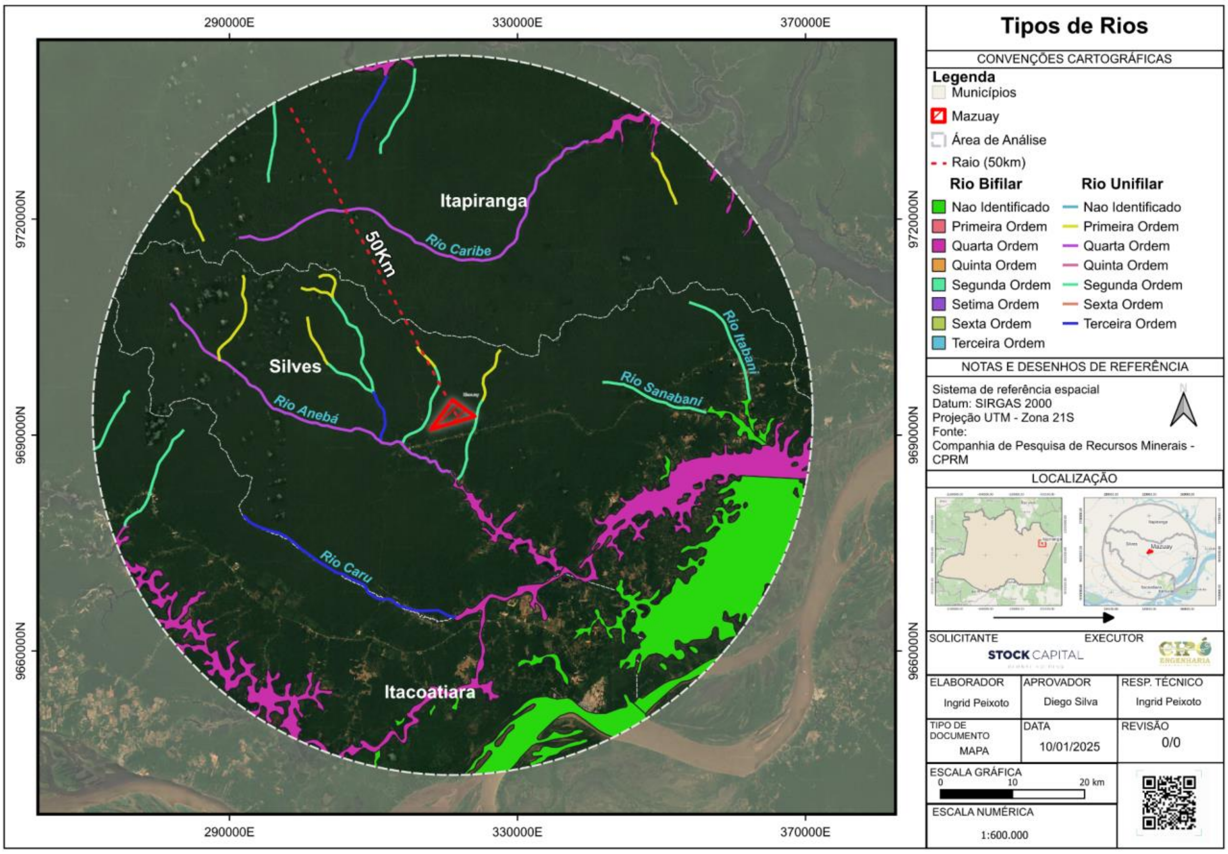


Unifilar and Bifilar River Types

The hydrographic analysis of the Gleba Mazuay region revealed the presence of different types of rivers, classified as unifilar and bifilar. This differentiation is crucial for understanding the local hydrological regime, as it directly influences water availability, surface runoff dynamics, and the interaction between the water bodies and the surrounding terrestrial environment.

Within a 50 km radius, we find fourth-order bifilar rivers and first-, second-, third-, and fourth-order unifilar rivers. Unifilar rivers are characterized by having a single drainage channel and are more common in areas with gentle topography and low slopes. In contrast, bifilar rivers feature two drainage channels, generally resulting from geomorphological processes that promote the formation of parallel or braided riverbeds.

The protection of these watercourses in the Gleba Mazuay area is essential to ensuring the maintenance of water quality and the stability of local ecosystems. The proximity of internal roads and other forms of human intervention may pose a risk to the integrity of these rivers, particularly concerning excessive sedimentation and water contamination from agricultural and industrial activities.



1.13.2.6 Mining

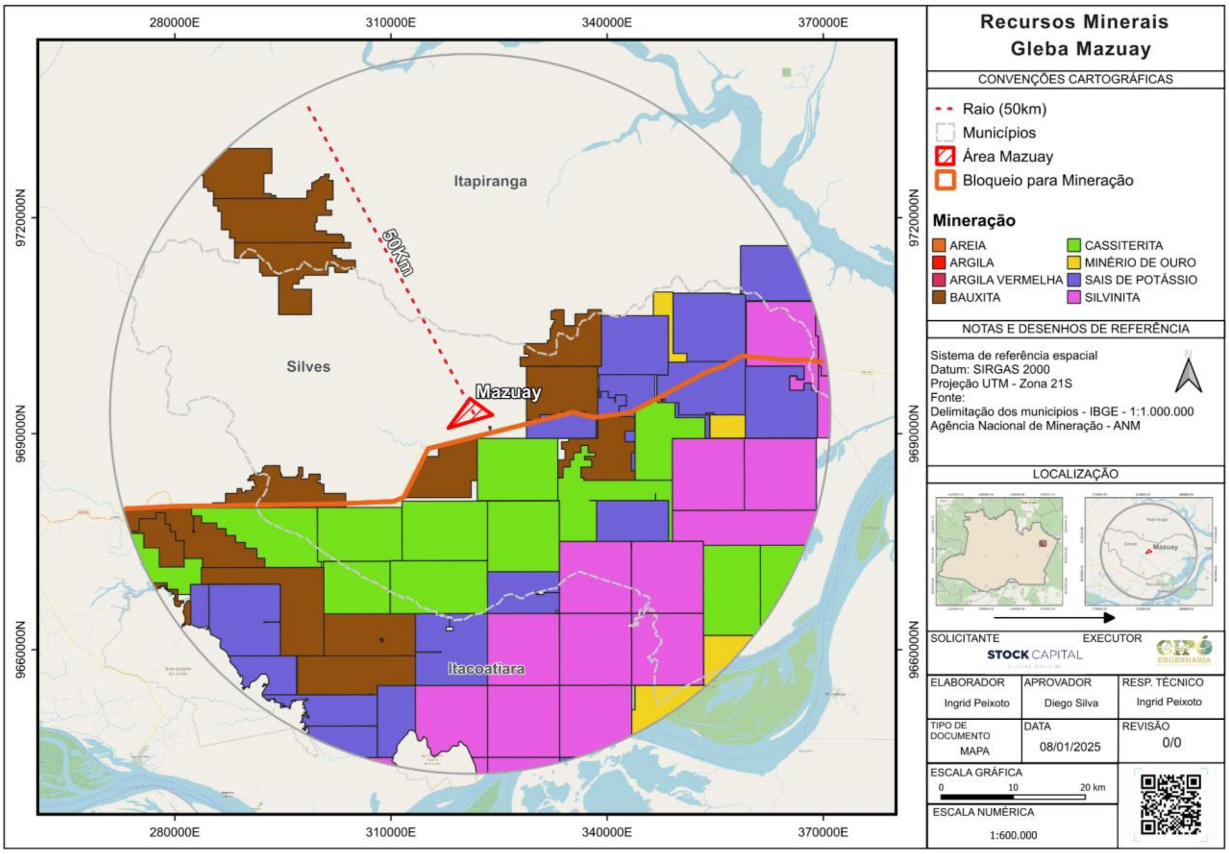
The northwest region of the Gleba Mazuay is rich in mineral resources, such as bauxite, cassiterite, potassium salts, gold, and sylvinite, according to data from the National Mining Agency (ANM). These minerals are primarily intended for industrial purposes, reflecting the region's strategic economic potential. However, mineral exploration requires detailed analysis of the environmental and social impacts.

The extraction of bauxite, for example, is known to cause significant deforestation, removal of the topsoil layer, and changes in water quality due to the release of sediments and chemicals (Silva & Souza, 2022). Additionally, areas mining cassiterite and gold often face issues of contamination by heavy metals, such as mercury, which can affect water bodies and aquatic fauna.

Another relevant point is the strategic location of the mineral deposits near transportation infrastructure, such as the AM 363 highway, which facilitates the flow of production. However, this also increases anthropogenic pressure, creating challenges related to sustainable land management and the protection of adjacent forested areas.

According to Carvalho et al. (2023), the implementation of environmental monitoring systems and programs for the recovery of degraded areas is crucial to minimize negative impacts. The rehabilitation of mined areas may include reforestation with native species, habitat reconstruction, and the implementation of technologies that reduce the emission of polluting waste.

The proximity of mining to native vegetation and local water resources requires an integrated approach involving government agencies, mining companies, and local communities. Studies indicate that community participation in licensing and monitoring processes can significantly contribute to the sustainability of operations (Barbosa & Lima, 2021).

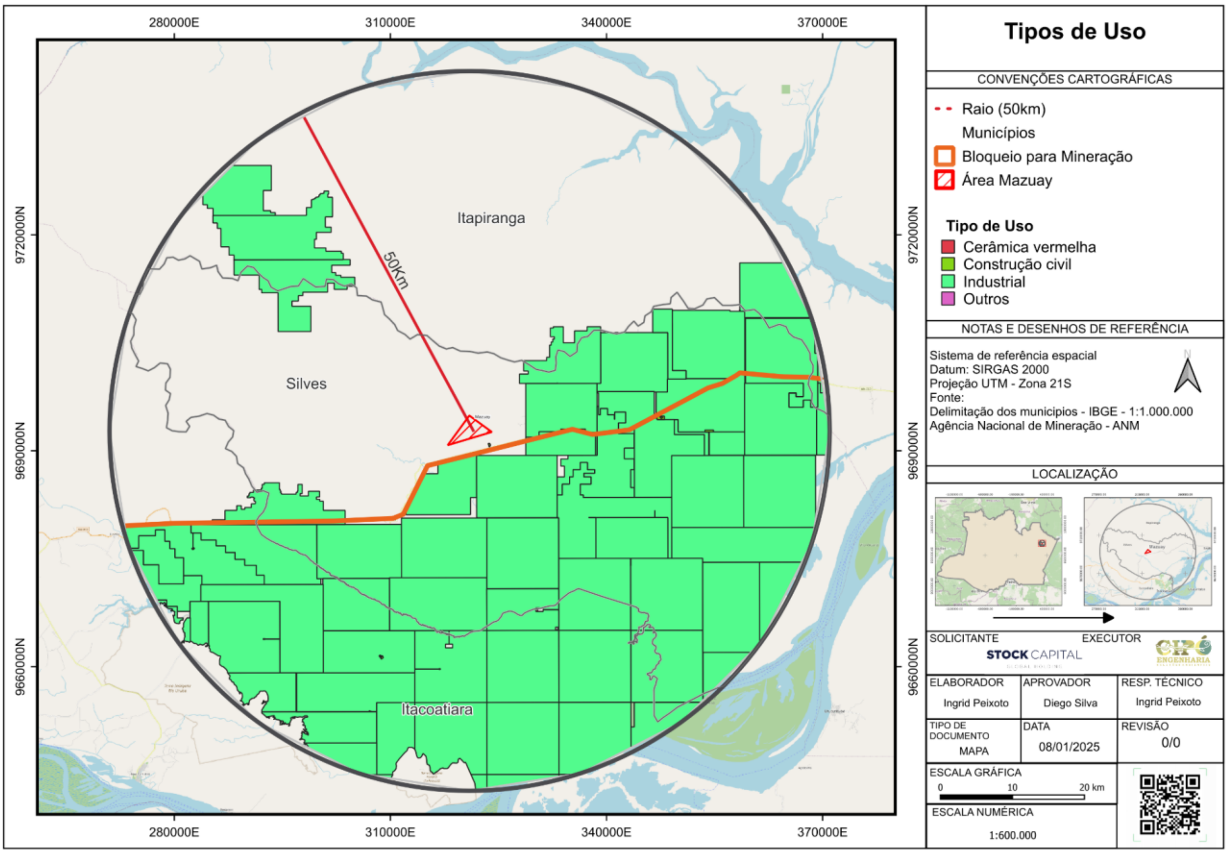


Characterization of Mining by Type of Use

The analysis of the mining map in the Gleba Mazuay region reveals that, within a 50 km radius, the main mining activity is focused on industrial purposes. The areas most impacted by this activity are located in the southern, eastern, and northwestern portions of the analyzed perimeter. However, within the limits of Gleba Mazuay, there are no records of mining exploitation.

The predominance of industrial mining in the region may be linked to the local geological characteristics, which favor the extraction of minerals used in various production processes, such as metallurgy, chemistry, and cement manufacturing. Additionally, the development of these activities may lead to significant environmental impacts, such as changes to the terrain, soil degradation, and contamination of water bodies due to the runoff of waste.

The absence of mining within the Gleba Mazuay suggests that the land use on the property is focused on other activities, such as agriculture and livestock. However, the proximity of industrial mining may pose challenges for the environmental management of the Gleba, requiring constant monitoring of water, air, and soil quality to prevent negative impacts.



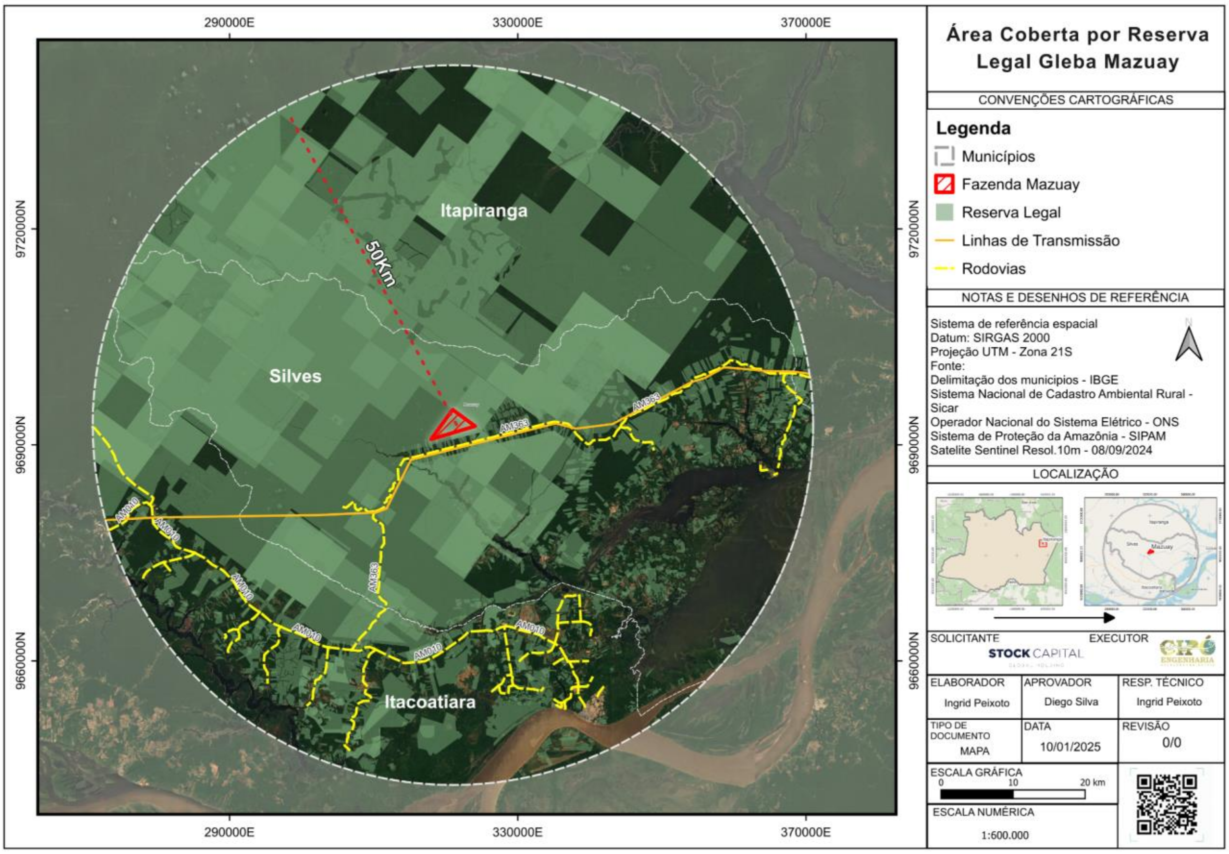
1.13.2.7 Legal Reserve

The legal reserve present within a 50 km radius around the Gleba Mazuay, with a focus to the north of the AM 363 highway, represents an important strategy for environmental conservation and compliance with the Brazilian Forest Code (Law No. 12,651, 2012). Legal reserves aim to maintain native vegetation cover, ensuring the preservation of biodiversity, water balance, and climatic stability in the region.

In the context of the Gleba Mazuay, the predominance of legal reserve areas implies restrictions on land use for agricultural or economic exploitation activities that may compromise natural resources. This is particularly relevant in a region with high environmental sensitivity, where anthropogenic pressures, such as mining and infrastructure expansion, can jeopardize ecosystem services.

Studies by Silva et al. (2023) highlight that legal reserve areas also play a crucial role in the connectivity between forest fragments, allowing for gene flow of both fauna and flora species.

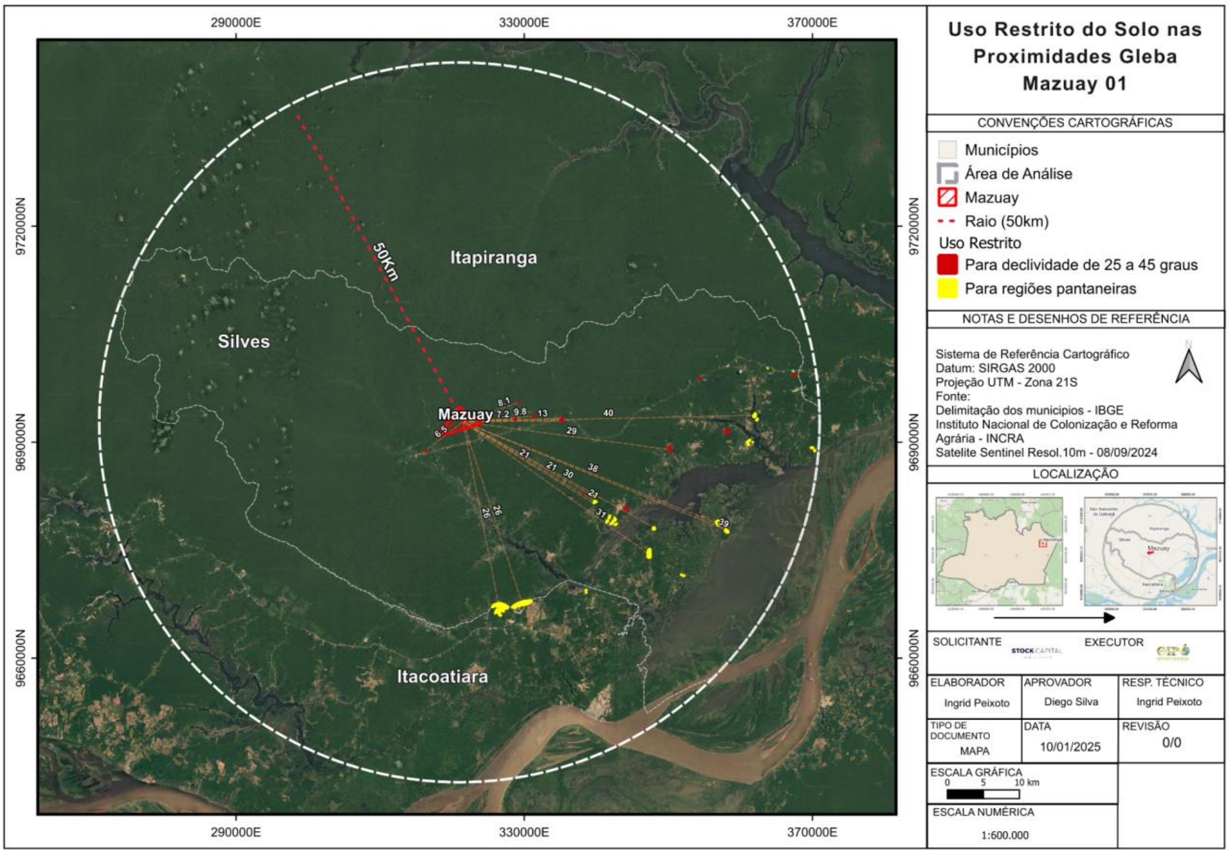
Another important aspect is the role of legal reserves in protecting water resources. The native vegetation present in these areas helps with water infiltration into the soil, reduces erosion, and minimizes the sedimentation of rivers and streams. In the Gleba Mazuay, the proximity to springs and bodies of water reinforces the importance of keeping legal reserves intact to preserve water quality and availability in the region.

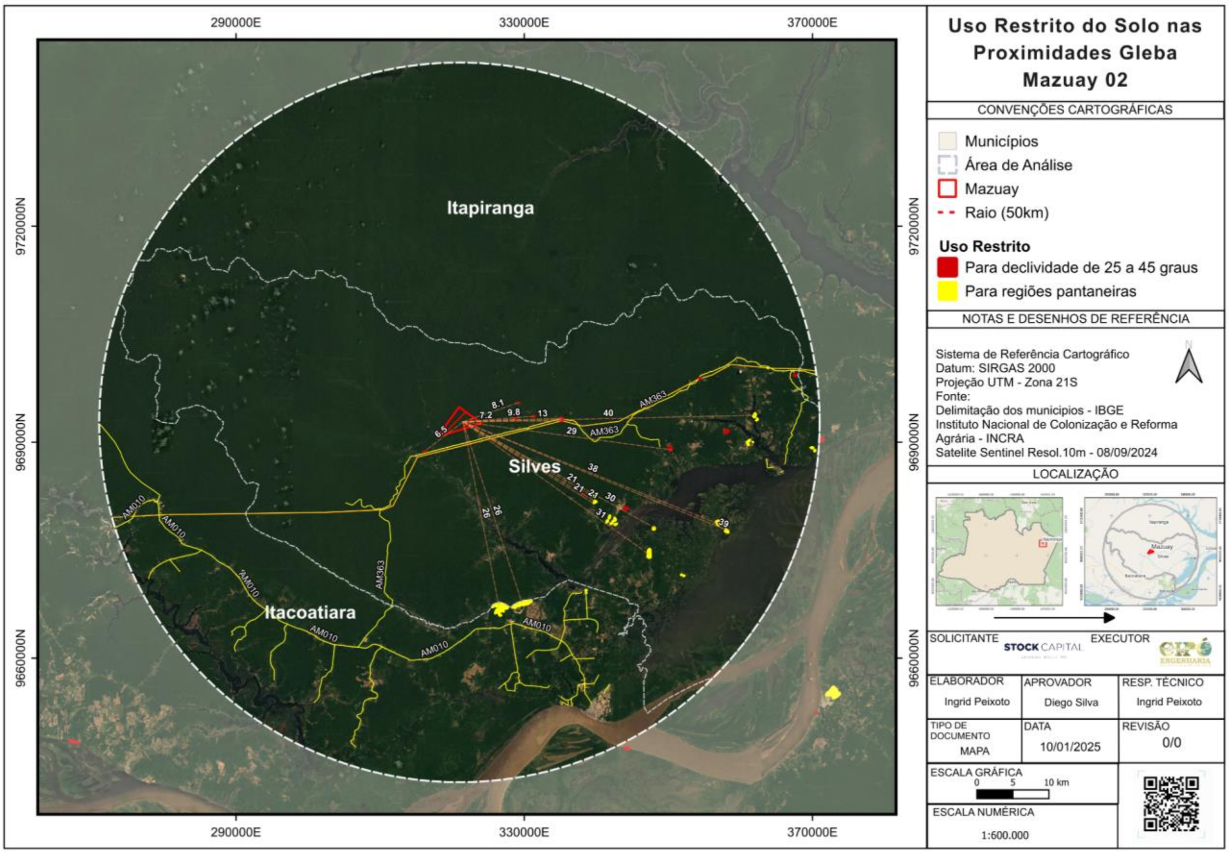


1.13.2.8 Restricted Soil Use

The restricted soil use map reveals areas classified as unsuitable for intensive occupation due to steep slopes, ranging from 25 to 45 degrees, and the presence of swampy regions. These characteristics significantly limit agricultural and infrastructure use in these areas, ensuring the preservation of crucial ecosystem services.

Regions with steep slopes play a vital role in maintaining soil stability, reducing erosion and surface runoff, factors that could result in sedimentation of the adjacent rivers and streams. Additionally, swampy areas act as important carbon sinks and water regulators, providing habitat for unique biodiversity and contributing to water quality in the local watershed (RADAMBRASIL Project, 2023).





1.13.2.9 Native Vegetation Remnants in Gleba Mazuay and Surroundings

The map of native vegetation remnants in Gleba Mazuay and its surroundings reveals that almost the entire analyzed area, within a 50 km radius, has preserved forest cover. The significant presence of native vegetation may be associated with factors such as environmental restrictions, access difficulties, and conservation regulations.

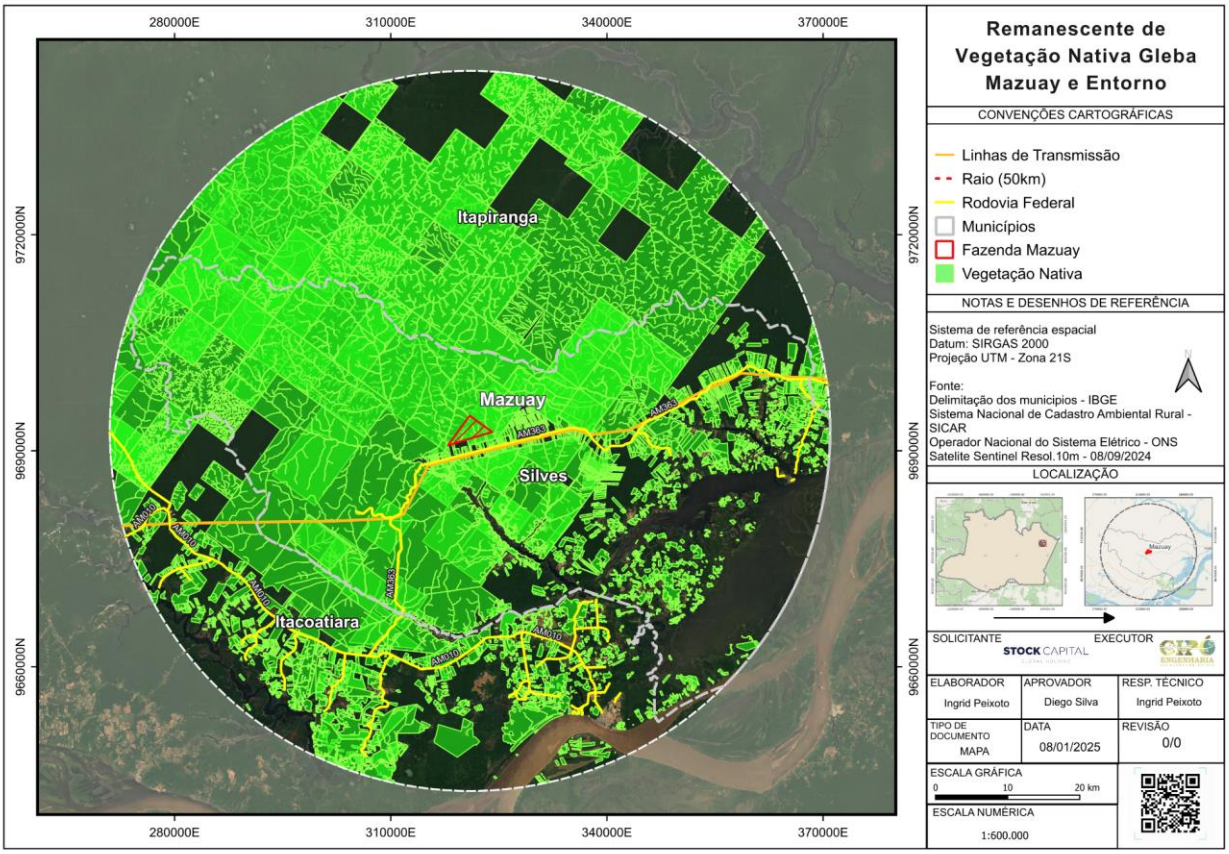
The preservation of these native vegetation areas plays a crucial role in microclimate regulation, water resource protection, and biodiversity conservation. According to Ferreira et al. (2021), the Amazon forests perform essential functions in the hydrological cycle, influencing evapotranspiration processes and contributing to the maintenance of rainfall patterns in the region. Native vegetation also serves as a natural barrier against erosive processes and helps stabilize soils, reducing the risk of sedimentation in water bodies.

In the southeastern portion of the analyzed area, a reduction in forest remnants is observed, which may indicate impacts from human occupation and agricultural activities. According to Lima et al. (2020), the conversion of native vegetation for agricultural use can result in habitat fragmentation and the loss of ecosystem services, compromising the region's environmental stability.

Another relevant aspect is the relationship between native vegetation and ecological connectivity. Continuous forest areas, such as those observed in Gleba Mazuay, enable fauna movement and facilitate gene flow between plant and animal populations. This is essential for maintaining biodiversity and for ecosystem resilience in the face of environmental pressures and climate change (Nobre et al., 2019).

Furthermore, the conservation of native vegetation is directly linked to mitigating greenhouse gas emissions. Studies indicate that Amazonian forests store large amounts of carbon, functioning as natural carbon sinks and helping regulate the global atmospheric balance (Silva & Barbosa, 2022). On the other hand, the degradation of these areas can release large volumes of CO_2 into the atmosphere, exacerbating climate change.

Given this context, maintaining the native vegetation remnants in Gleba Mazuay and its surroundings should be a priority to ensure ecological stability and the sustainability of natural resources. Conservation policies, such as sustainable forest management and the implementation of ecological corridors, can be effective strategies to protect these ecosystems and minimize the impacts of anthropogenic expansion in the region.

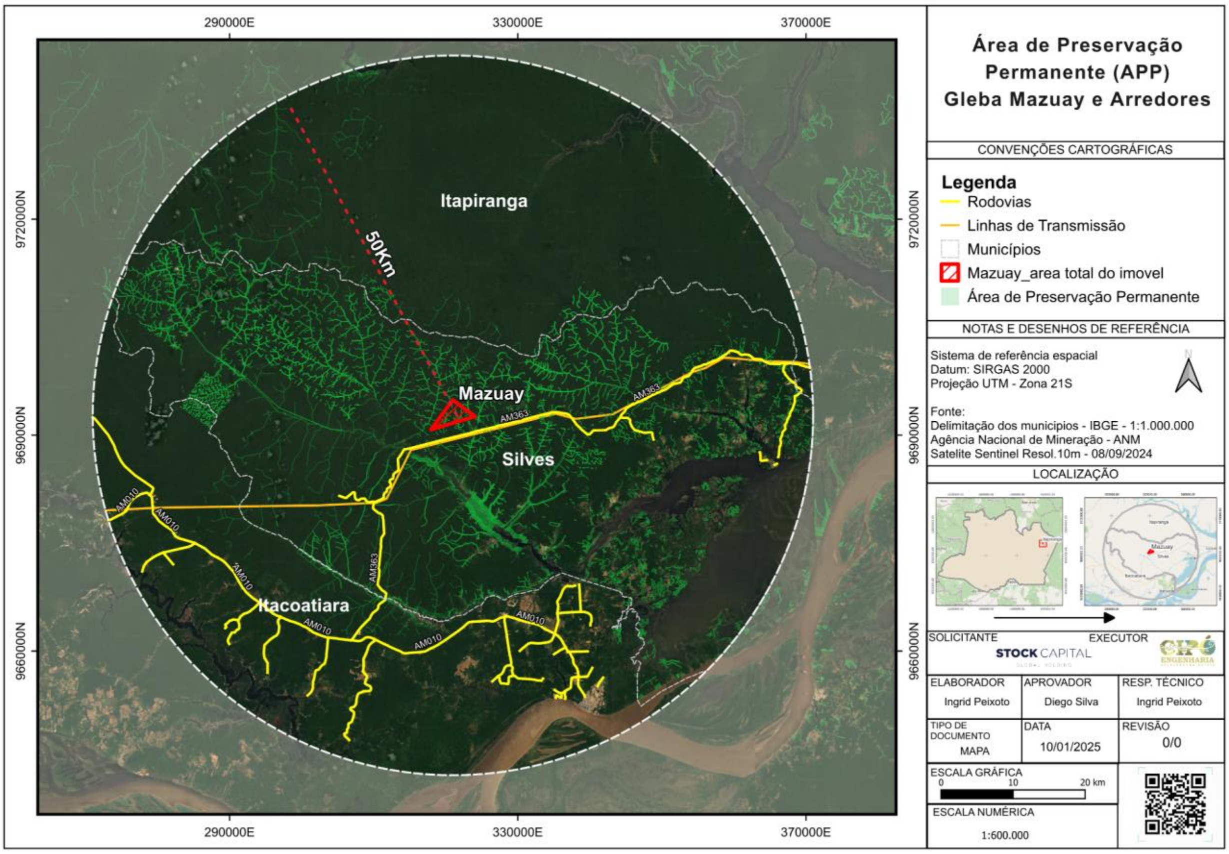


1.13.2.10 Permanent Preservation Area (PPA)

The map analysis reveals a concentration of Permanent Preservation Areas (PPAs) in the municipality of Silves, with a strong influence on Gleba Mazuay when observing the 50 km radius. PPAs are legally protected areas whose primary function is to conserve water resources, reduce soil erosion, and ensure ecological stability (Brazil, 2012).

According to Oliveira et al. (2021), maintaining these areas reduces the vulnerability of local ecosystems and minimizes anthropogenic impacts, such as river siltation and the contamination of water sources. In the context of Gleba Mazuay, the significant presence of PPAs indicates the existence of water bodies, hilltops, and steep slopes that require legal protection. These areas play an essential role in maintaining biodiversity and protecting springs and watercourses, preventing processes such as siltation and contamination from human activities (Sparovek et al., 2019).

The sustainable management of PPAs is essential to ensuring ecosystem resilience. Studies indicate that the degradation of these areas can compromise groundwater recharge and increase vulnerability to drought, especially in regions with intense agricultural activity (MMA, 2021). Therefore, the implementation of actions such as reforestation, fencing to prevent trampling by animals, and continuous monitoring through geospatial technologies is recommended.



1.13.3 Characterization of the Study Area – Anthropogenic Environment

1.13.3.1 Permanent Preservation Areas (PPAs) in Gleba Mazuay and Their Environmental Impacts

An analysis of PPAs in Gleba Mazuay and its immediate surroundings reveals that the area is situated in a sensitive environmental context due to the increasing number of access roads in the eastern, northern, and northwestern surroundings of the Gleba, raising significant environmental concerns.

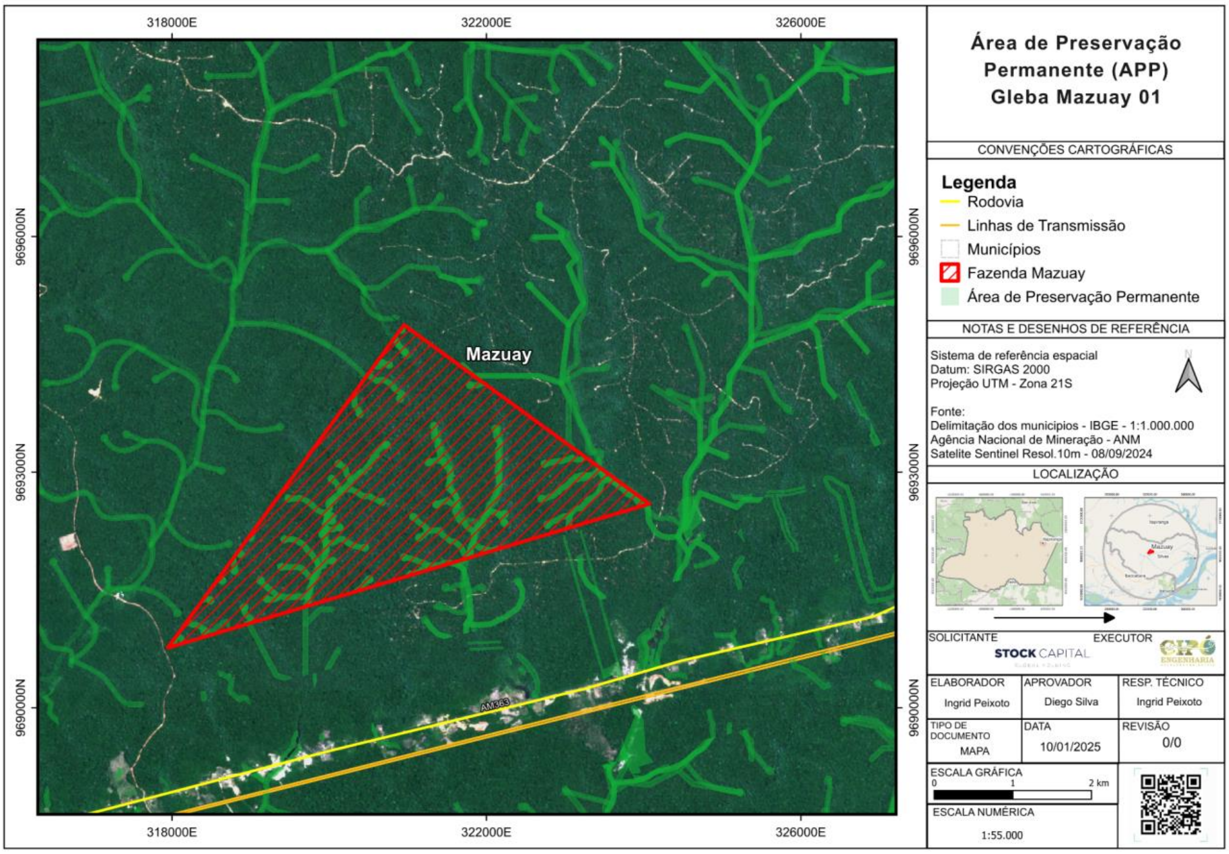
Brazilian environmental legislation, particularly the Forest Code (Law No. 12.651/2012), establishes the necessity of maintaining PPAs to ensure ecological stability and mitigate negative impacts on water resources, biodiversity, and ecosystem dynamics (BRASIL, 2012). In the case of Gleba Mazuay, the high concentration of PPAs is beneficial for conservation; however, the expansion of access roads around the property may pose a risk to the integrity of these environments.

Access roads are pathways used for internal transportation within rural properties, often opened to facilitate production flow or access to different areas of the Gleba. However, if not sustainably planned, they can have adverse effects, such as habitat fragmentation, soil compaction, erosion, and sedimentation of nearby water bodies (FERREIRA et al., 2019). Additionally, the creation of new roads may facilitate human access to sensitive areas, increasing the likelihood of illegal activities such as deforestation and poaching (SILVA et al., 2020).

One of the main impacts resulting from the increase in carriers in the Gleba Mazuay region is the alteration of the local hydrological regime. Studies show that soil compaction caused by the constant traffic of vehicles reduces water infiltration, increasing surface runoff and, consequently, soil erosion (OLIVEIRA et al., 2018). The presence of Permanent Preservation Areas (APPs), which play a crucial role in protecting springs and riverbanks, may be compromised if there is no adequate control of these internal road infrastructures.

Another relevant aspect is the ecological fragmentation resulting from the construction of new roads within preserved natural areas. Fragmentation reduces connectivity between habitats, hindering fauna movement and promoting edge effects, which can significantly alter vegetation composition and structure (MIRANDA et al., 2021). As a result, there is a decline in local biodiversity and an increase in the vulnerability of endangered species that depend on large continuous forest areas for survival.

Given this scenario, it is essential to implement mitigation strategies to minimize the impacts of carriers in the Gleba Mazuay. Among the recommended measures, the adoption of sustainable soil management practices stands out, such as the construction of drainage structures to reduce erosion and the careful planning of access roads to avoid unnecessary fragmentation of the landscape (SANTOS et al., 2022). Additionally, it is crucial that the creation of new roads within the property undergoes a prior environmental study, ensuring that the opening of carriers does not compromise the ecological functions of the existing Permanent Preservation Areas (APPs).

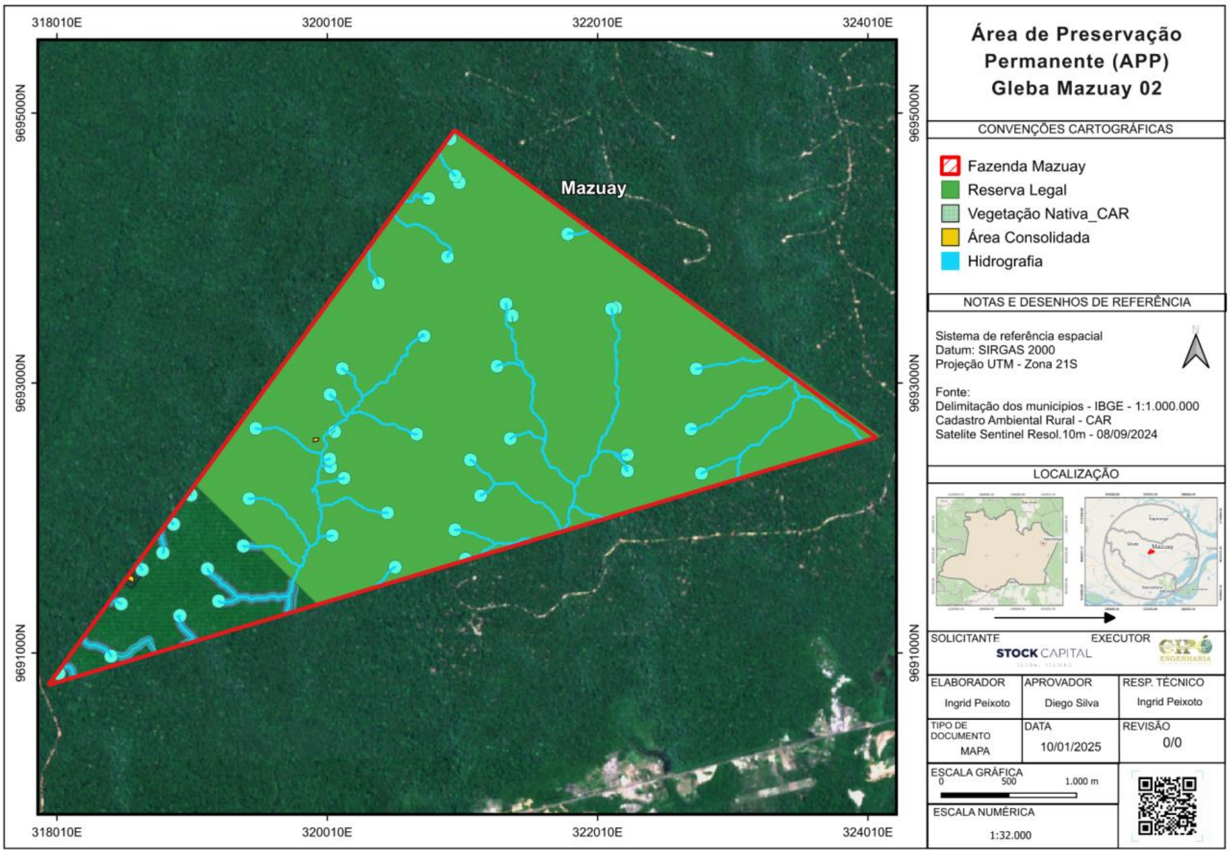


1.13.3.2 Land Use and Cover

The map presents land use and cover in the Gleba Mazuay, highlighting consolidated areas, native vegetation, legal reserve, and hydrography. This type of analysis allows for understanding territorial dynamics and the impacts of human activities on the environment (Ferreira et al., 2020).

In Gleba Mazuay, the predominance of native vegetation suggests a scenario of low human intervention, with limited economic activities. Additionally, the identification of spring and water eye points reinforces the importance of environmental preservation, as these areas are sensitive to water and erosion degradation processes (Batista et al., 2018).

The division between native vegetation and legal reserve within Gleba Mazuay is in compliance with the Forest Code, which requires maintaining a fraction of the rural property as an environmental reserve (Brazil, 2012). This delimitation is essential to ensure the conservation of biodiversity and the maintenance of ecosystem services. The adoption of sustainable management practices, such as agroforestry systems and the recovery of degraded areas, is recommended to promote the sustainability of the property and mitigate environmental impacts.



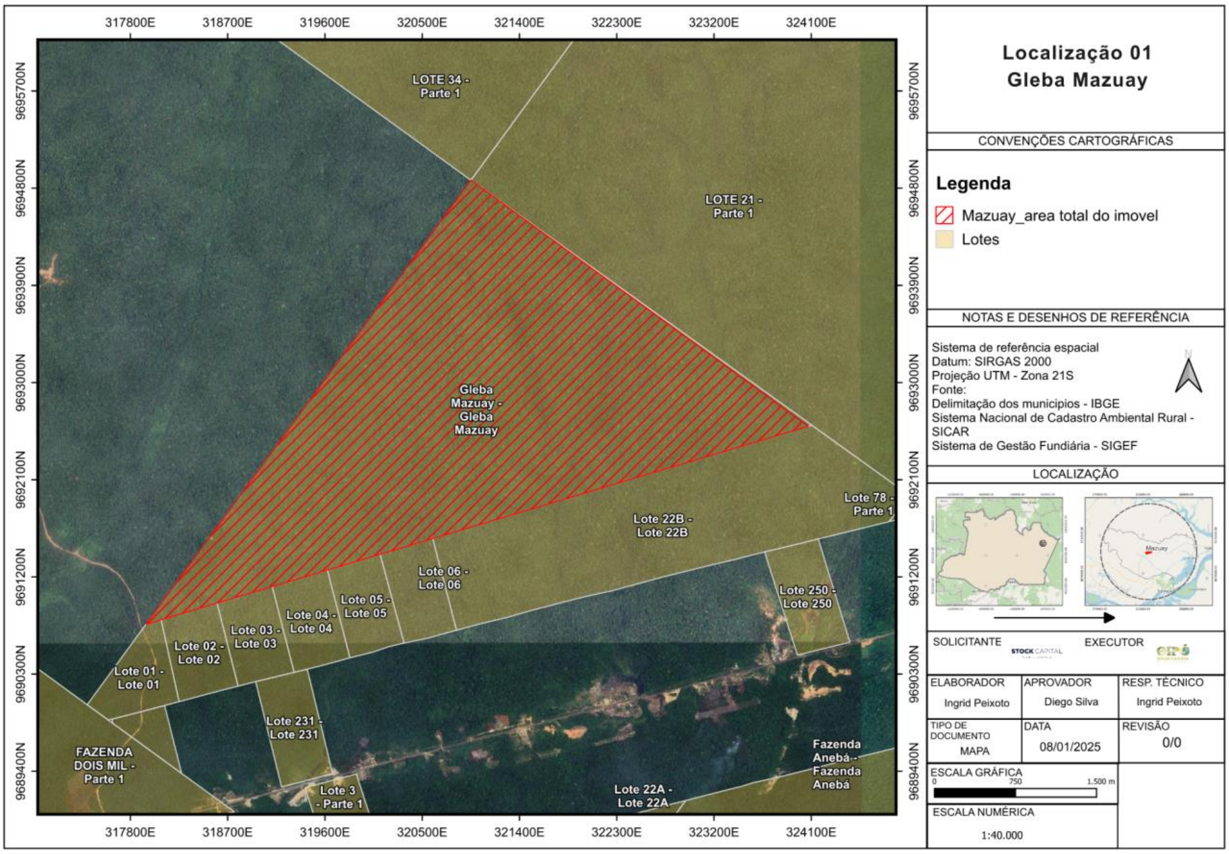
1.13.3.3 Lot Division

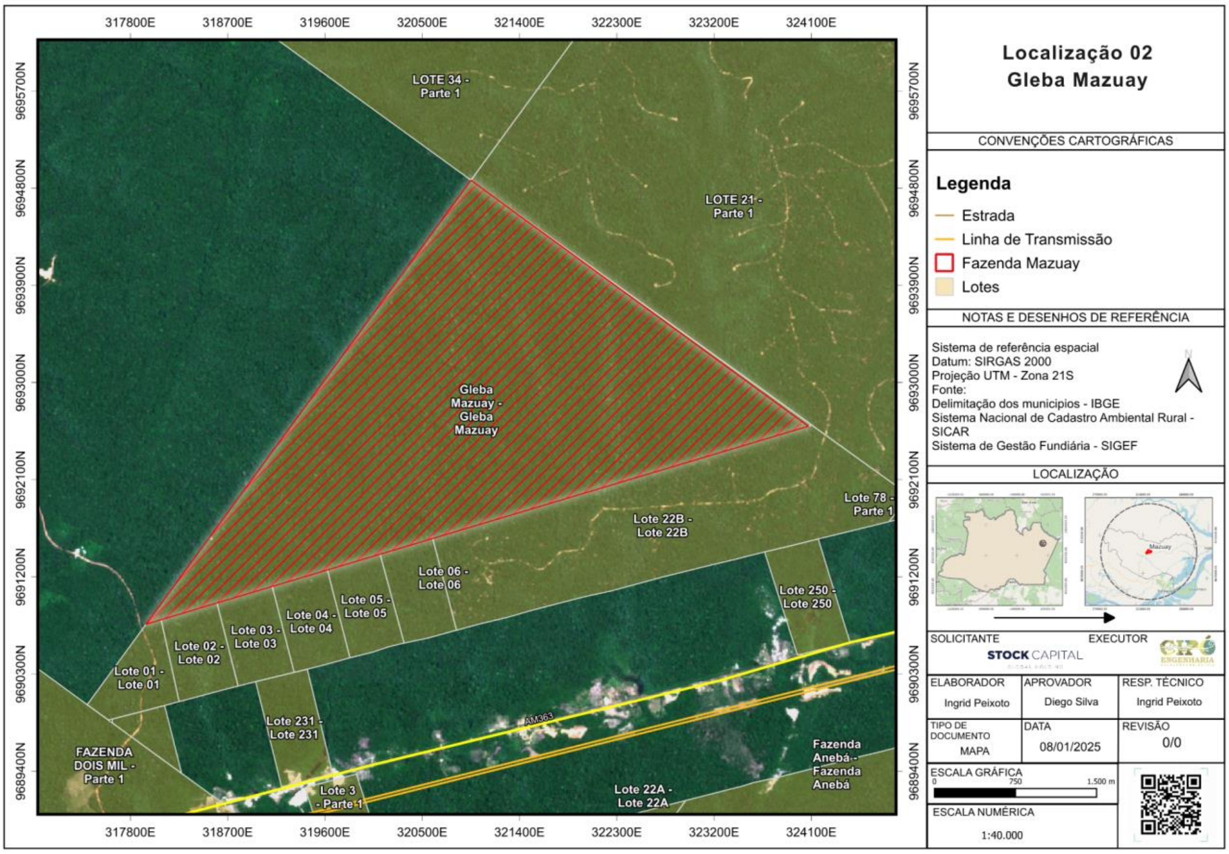
The plan of Gleba Mazuay shows a strategic division of the lots, with emphasis on lots 21 and 22B, where roads are opened that coincide with areas of bauxite and cassiterite occurrence. This infrastructure may indicate mining activities in the early stages or in planning. The construction of roads in these lots can facilitate access to the mineral deposits, but it also presents significant environmental risks, such as increased deforestation, habitat fragmentation, and soil erosion.

Additionally, the proximity of these lots to the AM 363 highway strengthens their economic potential due to the ease of transporting extracted resources. However, environmental mitigation measures such as engineering techniques are important to reduce road impacts and rehabilitate degraded areas.

Another relevant aspect is the existence of a community approximately 2 km from the aforementioned lots, which includes structures such as houses, warehouses, and a medical service post. Studies by Santos and Pereira (2020) highlight the importance of dialogue between mining companies and local communities to promote sustainable development and reduce potential conflicts.

The division of lots in Gleba Mazuay, therefore, represents a challenge in terms of territorial management and environmental conservation. It is essential that the activities undertaken are aligned with public policies for land-use planning and environmental protection.





Identification of Access Roads Near Gleba Mazuay

The mapping of access roads in the Gleba Mazuay area reveals a significant increase in these unpaved roads between February and September 2024. The analysis of high-resolution Sentinel satellite images shows that these new roads are mostly concentrated in the eastern and northern portions of the property, connecting it to other rural areas and possible natural resource exploitation points. This expansion suggests an accelerated process of opening rural roads. These structures may have significant environmental impacts, including habitat fragmentation, erosion, and changes in the local hydrological regime (FERREIRA et al., 2019). The expansion of access roads should be monitored to prevent degradation of native vegetation and the facilitation of illegal deforestation.

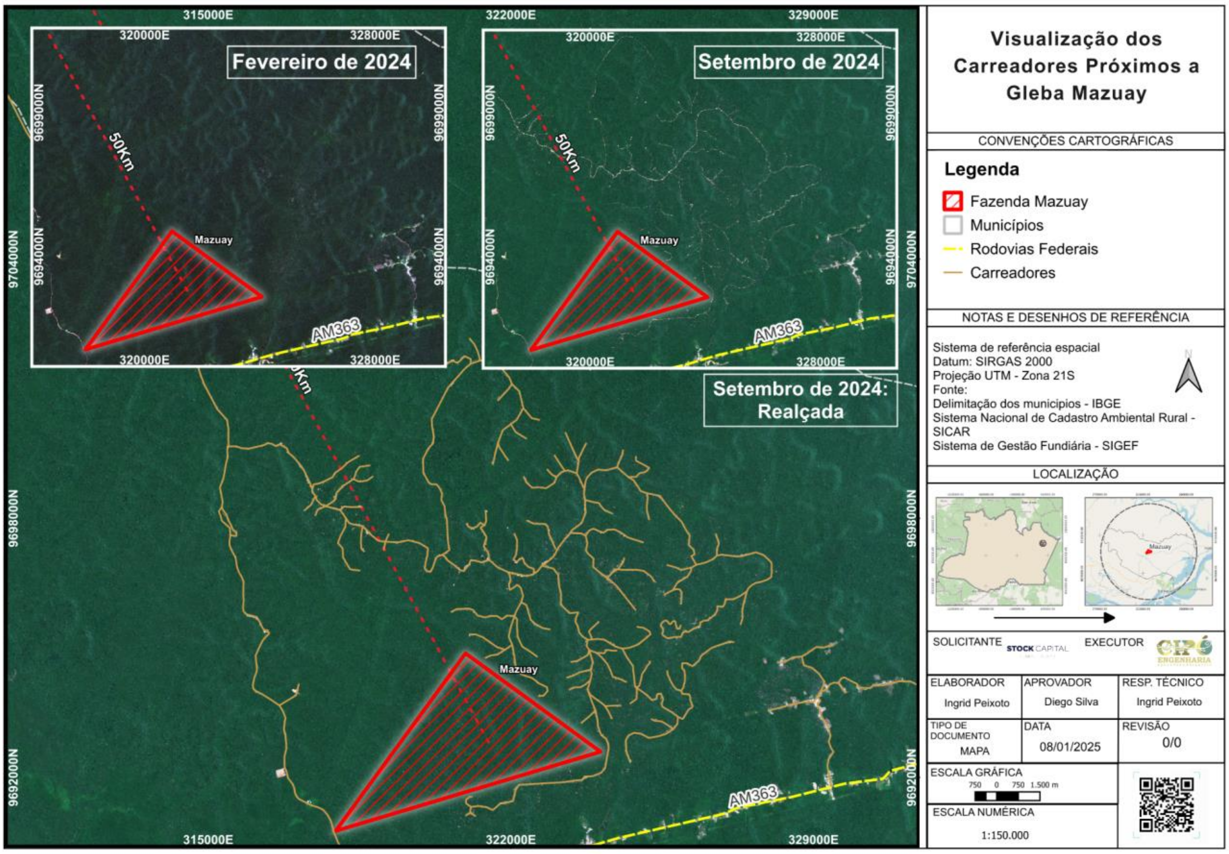
The opening and expansion of access roads are processes typically associated with the expansion of agricultural, logging, and mining activities. However, their unchecked proliferation can lead to significant environmental impacts, including habitat fragmentation, soil compaction, increased erosion, and sedimentation in water bodies (FERREIRA et al., 2019). The fragmentation of forests resulting from the construction of access roads can reduce connectivity between remaining vegetation, hindering species that depend on habitat continuity for movement and reproduction.

Moreover, the soil compaction resulting from constant vehicle traffic on these roads alters its porosity and permeability, hindering water infiltration and increasing surface runoff. his phenomenon amplifies erosion and the transport of sediments to nearby rivers and creeks, potentially compromising water quality and aquatic biodiversity (OLIVEIRA et al., 2018). In the specific case of Gleba Mazuay, this impact may be even more concerning due to the presence of APPs that play a fundamental role in protecting the region's water resources.

The analysis of these access roads also allows for inferences about the vulnerability of the soil in the area. In regions with sandy or poorly cohesive soils, the construction of unpaved roads can accelerate land degradation, making vegetation recovery slower and more costly (SILVA et al., 2020). In the Amazon context, where rainfall is frequent and abundant, improper maintenance of these roads can lead to sedimentation of watercourses and the formation of gullies, phenomena that compromise soil stability and increase environmental recovery costs.

Another critical aspect of the expansion of roads is their potential to facilitate access to previously isolated forest areas. This phenomenon, widely documented in environmental studies, has been linked to the increase in illegal deforestation and the predatory exploitation of timber in the Amazon (MIRANDA et al., 2021). In the case of the Gleba Mazuay, the opening of new roads in the eastern and northern portions may pose a risk to the conservation of native vegetation, requiring monitoring and control actions by environmental authorities.

Thus, the evolution of roads near Gleba Mazuay reinforces the need for an integrated approach to environmental management in the area. The uncontrolled expansion of these roads could compromise the integrity of local ecosystems and intensify the negative impacts of human occupation on biodiversity and natural resources.



1.13.3.4 Geological Structures and Arcs

The presence of concealed geological faults in the analyzed region, intersecting the AM-363 highway and located near Gleba Mazuay, indicates a potentially unstable geological substrate.

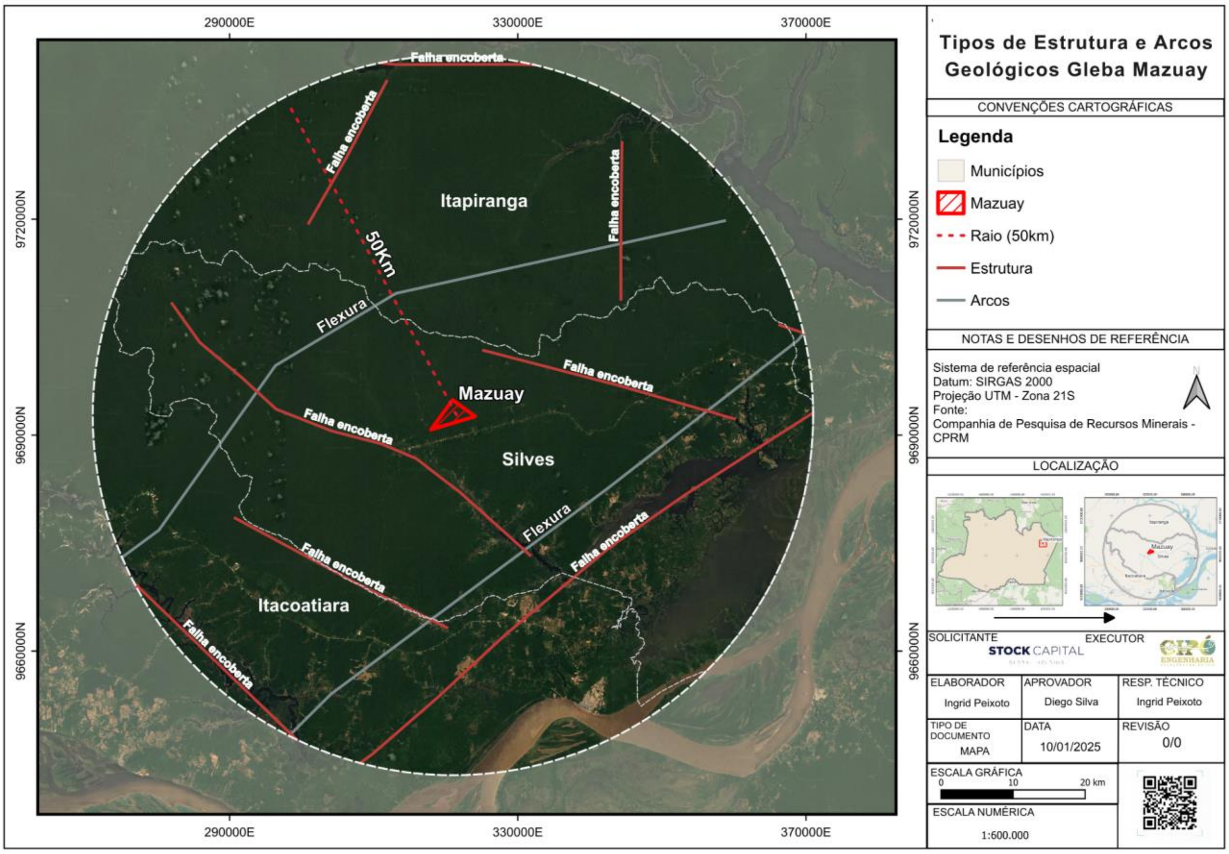
According to Santos et al. (2022), these faults may influence subsurface hydrodynamics, affecting groundwater recharge and availability in the region. Additionally, the identified flexural arch may have structural implications for soil behavior, underscoring its relevance for future studies on land-use planning and environmental conservation.

The geological analysis of Gleba Mazuay identified concealed geological faults and flexural-type arches, geostructural factors that may affect terrain stability and the region's hydrogeological dynamics. The presence of three concealed geological faults within the municipality of Silves, along with three others in Itapiranga and Itacoatiara, highlights a geological setting that demands close attention, particularly due to the proximity of these structures to the AM-363 highway.

According to Santos et al. (2022), concealed geological faults are structural discontinuities in the subsurface that can act as preferential pathways for groundwater flow, affecting aquifer recharge and water distribution in the region. Additionally, these structures may be associated with subsidence processes and soil instability, potentially posing a risk to road infrastructure and agricultural enterprises in the area.

The presence of the flexural arch within the analyzed territory, within a 50 km radius, is another relevant factor, as it indicates a tectonic deformation zone that may influence landform characteristics and surface drainage patterns. In Amazonian lowland regions, where geomorphology is strongly influenced by tectonic and hydrodynamic processes, identifying these structural elements is essential for land-use planning and sustainable environmental management (Fernandes et al., 2021).

The interaction between geological faults and the region's hydrodynamic system must be monitored to prevent adverse impacts, such as alterations in river and igarapé flow or the compromise of slope stability. A detailed understanding of these structures is crucial for guiding environmental conservation efforts and minimizing geotechnical risks that could affect the integrity of the ecosystem and human activities in the Gleba Mazuay area.



1.13.4 Characterization of the Study Area – Biotic Environment

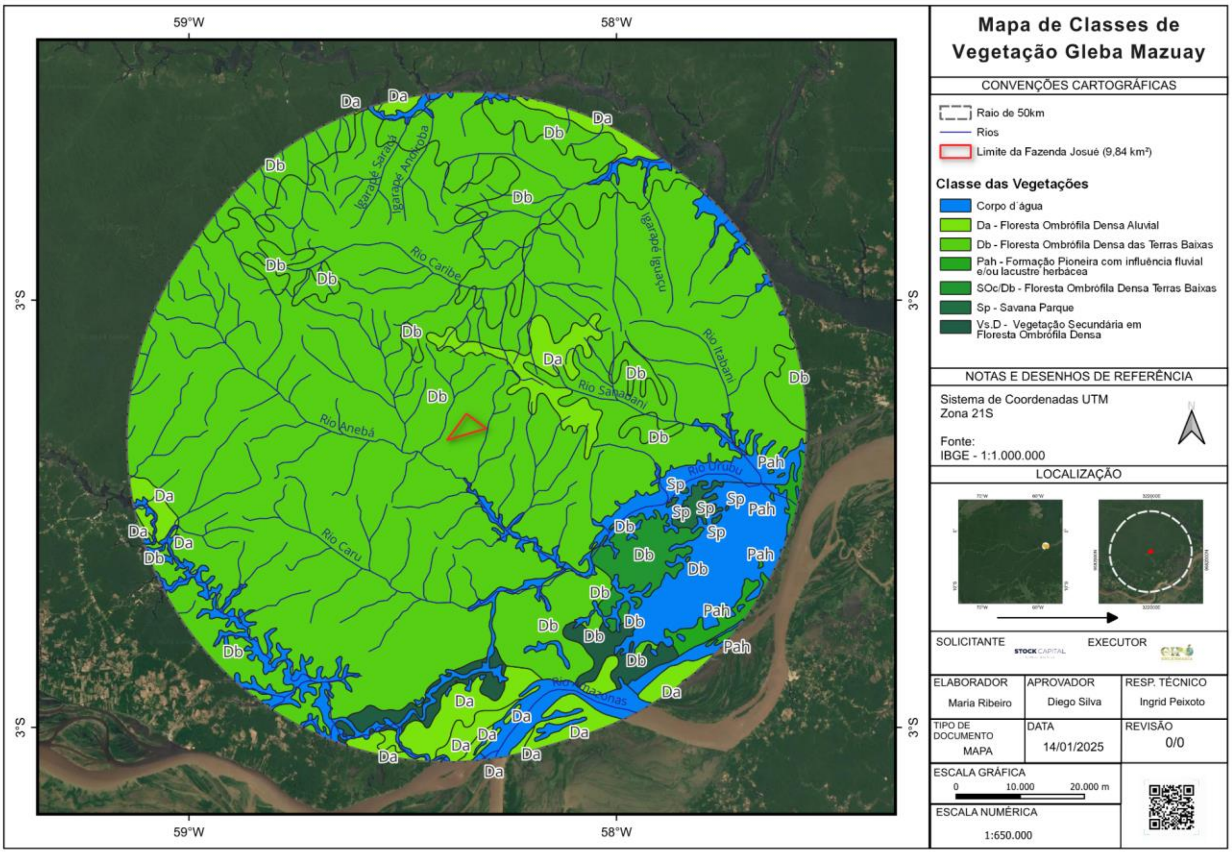
1.13.4.1 Vegetation

The analysis of vegetation classes around Gleba Mazuay revealed a predominance of Lowland Dense Ombrophilous Forest, with other vegetation types such as Alluvial Dense

Ombrophilous Forest and Pioneer Formation with Fluvial and/or Lacustrine Herbaceous Influence in the flooded areas. Park Savannas and Secondary Vegetation were also identified, along with water bodies such as the Amazon River and the Urubu River.

The region's ecological diversity is marked by a variety of forests, savannas, and floodplains, reflecting its environmental complexity. Natural regeneration processes coexist with anthropogenic impacts, such as logging and wildfires.

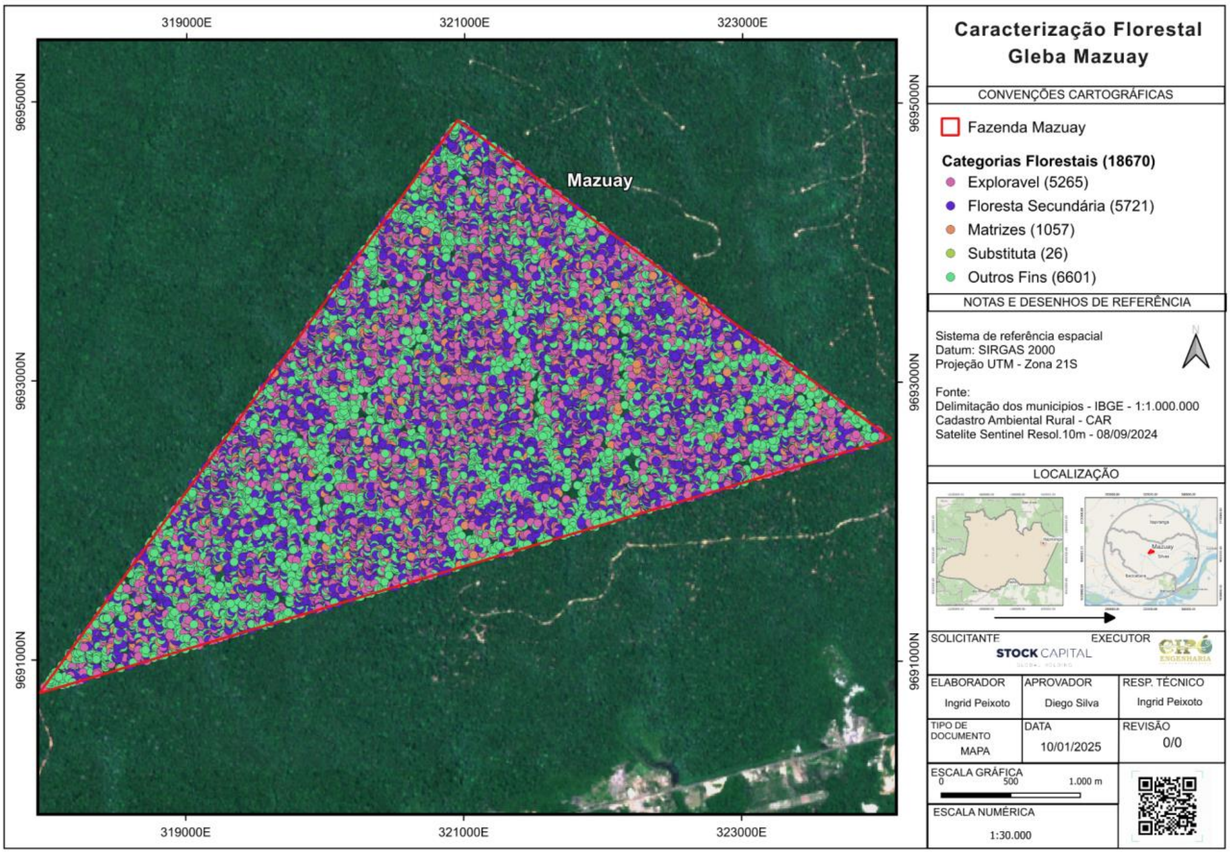
According to IBGE (2004), Dense Ombrophilous Forest is characterized by an ombrophilous climate, with high humidity levels throughout the year and average temperatures ranging from 22°C to 25°C. This vegetation formation extends across the Amazon and the Atlantic coast, characterized by large trees on alluvial terraces and tertiary plateaus, and medium-sized trees on maritime slopes. Typical genera include Hevea, Bertholletia, and Dinizia.



The forest characterization of Gleba Mazuay, through a 100% Forest Inventory, identified 18,670 trees distributed across several categories: Exploitable (52,265 specimens), Secondary Forest (5,721 specimens), Matrices (1,057 specimens), Substitute (26 specimens), and Other Purposes (6,601 specimens).

The high density of exploitable trees suggests significant potential for sustainable forest management. However, it is essential to adopt best management practices, such as selective cutting and monitoring the regeneration of native species, to ensure biodiversity conservation and avoid negative impacts (Ribeiro et al., 2020).

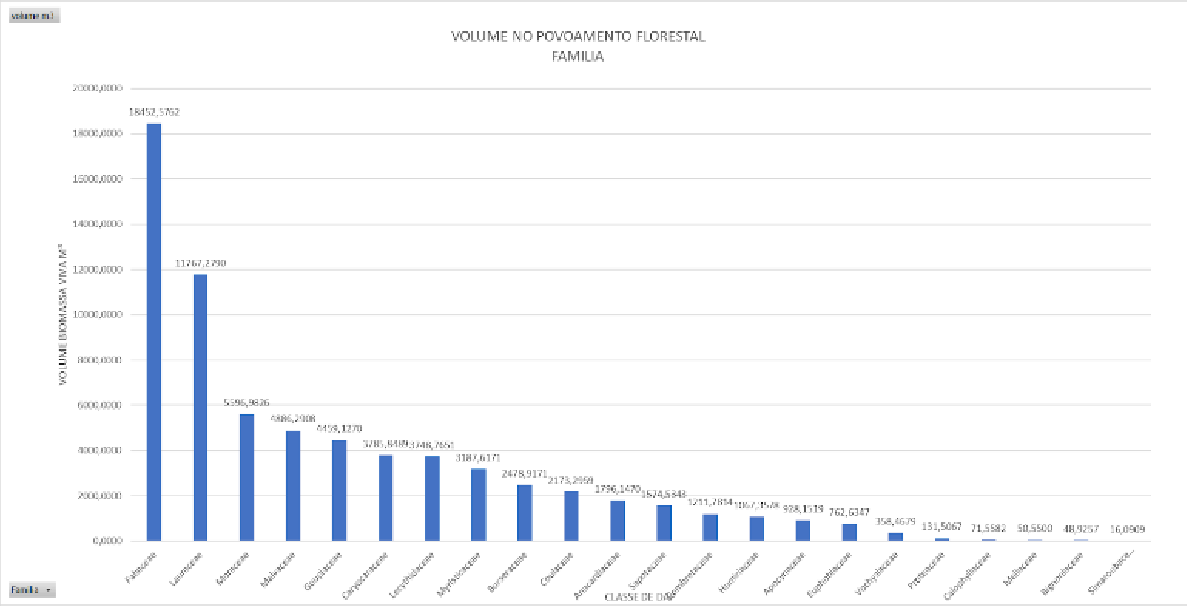
The presence of forest matrices and secondary forest areas indicates potential for natural vegetation recovery. Encouraging natural regeneration and adopting reforestation strategies can increase the resilience of the vegetation and ensure the sustainability of the forest landscape over time.



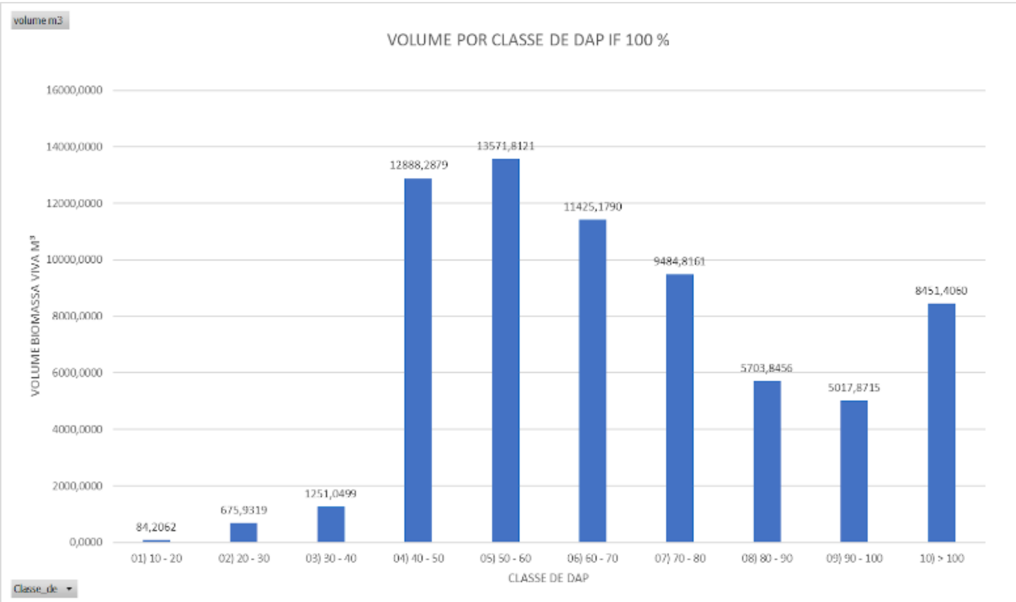
Based on the collected information and the provided table, we can perform a detailed analysis of the forest population in Gleba Mazuay. The table shows the distribution of several plant families, with a significant number of specimens, such as Fabaceae (3,839 specimens), Lauraceae (1,181 specimens), and Lecythidaceae (1,696 specimens).

The high density of exploitable trees, particularly in the Fabaceae family, suggests significant potential for sustainable forest management, as highlighted by Ribeiro et al. (2020). However, it is essential to adopt appropriate management practices to preserve biodiversity and ensure the regeneration of native species.

The diversity of species found in Gleba Mazuay is consistent with patterns observed in other regions of the Amazon, where Dense Ombrophilous Forest predominates. According to Nascimento et al. (2021), fragmentation of ecological corridors and resource exploitation may compromise connectivity and population dynamics of species. Therefore, a forest management plan that strikes a balance between economic exploitation and environmental preservation is crucial for the long-term sustainability of the forest landscape in Gleba Mazuay.



Based on the collected information, we analyzed the diameter classes of trees in Gleba Mazuay and compared them with the region of Silves, Amazonas, and the Amazon biome. The diversity of diameter classes indicates significant potential for carbon storage and sequestration.



The forest characterization identified a high density of trees, especially those with larger diameters, which are important for carbon storage. Large-diameter trees accumulate more biomass, making them more efficient at sequestering CO_2. The diversity of species found in Gleba Mazuay is consistent with the patterns observed in other regions of the Amazon, where Dense Ombrophilous Forest predominates.

According to studies, the Dense Ombrophilous Forest in the Amazon presents a variety of diameter classes, with large trees predominating in some areas. The region's biodiversity contributes to the sustainability of the ecosystem and its carbon sequestration capacity (Nascimento et al., 2021).

The presence of various diameter classes in Gleba Mazuay increases the efficiency of CO_2 sequestration and the reduction of greenhouse gases (GHGs). The diversity of diameter classes plays a significant role in carbon stock and sequestration rates. Larger trees store significant amounts of carbon in their biomass, and conserving these trees is crucial for maintaining this sequestered carbon over the long term. Sustainable forest management strategies, such as promoting natural regeneration and reforestation, enhance carbon stock and improve the resilience of the vegetation.

Based on the collected data and the analysis of diameter classes, Gleba Mazuay demonstrates significant potential for generating carbon credits. The high density of exploitable trees and the diversity of diameter classes ensure efficiency in carbon sequestration.

Brazil is home to one of the largest preserved forests in the world, and reducing global temperatures by 1.5°C is essential to meet climate goals. The country stands out for maintaining much of the Amazon intact. Studies from the University of São Paulo indicate that species such as Feijão-do-mato, Guapuruvu, Pau-jacaré, Jacarandá-da-bahia, and Jatobá are highly efficient in absorbing CO_2.

Grouping species into ecological groups is crucial for forest management, facilitating conservation and forest recovery, as emphasized by R. B. de A. Lima et al. (2011). Understanding successional groups, according to Ferraz et al. (2004), can be applied to conservation, sustainable management, and the rehabilitation of degraded areas.

Understanding ecological succession is vital for the regeneration of biological communities in fragmented landscapes. The table below lists species observed in the southern Amazon region, detailing phenological formations, uses, families, and scientific names.

Common Name	Scientific Name	Family	Phenology	Uses
Abiurana	*Pouteria caimito*	Sapotaceae	Fruit-bearing: December-March	Wood, Edible fruits
Abiurana-guajará	*Abiu_heterophytum*	Sapotaceae	Fruit-bearing: November-February	Edible fruits
Acariquara	*Minquartia guianensis*	Olacaceae	Flowering: October-December	Wood
Amapá	*Parahancornia fasciculata*	Apocynaceae	Flowering: May-July	Medicinal latex
Amapá-doce	*Brosimum parinarioides*	Moraceae	Flowering: June-August	Wood
Andiroba	*Carapa guianensis*	Meliaceae	Fruit-bearing: January-March	Medicinal oil, Cosmetics
Angelim-do-campo	*Vatairea guianensis*	Fabaceae	Fruit-bearing: March-June	Wood
Angelim-fava	*Parkia gigantocarpa*	Fabaceae	Fruit-bearing: April-June	Wood
Angelim-pedra	*Hymenolobium excelsum*	Fabaceae	Fruit-bearing: September-November	Wood
Angelim-rajado	*Peltogyne confertiflora*	Fabaceae	Flowering: August-October	Wood

Angelim-vermelho	*Dinizia excelsa*	Fabaceae	Fruit-bearing: July-September	Wood
Arara-tucupi	*Sclerolobium paniculatum*	Fabaceae	Fruit-bearing: August-October	Wood
Arurá-branco	*Geissospermum vellosii*	Apocynaceae	Flowering: November-January	Wood, Medicinal
Breu-branco	*Protium heptaphyllum*	Burseraceae	Flowering: April-June	Resin, Medicinal
Breu-vermelho	*Protium spp.*	Burseraceae	Flowering: March-May	Resin, Wood
Caju	*Anacardium occidentale*	Anacardiaceae	Flowering: September-November	Edible fruits, Nuts
Castanharana	*Eschweilera spp.*	Lecythidaceae	Flowering: November-January	Wood
Castanha-sapucaia	*Lecythis pisonis*	Lecythidaceae	Fruiting: March-May	Edible nuts
Cedrinho	*Cedrela odorata*	Meliaceae	Flowering: December-February	Wood
Copaíba	*Copaifera spp.*	Fabaceae	Flowering: July-September	Medicinal oil
Coração-de-negro	*Simarouba amara*	Simaroubaceae	Flowering: October-December	Wood
Cumarú	*Dipteryx odorata*	Fabaceae	Fruiting: August-October	Perfume, Wood
Cumarú-vermelho	*Dipteryx micrantha*	Fabaceae	Flowering: July-September	Wood, Perfume
Cupiúba	*Goupia glabra*	Celastraceae	Fruiting: October-December	Wood
Fava-amargosa	*Vatairea macrocarpa*	Fabaceae	Flowering: August-October	Wood
Faveira	*Parkia spp.*	Fabaceae	Fruiting: May-June	Wood
Guariúba	*Clarisia racemosa*	Moraceae	Flowering: September-November	Wood, Handicrafts
Ipê	*Tabebuia spp.*	Bignoniaceae	Flowering: August-October	Wood, Ornamental
Jacaréuba	*Calophyllum brasiliense*	Clusiaceae	Flowering: November-January	Wood, Medicinal
Jarana	*Holocalyx balansae*	Fabaceae	Flowering: July-September	Wood
Jatobá	*Hymenaea courbaril*	Fabaceae	Fruiting: May-July	Wood, Edible fruits
Jutaí-pororoca	*Dialium guianense*	Fabaceae	Flowering: February-April	Wood
Louro-amarelo	*Ocotea spp.*	Lauraceae	Flowering: November-January	Wood, Essential oil
Louro-aritu	*Aniba rosaeodora*	Lauraceae	Flowering: December-February	Essential oil, Wood
Louro-chumbo	*Ocotea cymbarum*	Lauraceae	Flowering: November-January	Wood, Perfume
Louro-faia	*Ocotea spp.*	Lauraceae	Flowering: August-October	Wood, Essential oil
Louro-gamela	*Ocotea puberula*	Lauraceae	Flowering: September-November	Wood, Essential oil
Louro-itaúba	*Mezilaurus itauba*	Lauraceae	Flowering: July-September	Wood
Louro-preto	*Ocotea catharinensis*	Lauraceae	Flowering: August-October	Wood
Louro-rosa	*Aniba rosaeodora*	Lauraceae	Flowering: January-March	Essential oil, Wood
Maçaranduba	*Manilkara huberi*	Sapotaceae	Fruiting: April-June	Wood
Mandioqueira	*Qualea paraensis*	Vochysiaceae	Flowering: September-November	Wood
Maparajuba	*Margaritaria nobilis*	Euphorbiaceae	Flowering: October-December	Wood
Marupá	*Simarouba amara*	Simaroubaceae	Flowering: August-October	Wood, Medicinal
Matamatá-preto	*Eschweilera coriacea*	Lecythidaceae	Flowering: October-December	Wood
Melancieira	*Alchornea triplinervia*	Euphorbiaceae	Flowering: October-December	Wood, Medicinal
Muiracatiara	*Astronium lecointei*	Anacardiaceae	Flowering: September-November	Wood
Muirapiranga	*Brosimum paraense*	Moraceae	Flowering: August-October	Wood
Mururé	*Brosimum acutifolium*	Moraceae	Fruiting: March-May	Wood, Latex
Paricarana	*Apeiba tibourbou*	Malvaceae	Flowering: June-August	Wood
Pau-rosa	*Aniba rosaeodora*	Lauraceae	Flowering: November-January	Essential oil, Wood
Pequiá	*Caryocar villosum*	Caryocaraceae	Fruiting: December-February	Wood, Edible fruits
Pequiá-marfim	*Caryocar glabrum*	Caryocaraceae	Fruiting: January-March	Wood, Edible fruits
Pequiarana	*Aspidosperma desmanthum*	Apocynaceae	Flowering: October-December	Wood

Preciosa	*Aniba canelilla*	Lauraceae	Flowering: March-May	Wood, Essential oil
Seringueira	*Hevea brasiliensis*	Euphorbiaceae	Flowering: June-August	Latex production
Sorva	*Couma utilis*	Apocynaceae	Fruiting: April-June	Latex, Edible fruits
Sucupira-amarela	*Pterodon pubescens*	Fabaceae	Flowering: October-December	Wood, Medicinal
Sucupira-preta	*Bowdichia nitida*	Fabaceae	Flowering: August-October	Wood, Medicinal
Sucupira-vermelha	*Pterodon emarginatus*	Fabaceae	Flowering: September-November	Wood, Medicinal
Tanibuca	*Terminalia amazonia*	Combretaceae	Fruiting: October-December	Wood
Tauari-branco	*Couratari guianensis*	Lecythidaceae	Flowering: June-August	Wood
Tauari-cachimbo	*Couratari spp.*	Lecythidaceae	Fruiting: July-September	Wood
Tauari-vermelho	*Cariniana legalis*	Lecythidaceae	Flowering: August-October	Wood
Táxi	*Scleronema micranthum*	Lecythidaceae	Flowering: September-November	Wood
Taxi-amarelo	*Pouteria torta*	Sapotaceae	Fruiting: April-June	Wood
Taxi-preto	*Dendropanax arboreus*	Araliaceae	Flowering: November-January	Wood
Taxi-vermelho	*Pouteria spp.*	Sapotaceae	Flowering: January-March	Wood
Tento	*Ormosia spp.*	Fabaceae	Fruiting: August-October	Wood, Ornamental
Timborana	*Enterolobium schomburgkii*	Fabaceae	Flowering: July-September	Wood
Ucuuba	*Virola surinamensis*	Myristicaceae	Flowering: September-November	Wood, Vegetable oil
Uxi-liso	*Endopleura uchi*	Humiriaceae	Fruiting: March-May	Wood, Edible fruits
Uxi-torrado	*Euxylophora paraensis*	Rutaceae	Flowering: May-July	Wood
Violeta	*Duguetia spp.*	Annonaceae	Flowering: June-August	Wood, Ornamental

Therefore, Gleba Mazuay is economically viable for carbon projects, offering both environmental and economic benefits by ensuring biodiversity conservation and the sustainability of forest management.

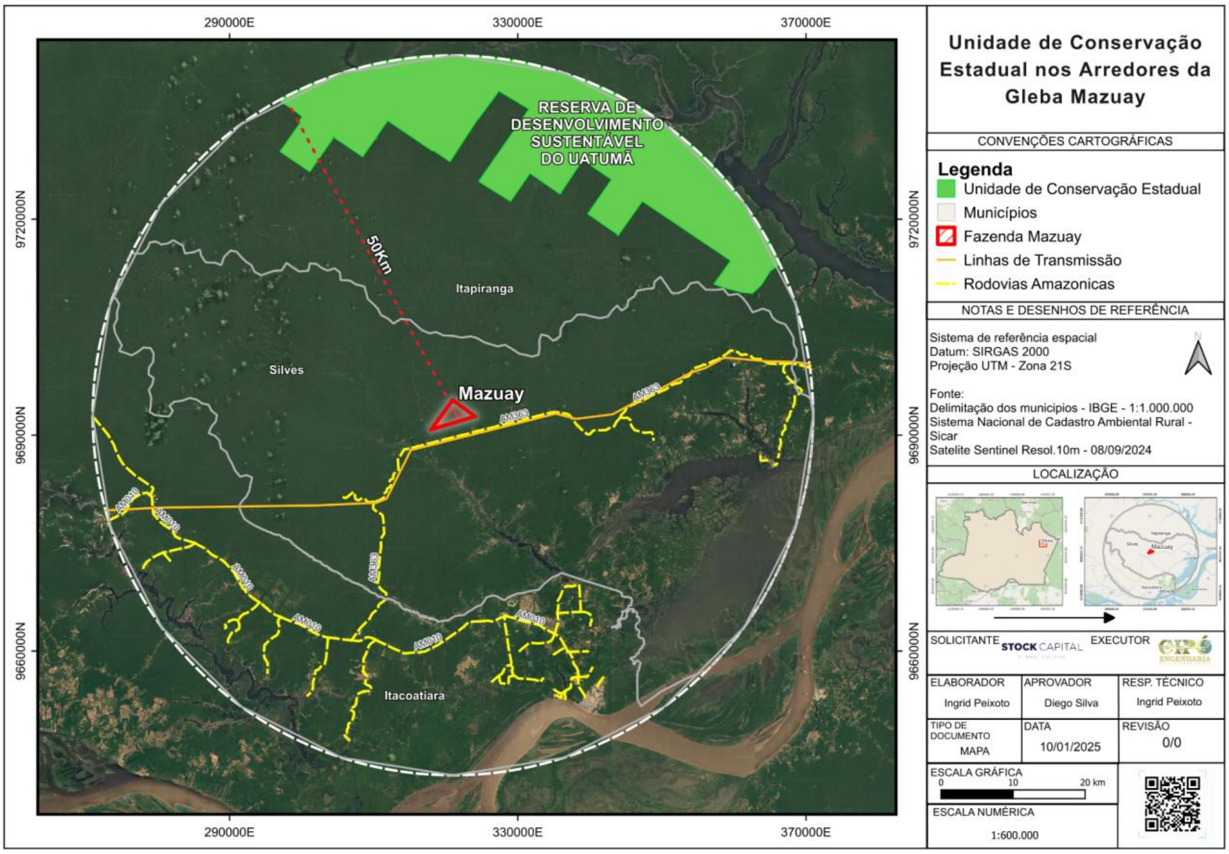
1.13.5 Protected Areas

1.13.5.1 State Conservation Unit

The Uatumã Sustainable Development Reserve (RDS), located in the municipality of Itapiranga, is a state conservation unit of great relevance in the region. RDS areas are designated to reconcile environmental conservation with the sustainable use of natural resources by traditional populations living within them. According to Barreto et al. (2021), this strategy not only promotes the protection of biodiversity but also strengthens local economies based on sustainable practices such as managed fishing, agro-extraction, and community-based tourism.

The proximity of Gleba Mazuay to the Uatumã RDS reinforces the importance of integrated conservation strategies that consider regional environmental and socioeconomic dynamics. Additionally, these protected areas play a crucial role in mitigating the impacts of climate change, acting as important carbon sinks and helping regulate hydrological regimes.

According to recent studies, public-private partnerships in these areas can generate mutual benefits, encouraging scientific research, the restoration of degraded areas, and the valorization of traditional knowledge (Oliveira & Mendes, 2022).

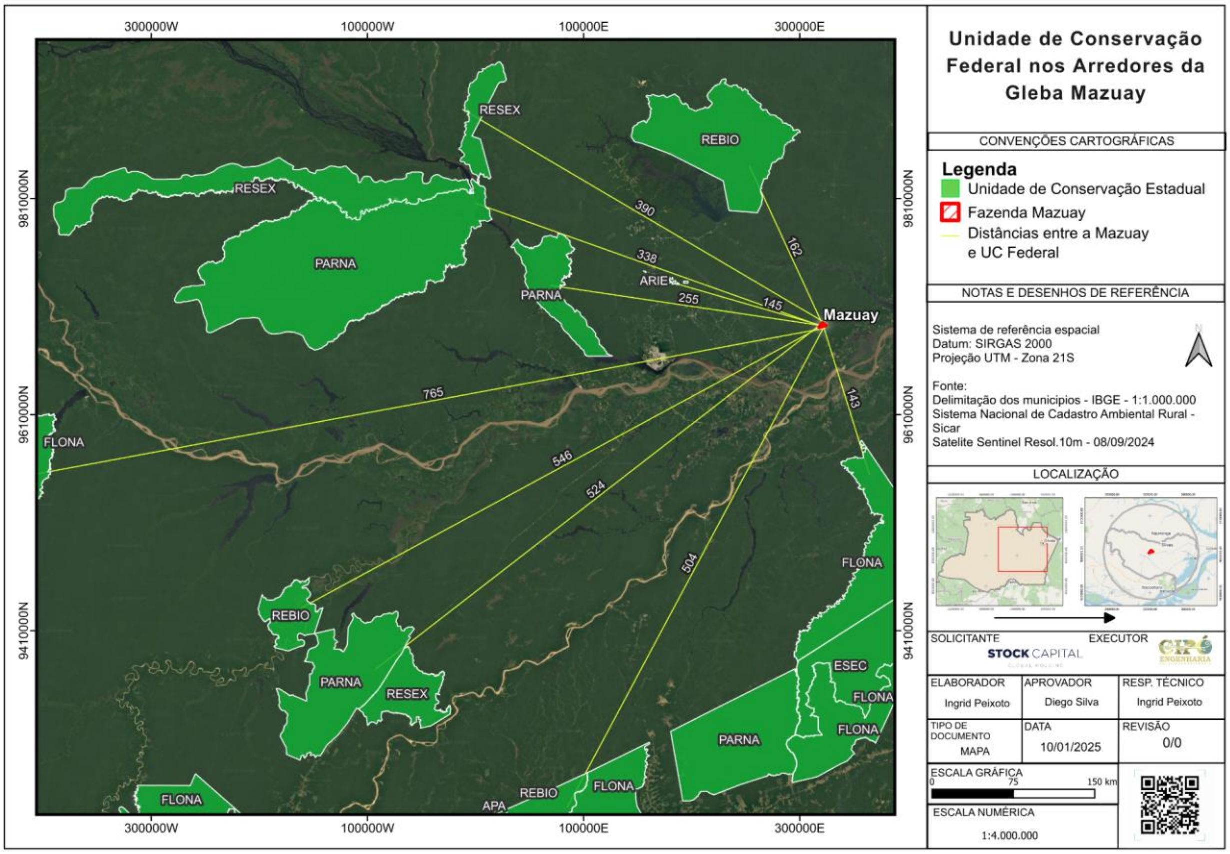


1.13.5.2 Federal Conservation Units Near Gleba Mazuay

The analysis of the federal Conservation Units (UCs) located near Gleba Mazuay provides an understanding of the influence these protected areas have on the region and their potential ecological, economic, and social interactions. The categories of UCs present include Biological Reserves (REBIO), Areas of Relevant Ecological Interest (ARIEL), National Parks (PARNA), Extractive Reserves (RESEX), Ecological Stations (ESEC), and National Forests (FLONA). All these units are located to the west of the Gleba, with FLONA being the closest at a distance of 143 km, followed by ARIEL (145 km) and REBIO (162 km).

The proximity of these conservation units may have both positive impacts and challenges for Gleba Mazuay. Environmentally, the presence of these UCs promotes biodiversity conservation and the preservation of water resources. However, it may also impose land-use restrictions and affect the development of economic activities that could impact protected ecosystems. FLONA, for example, allows sustainable forest management, while REBIO has stricter restrictions, being focused on the integral protection of fauna and flora.

The influence of these UCs on Gleba Mazuay may also be reflected in climate regulation, maintenance of ecological corridors, and protection of springs and watercourses. Furthermore, the existence of these units may directly affect human settlement and agriculture and livestock activities in the region, requiring harmonization between rural production and environmental conservation.



1.13.5.3 Indigenous Lands

According to Article 231 of the Brazilian Federal Constitution of 1988, Indigenous Lands to be regularized by the Government must meet the following criteria:

1) Be permanently inhabited by indigenous peoples;
2) Be essential for carrying out their productive activities;
3) Be indispensable for the preservation of the resources necessary for their well-being; and
4) Be fundamental for the physical and cultural reproduction of indigenous communities.

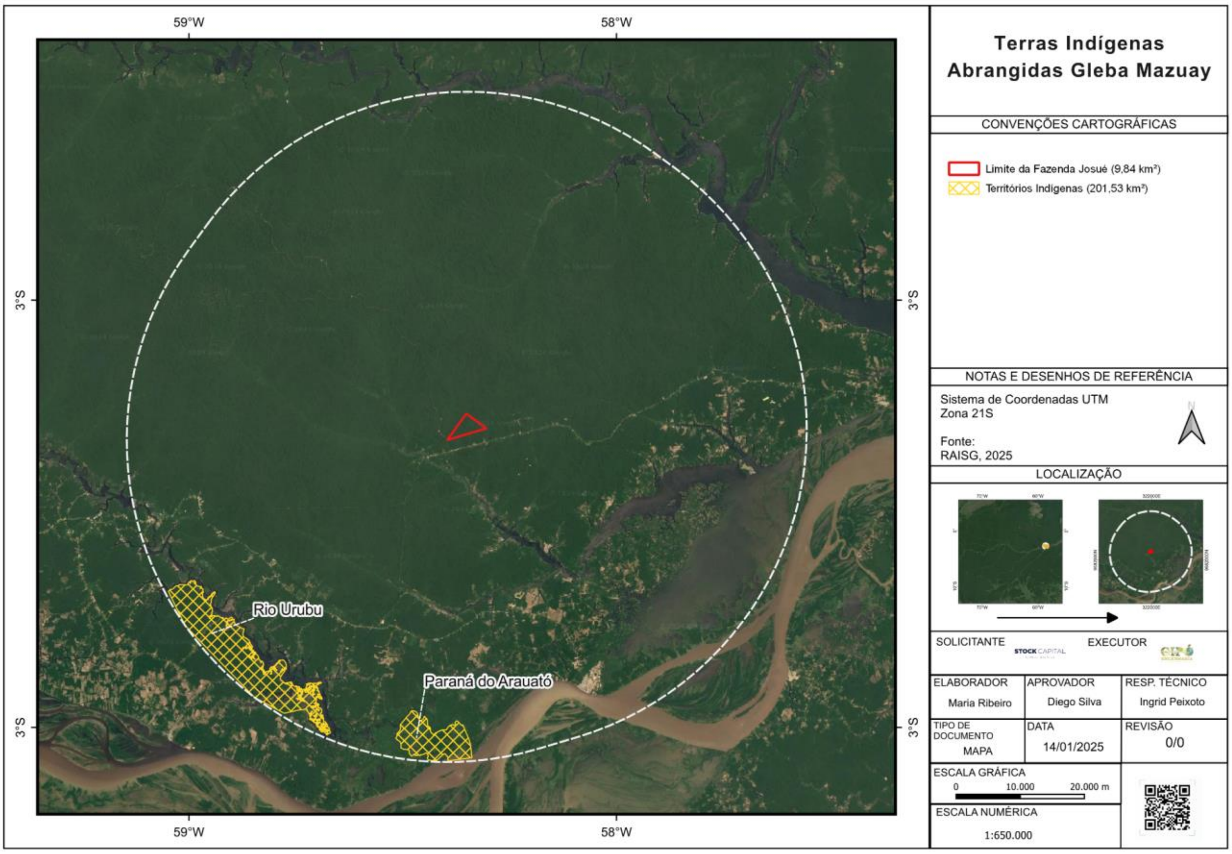
Currently, the process of administrative demarcation of Indigenous Lands, conducted by the National Indian Foundation (FUNAI), must follow the procedures established by Decree No. 1.775/1996.

The information presented results from a compilation of surveys conducted from various sources, providing a comprehensive overview of the population dynamics and the situation of Indigenous Lands (TIs) in Brazil.

In the study area – within a 50 km radius – two Indigenous Lands were found:

I) Rio Urubu Indigenous Land with an Official Area of 27,065.21 hectares and a population of 949. Legal Jurisdiction: Amazon Legal (IBGE, 2025).

II) Paraná do Jauató Indigenous Land with an Official Area of 5,966.50 hectares and a population of 103. Legal Jurisdiction: Amazon Legal (IBGE, 2025).



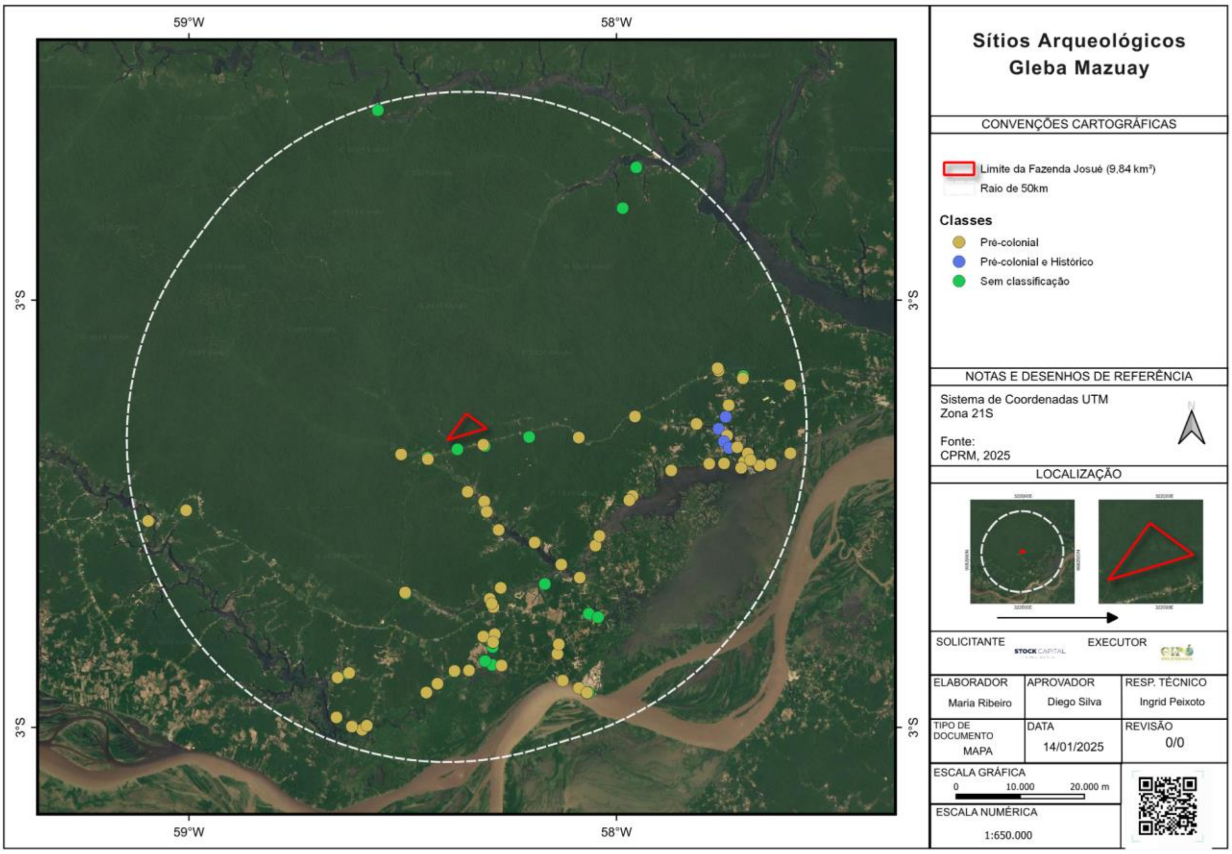
1.13.5.4 Archaeological Sites

The archaeological sites present on the Gleba Mazuay map, according to data provided by the National Institute of Historic and Artistic Heritage (IPHAN), are strategically distributed within the delimited area, with a significant concentration along a watercourse. This distribution suggests that the ancient inhabitants of the region relied heavily on proximity to rivers for survival, either for the acquisition of natural resources, transportation, or as a settlement factor for their communities. Furthermore, the linear arrangement of the sites may indicate the existence of ancient paths or routes, reinforcing the hypothesis that this area played an important role in the dynamics of occupation over time.

The sites have been classified into three main categories by IPHAN: pre-colonial, pre-colonial and historical, and unclassified. The predominance of pre-colonial sites suggests that the region was inhabited by indigenous peoples long before the arrival of colonizers, and may contain archaeological remains such as pottery, stone tools, and housing structures. Sites classified as both pre-colonial and historical indicate a continuity of occupation, possibly reflecting interactions between indigenous peoples and the first Europeans who arrived in the region. These locations can provide valuable insights into cultural and environmental changes over the centuries.

The geographical factor is crucial for understanding the distribution of these sites. The presence of dense vegetation and proximity to rivers suggest that the forest may have acted as both a natural barrier and a source of essential resources. The delineation of Gleba Mazuay on the map shows that the archaeological sites are located in an area that may have undergone modifications over time, either through agricultural activities or economic exploitation. This highlights the need for preservation measures to ensure that the archaeological heritage is not compromised.

In this context, the sites of Gleba Mazuay offer a vast field of possibilities for future studies. More detailed research could reveal new patterns of occupation and provide valuable data on the region's history. The relationship between the sites and the environment could be further explored through geospatial analyses, helping to better understand the settlement choices of ancient peoples. Furthermore, conservation initiatives, guided by IPHAN, are essential to ensure that these sites are protected and studied without external interference.



1.13.6 Analysis of Vegetation Indices and the Fraction of Photosynthetically Active Radiation Using Remote Sensing

Life on Earth is intrinsically dependent on the health and vitality of vegetation, composed of plants and trees. These elements provide essential resources for life, where changes in the quantity, quality, and distribution of vegetation can significantly impact various aspects, including human health, the environment, and the economy.

To understand vegetation changes and their impacts, various techniques are employed, including satellite-based remote sensing. NASA satellites, such as MODIS, Landsat, and Sentinel-2, provide high-resolution multispectral imagery that enables mapping vegetation cover, quantifying plant biomass, monitoring vegetation health and vigor, detecting changes in vegetation cover over time, and identifying areas of deforestation and forest degradation.

Vegetation index maps, such as NDVI (Normalized Difference Vegetation Index), EVI (Enhanced Vegetation Index), LAI or IAF (Leaf Area Index – Índice de Área Foliar, in Portuguese), and FPAR (Fraction of Absorbed Photosynthetically Active Radiation), are essential tools for vegetation monitoring. They translate light reflectance from plants into numerical values that indicate the quantity and health of vegetation in a specific area, providing valuable data for estimating carbon fluxes in ecosystems. The relationship between these indices and the carbon cycle is complex and multifaceted, with significant implications for the carbon credit market.

Vegetation indices can be used to estimate Gross Primary Productivity (GPP or PPB), which represents the amount of carbon fixed by vegetation through photosynthesis and autotrophic respiration (AR). The Net Primary Productivity (NPP) is the difference between GPP and autotrophic respiration, determining the net carbon flux, which can be positive (sequestration) or negative (emission).

In this context, forests play a fundamental role in regulating the global climate by sequestering and storing carbon and as indicators for estimating carbon flux in the region, and vegetation indices are used to monitor REDD+ (Reducing Emissions from Deforestation and Forest Degradation) projects, which generate carbon credits, where accurate quantification of carbon flux is crucial to ensure the reliability of the carbon credits.

Therefore, this study aims to analyze, through remote sensing and geostatistical techniques, the characteristics of vegetation cover in spatial, temporal, and seasonal terms for the period from 2000 to 2023, using images from the MODIS sensor aboard the Aqua and Terra satellites with spatial resolution ranging from 500 meters to 5 km, performing a statistical comparative analysis of the vegetation indices NDVI (Normalized Difference Vegetation Index), EVI (Enhanced Vegetation Index), IAF or LAI (Leaf Area Index), and FPAR (Fraction of Absorbed Photosynthetically Active Radiation), covering a period of 23 years. The analysis takes into account the seasonality of the seasons and the descriptive statistics of each index, including mean, variance, standard deviation, median, minimum, and maximum.

1.13.6.1 Conceptual Description

According to VASHUM and JAYAKUMAR (2012), carbon, present in the atmosphere as carbon dioxide (CO_2), makes up about 0.04% of its composition. Despite its low concentration, CO_2 has gained prominence as one of the main greenhouse gases (GHG), with the potential to significantly influence global climate.

Anthropogenic activities, such as the intensification of industrialization, rampant deforestation, forest degradation, and the burning of fossil fuels, have resulted in a considerable increase in CO_2 levels in the atmosphere. This accumulation imbalances the natural carbon cycle, raising global temperatures and intensifying the effects of climate change, as noted by LU et al. (2014).

In this context, forests emerge as important allies in the fight against climate change. Through photosynthesis, trees absorb CO_2 from the atmosphere, storing it in their biomass (trunk, branches, leaves, and roots). This natural carbon sequestration process by forests makes them crucial for mitigating the effects of climate change.

The Intergovernmental Panel on Climate Change (IPCC) identifies five carbon reservoirs in forests:

- Aboveground biomass: the largest carbon reserve, composed of trunks, branches, and leaves.
- Belowground biomass: roots and mycorrhizae store a significant amount of carbon.
- Litter: leaves and branches fallen on the soil contribute to the carbon stock.
- Woody debris: trunks and branches in various stages of decomposition.
- Soil organic matter: a key component for soil fertility and an important carbon reservoir.

Estimating forest biomass is crucial for monitoring and quantifying carbon lost during deforestation, assessing the carbon sequestration potential of forests, and formulating effective policies for the protection and sustainable management of forests (IPCC, 2021; Vashum & Jayakumar, 2012; Brown & Lugo, 1990).

The carbon stock in a forest is typically estimated from forest biomass. Various methods, such as forest inventories and remote sensing, are used to quantify biomass and, consequently, the carbon stock (Fensholt & Proud, 2012).

Studies indicate that the carbon concentration in tree parts ranges from 45% to 50% of dry biomass. This information is essential for calculating the forest carbon stock from the estimated biomass.

In this study, data were collected for NDVI (Normalized Difference Vegetation Index), EVI (Enhanced Vegetation Index), IAF or LAI (Leaf Area Index), and FPAR (Fraction of Absorbed Photosynthetically Active Radiation), which are described as follows:

– NDVI (Normalized Difference Vegetation Index): ROUSE et al. (1973) proposed NDVI "as a new technique to assess vegetation cover," where NDVI is a classic indicator of vegetation health and vigor, based on the difference between the red and near-infrared bands. It is widely used to estimate forest biomass, as it has a positive correlation with the amount of chlorophyll present in the leaves, which is related to photosynthesis and vegetation productivity. The advantage of NDVI is that it is relatively easy to calculate and widely available in satellite images. Its disadvantage is that it is sensitive to factors unrelated to biomass, such as land cover and atmospheric conditions (Tucker, C. J. 1979; Rouse, J. W et al. 1974).

– EVI (Enhanced Vegetation Index): HUETE et al. (1995) developed a three-band model to estimate vegetation canopy cover, similar to NDVI, but it incorporates the blue band to minimize atmospheric and soil effects. It was developed to minimize the effects of non-biomass factors that influence NDVI. It shows a stronger correlation with forest biomass compared to NDVI. Its advantage is that it is less sensitive to factors such as land cover and atmospheric conditions, while its disadvantage is that it requires additional sensor bands, which are not always available on all satellites (Huete, A. et al. 2002; Justice, C. O. et al. 1998).

– IAF or LAI (Leaf Area Index): WATSON (1969) defined IAF as the index of leaf area in relation to light absorption and productivity. It estimates the amount of leaf area per unit of ground area, a crucial parameter for photosynthesis and carbon sequestration. It is directly related to leaf biomass, which is an important component of total forest biomass. Its advantage is that it provides a more accurate estimate of leaf biomass compared to NDVI or EVI, while its disadvantage is that it requires data from specific sensors, such as LiDAR, which can be more expensive and less accessible (Asner, G. P., Scurlock, J. M., & Hicke, J. A. 2003; Zhao, F. et al. 2016).

– FPAR (Fraction of Absorbed Photosynthetically Active Radiation): GAMON and PEARCY (1986) presented a method to estimate the fraction of photosynthetically active radiation absorbed (FPAR) from the leaf area index (LAI). It indicates the proportion of photosynthetically active radiation (PAR) absorbed by vegetation as an indicator of photosynthetic capacity. It is related to the amount of chlorophyll in the leaves and the photosynthetic capacity of the vegetation. Its advantage is that it provides an estimate of the photosynthetic capacity of vegetation, which is related to productivity and biomass, while its disadvantage is that it requires bio-physical models and remote sensing data, which can be complex and labor-intensive.

1.13.6.2 Relation with Carbon Flux

Vegetation indices can be used to estimate Gross Primary Productivity (GPP), which represents the amount of carbon fixed by vegetation through photosynthesis, as well as Autotrophic Respiration, which is the respiration of vegetation releasing CO2 back into the atmosphere, where biogeochemical models incorporate vegetation indices to estimate autotrophic respiration.

The Net Primary Production (NPP) is the difference between GPP and autotrophic respiration, which determines the net carbon flux, which can be positive (sequestration) or negative (emission).

In the same way, vegetation indices are used to monitor REDD+ (Reducing Emissions from Deforestation and Forest Degradation) projects, which generate carbon credits, where precise quantification of carbon flux is crucial to ensure the reliability of carbon credits (LU, D. et al. 2014).

1.13.6.3 Land Use and Land Cover Classification

The understanding of the importance of vegetation and the monitoring of its changes are essential to ensure human health, environmental sustainability, and economic development. Technological advances, such as satellite remote sensing, allow a more accurate and comprehensive analysis of global vegetation cover, providing subsidies for decision-making and actions for the protection and conservation of vegetation.

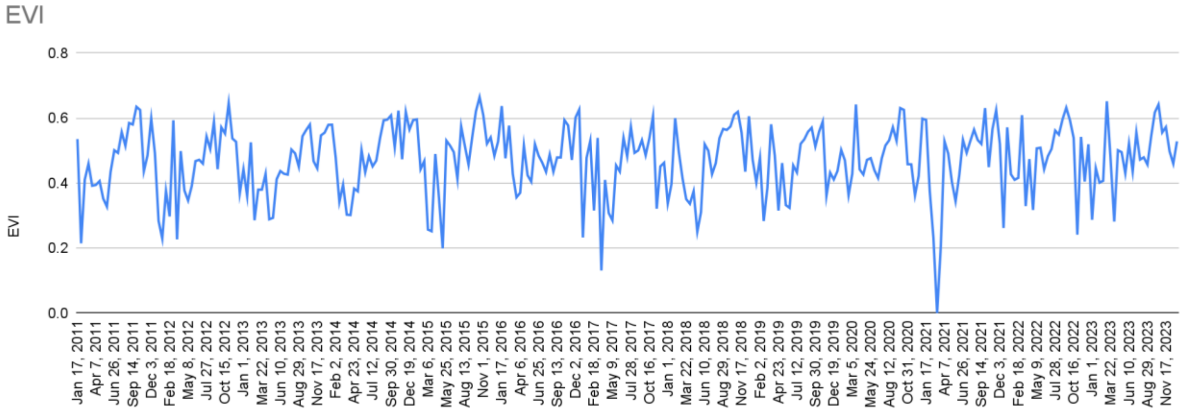
The MODIS Land Cover Classification dataset is used for the biophysical and biogeochemical parameterization of land cover for input into climate models, hydrological processes, and biogeochemical cycles on a global and regional scale. Examples of biogeophysical parameters linked to land cover include leaf area index, vegetation density, and fraction of photosynthetically active radiation (FPAR) absorbed by vegetation. The algorithm uses a hybrid decision tree/neural network classifier to categorize the Earth's surface into 17 categories, following the global vegetation database of the International Geosphere-Biosphere Programme, which defines 9 natural vegetation classes, 3 developed land classes, 2 mosaic land classes, and 3 non-vegetated land classes (snow/ice, barren land/rocks, water).

Previously, land cover datasets on a global and regional scale used coarse spatial resolution and high temporal frequency data from the Advanced Very High-Resolution Radiometer (AVHRR) instrument onboard the NOAA series of meteorological satellites. Most land cover datasets were based on the parameter of the Normalized Difference Vegetation Index (NDVI) derived from these data. NDVI generally quantifies the biophysical activity of the Earth's surface and, as such, does not provide direct land cover. The algorithm that produces the MODIS land cover dataset is based on information domains far beyond those used in previous efforts, including surface directional reflectance, texture, acquisition geometry, land surface temperature, and snow/ice cover.

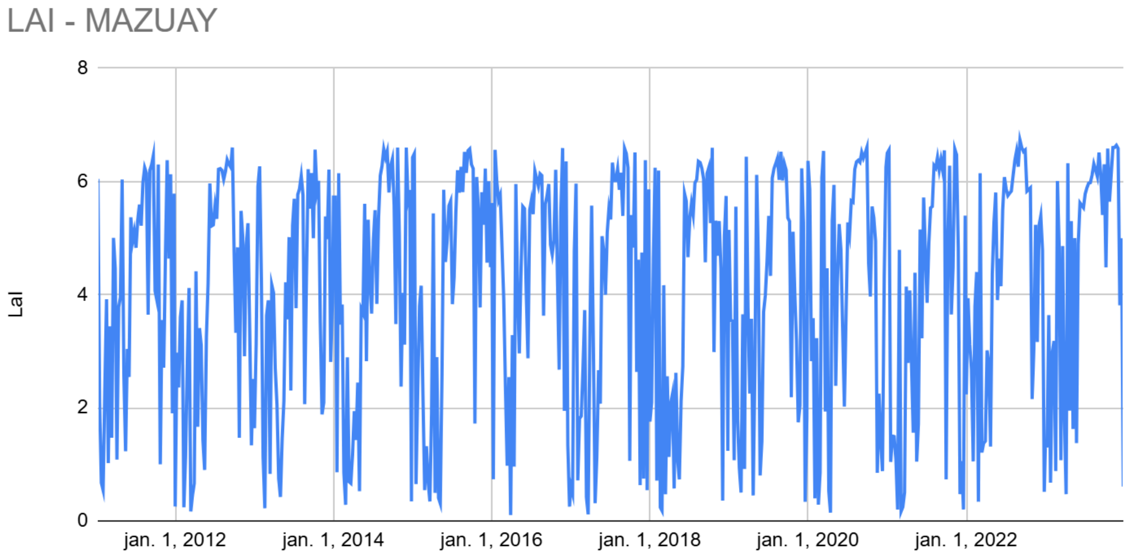
The MODIS land cover classification dataset also includes a land cover change parameter. The land cover change parameter quantifies subtle and progressive transformations of the land surface, as well as large and rapid changes. As such, it is not a conventional change product that only compares changes in land cover type at two points in time, but actually the parameter combines analyses of changes observed in multispectral and multitemporal satellite data vectors with models of vegetation change mechanisms to recognize both the type of change, such as desertification, or forest succession, or agricultural conversion, and its intensity.

Figure IV-1: Land cover and land use map, Source: MODIS Land Cover Classification product

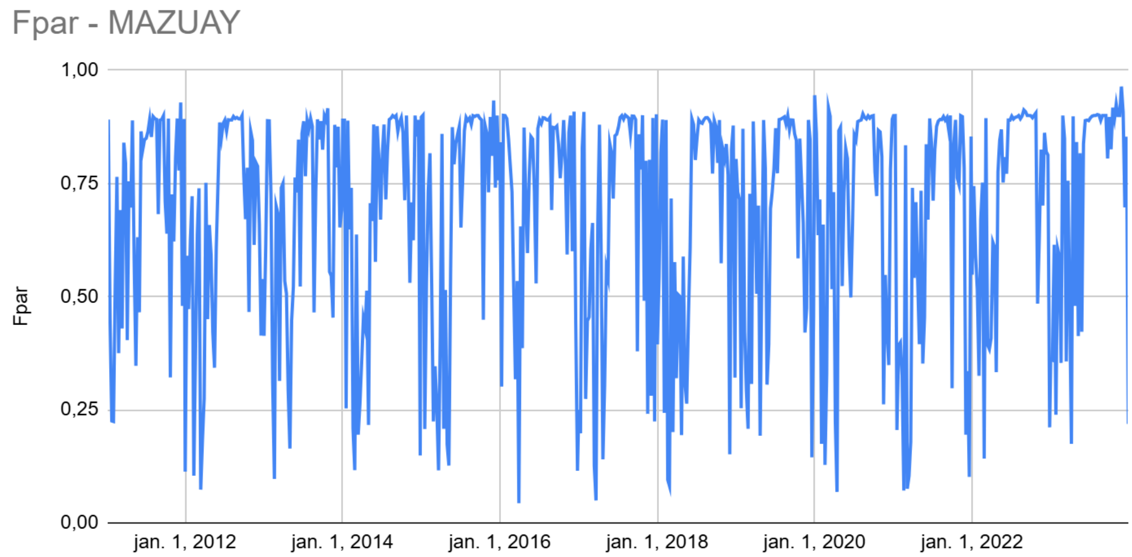
Mean EVI	Standard deviation EVI	Maximum EVI	Minimum EVI	Median EVI
0.4700227204	0.1059915525	0.6645445	0.0005556	0.4791851



Mean LAI (area fraction)	Standard deviation LAI	Maximum LAI	Minimum LAI	Median LAI
4.07	2.050754094	6.73	0.10	4.65

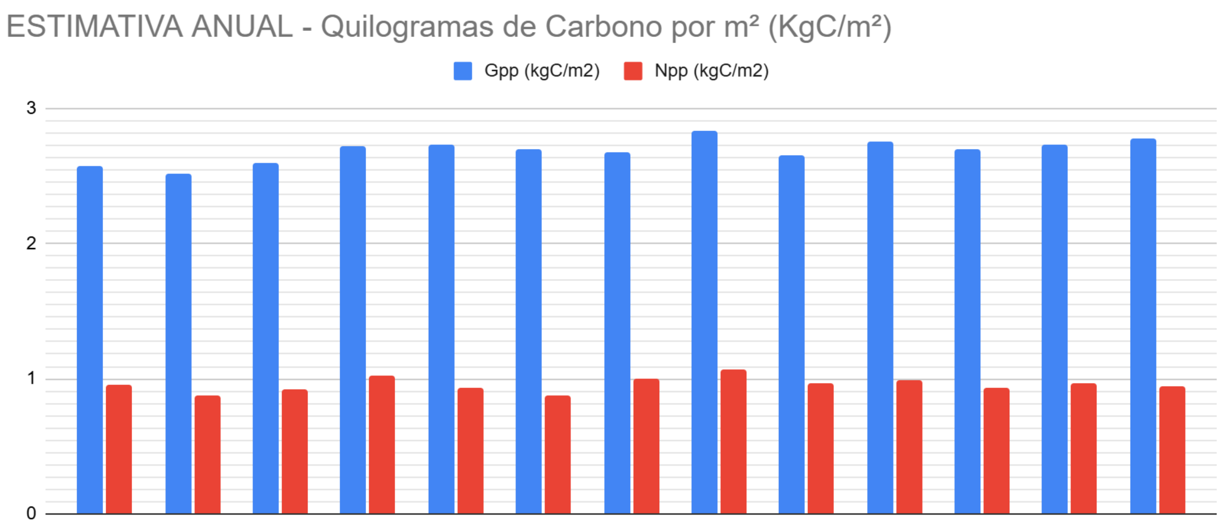


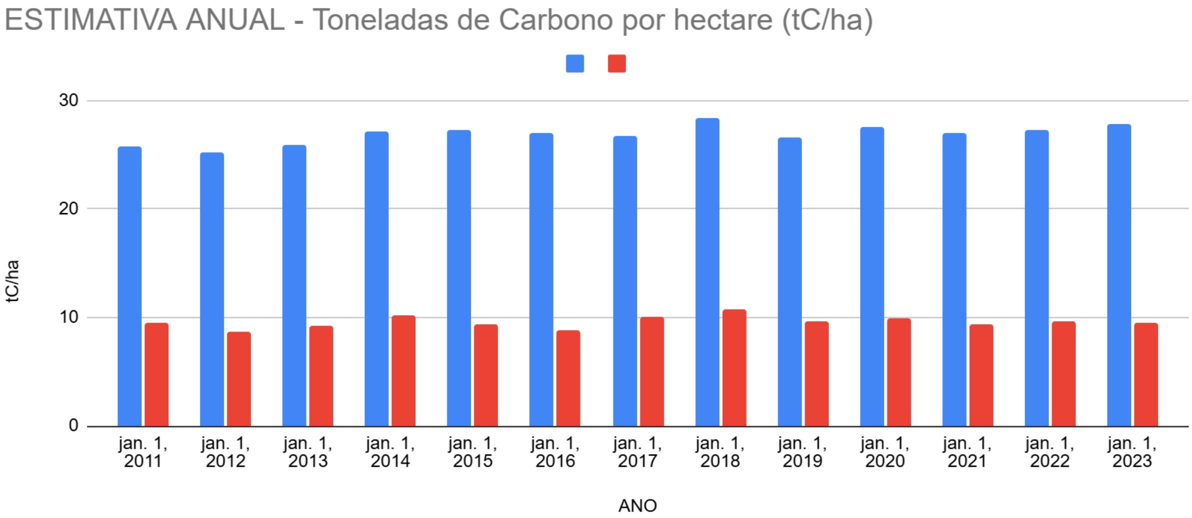
Mean FPAR (%)	Standard deviation FPAR	Maximum FPAR	Minimum PAR	Median FPAR
0.6846406544	0.2441932098	0.96412	0.04334	0.788895

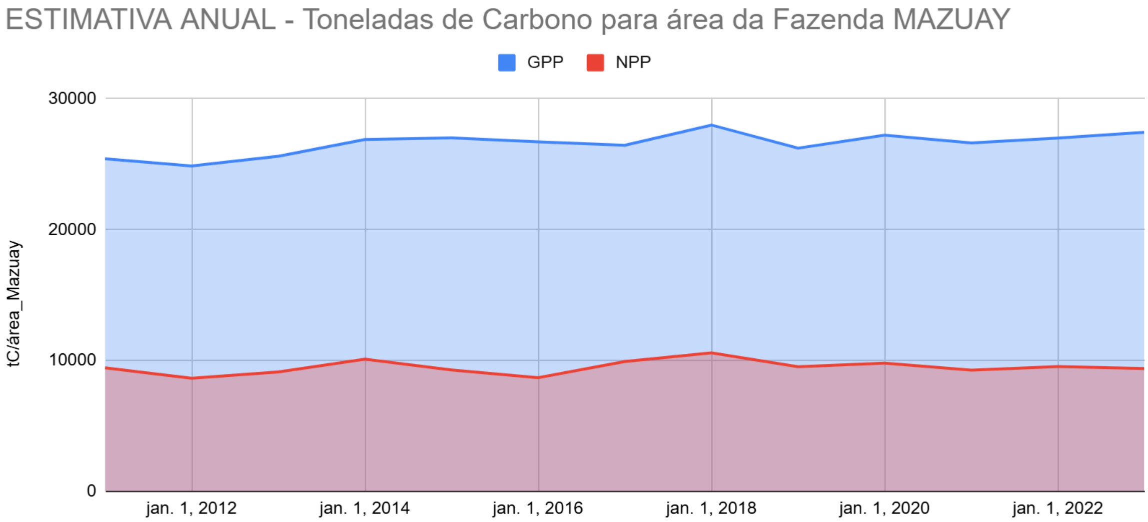


Mean GPP/Area	Standard deviation GPP/Area	Maximum GPP/Area	Minimum GPP/Area	Median GPP/Area
26582.3	865.6	28002.9	24879.0	26717.9

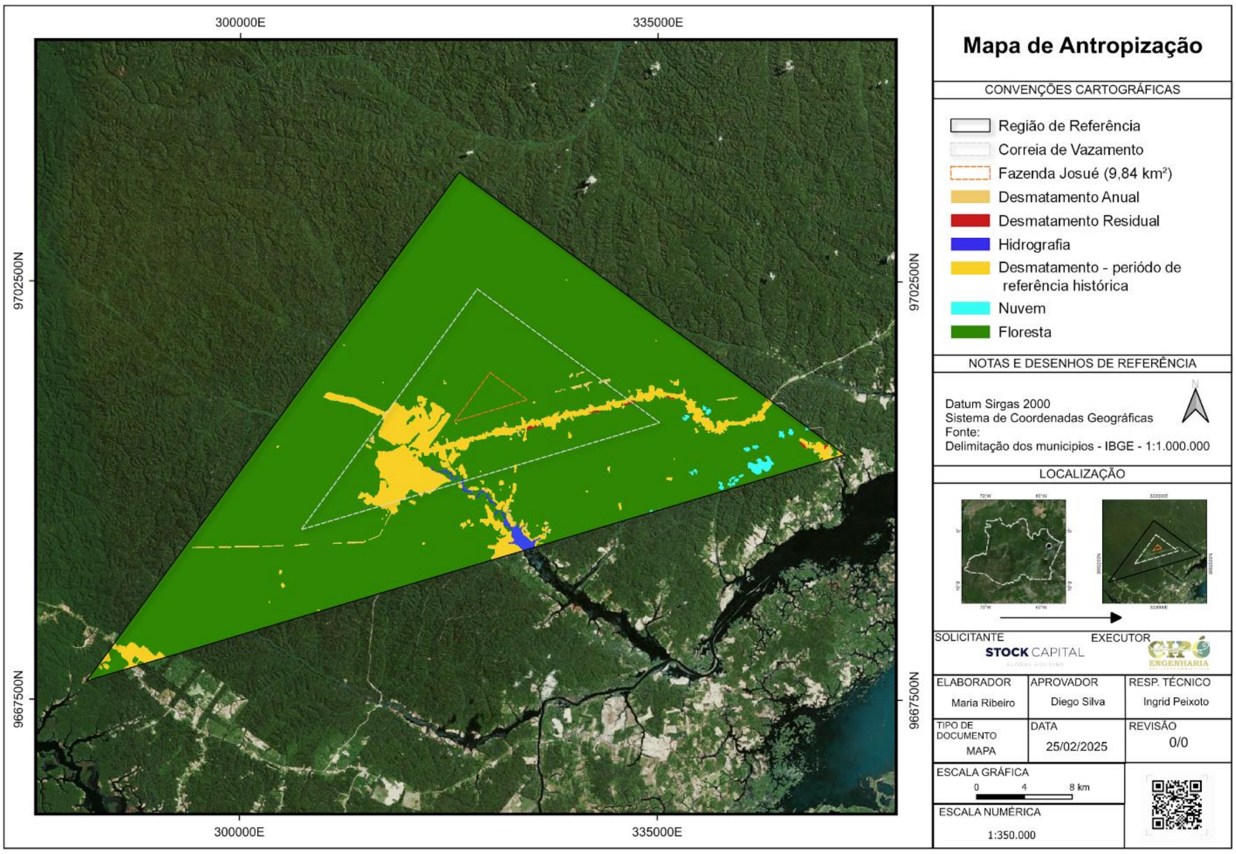
Mean NPP/Area	Standard deviation NPP/Area	Maximum NPP/Area	Minimum NPP/Area	Median NPP/Area
9480.4	536.5	10573.4	8641.4	9434.5




ESTIMATIVA ANUAL - Toneladas de Carbono por hectare (tC/ha)
30
20
10
0
tC/ha
jan. 1, 2011
jan. 1, 2012
jan. 1, 2013
jan. 1, 2014
jan. 1, 2015
jan. 1, 2016
jan. 1, 2017
jan. 1, 2018
jan. 1, 2019
jan. 1, 2020
jan. 1, 2021
jan. 1, 2022
jan. 1, 2023
ANO


ESTIMATIVA ANUAL - Toneladas de Carbono para área da Fazenda MAZUAY
GPP
NPP
30000
20000
10000
0
tC/área_Mazuay
jan. 1, 2012
jan. 1, 2014
jan. 1, 2016
jan. 1, 2018
jan. 1, 2020
jan. 1, 2022

1.14 Conditions Prior to Project Initiation



- **Type of ecosystem:** The vegetation analysis around the Gleba showed a predominance of Lowland Dense Ombrophilous Forest. Other identified vegetation classes include Alluvial Dense Ombrophilous Forest and Pioneer Formation with Fluvial and/or Herbaceous Lacustrine Influence, present in flooded areas. Park Savannas, Secondary Vegetation, and water bodies, such as the Amazon River and the Urubu River, were also found.

- **Current and historical land use:** The conditions prior to the start of the project are the same as the baseline scenario: agricultural projects in planning or implementation, with deforestation authorizations issued or requested from the environmental agency responsible for the state of Amazonas.

- **Current and previous environmental conditions of the project area:** According to the Köppen climate classification, the region of Gleba Mazuay presents an Aw-type climate, characterized as tropical humid with a dry season. The main characteristics of this climate are high temperatures throughout the year, with annual averages around 25°C to 27°C, and a well-defined dry season, usually occurring in winter, with a rainy period concentrated in the summer months. This climatic configuration favors the growth of vegetation adapted to periods of drought and intense rainfall.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The Stock Carbon Mazuay Project is in accordance with the main environmental laws and regulations applicable in Brazil, including: In addition to the Federal Constitution of 1988, Article 225, which addresses the ecologically balanced environment, with a diffuse right (right and duty) to preserve it for future generations.

National Environmental Policy, Law No. 6.938/81, Article 4 and its items, as environmental instruments:

1. Brazilian Forest Code (Law No. 12.651/2012): The project complies with the requirements for the preservation of native vegetation, including the maintenance of the Legal Reserve and Permanent Preservation Areas (APPs) on the property.

2. National System of Conservation Units (SNUC - Law No. 9.985/2000): Although Gleba Mazuay is not located within a conservation unit, the project adopts sustainable management practices aligned with the conservation objectives of the SNUC.

3. National Policy on Climate Change (Law No. 12.187/2009): The project contributes to climate change mitigation through conservation and forest management, aligning with the commitments and guidelines established in this policy.

4. Applicable state and municipal legislation: The project comply with the environmental requirements established by the legislation of the state of Amazonas and the municipality of Silves, where Gleba Mazuay is located.

– State Law No. 1.532, of 1982 – Regulation of the State Policy for the Prevention and Control of Pollution, improvement and recovery of the environment, and protection of natural resources.

– State Law No. 3.525, of 2010 – Establishes the Sustainable Development Council of Traditional Peoples and Communities of the State of Amazonas (CDSPCT/AM), within the organizational structure of the State Secretariat for the Environment and Sustainable Development.

– State Law No. 3.785, of 2012 – Establishes the environmental licensing regulations in the State of Amazonas.

– State Law No. 4.222, of 2015 – Establishes the State Technical Registry of Potentially Polluting Activities or Users of Environmental Resources, part of the National Environmental System (SISNAMA), and the Environmental Control and Inspection Fee (TCFA/AM) in accordance with Federal Law No. 6.938, of August 31, 1981, and its amendments, and provides other provisions.

- Federal Law No. 6.938, of August 31, 1981, and its amendments – and provides other provisions.
- State Law No. 4.457, of 2017 – Establishes the State Policy on Solid Waste of Amazonas (AM) and provides other provisions.
- Complementary State Law No. 187, of 2018 – Regulates the execution of Article 220 of the State Constitution, which establishes the State Environmental Council of the State of Amazonas (CEMAAM) and provides for the State Environmental Fund (FEMA) and other provisions.
- Ordinance No. 41.863, of 2020 – This ordinance establishes rules for the execution of the State Policy on Solid Waste.
- Normative Resolution CERH-AM No. 001, of 2016 – Establishes the Internal Regulations of the State Water Resources Council, as set forth in the Annex to this Normative Resolution.
- State Law No. 5.695, of 2021 – Amends, as specified, Law No. 4.222, of October 8, 2015, which "Establishes the State Technical Registry of Potentially Polluting Activities or Users of Environmental Resources, part of the National Environmental System (SISNAMA), the Environmental Control and Inspection Fee (TCFA/AM), in accordance with Federal Law No. 6.938, of August 31, 1981, and its amendments, and provides other provisions," and provides other provisions.
- State Law No. 5.755, of 2021 – Provides for the reorganization of the Sustainable Development Council of Traditional Peoples and Communities of the State of Amazonas, established by Law No. 3.525, of July 15, 2010, and provides other provisions.
- State Law No. 5.491, of 2021 – Amends the caput of Article 12, the sole paragraph of Article 13, §1 of Article 14, and the sole paragraph of Article 15 of Law No. 3.785, of July 24, 2012, which "Provides for environmental licensing in the State of Amazonas, repeals Law No. 3.219, of December 28, 2007, and provides other provisions."
- State Ordinary Law No. 6.014, of 2022 – Establishes an administrative term intended for the analysis and decision regarding the granting or renewal of environmental licenses, in accordance with Complementary Law No. 140/2011 and CONAMA Resolution No. 237/97.
- State Law No. 6.052, of 2022 – Recognizes the contribution of indigenous peoples to the preservation of forests, culture, folklore, customs, legends, gastronomy, handicrafts, and language.

In addition, the project is in compliance with the requirements of the VCS Program (Verified Carbon Standard), demonstrating its eligibility and adherence to international best practices for forest conservation projects and greenhouse gas emissions reduction.

Therefore, the Forest Conservation and Management Project of Gleba Mazuay is in full compliance with the applicable laws, statutes, and regulatory frameworks.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

If yes, provide the registration number and the date of project inactivity under the other GHG program.

Is the project active under the other program?

☐ Yes ☒ No

Project proponents, or their authorized representative, must attest that the project is no longer active in the other GHG program in the Registration Representation.

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

If yes, provide the program name(s), reason(s) and date for the rejection, justification of eligibility under the VCS Program, and any other relevant information.

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the VCS Program Definitions for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the VCS Program Definitions for definition of GHG-related environmental credit system.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☒ No

If yes to all:

Provide evidence of the public statement. Evidence must be provided in this section or in an appendix.

1.18 Sustainable Development Contributions

The project, developed and managed by Stock Capital Holding Ltda and Cipó Engenharia e Tecnologia de Biomassa, implements forest conservation and sustainable management activities in priority areas of the Legal Amazon, in Brazil. These activities include constant monitoring of deforestation and manipulation, the promotion of sustainable management practices, and the engagement of local communities in actions for the preservation and sustainable use of natural resources. The specific location of the activities covers areas of high biodiversity and significant carbon stocks, ensuring protection against conversion to unsustainable uses, such as agricultural and logging expansion.

How the project's activities contribute to sustainable development

- **Protection of carbon stocks and mitigation of climate change:** The project's activities prevent the release of large amounts of greenhouse gases (GHG) by avoiding deforestation and the manipulation of native forests. In addition, they promote natural regeneration in degraded areas.

- **Promotion of biodiversity:** By protecting critical natural habitats, the project contributes to the preservation of endangered species, ensuring the ecological health of forest areas rich in biodiversity.

- **Community engagement and sustainable income generation:** The project integrates local communities into sustainable management activities, providing training in agroforestry practices, silviculture, and ecotourism, creating economic alternatives to deforestation. These initiatives increase local economic resilience and promote human development.

Contributions to national sustainable development priorities

The project is aligned with Brazil's declared sustainable development priorities, including the commitments established in its Nationally Determined Contribution (NDC) to the Paris Agreement. The main contributions include:

- **Reduction of GHG emissions:** Supports Brazil's commitment to reduce GHG emissions by 37% by 2025 and by 43% by 2030 compared to 2005 levels.

- **Biodiversity conservation:** Contributes to the goal of protecting at least 30% of terrestrial areas and marine ecosystems by 2030, as stipulated by Target 12 of the National Adaptation Plan to Climate Change.

- **Improvement of the quality of life of local communities:** Supports the Sustainable Development Goals (SDGs), with emphasis on SDG 13 (Climate Action), SDG 15 (Life on Land), and SDG 8 (Decent Work and Economic Growth).

Monitoring and Reporting

The project's contributions to sustainable development are monitored through key indicators, such as:

- Annual reductions in GHG emissions, calculated in accordance with VCS standards.

- Biodiversity monitoring, with periodic reports on protected species.

- Socioeconomic indicators, such as the number of families benefited, generation of alternative income, and trainings conducted.

The results will be reported annually and made available to stakeholders and validation and verification organizations, ensuring transparency and alignment with national and global sustainable development goals.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Considering that the project's activities will be focused on reducing GHG emissions caused by planned deforestation, the leakage management plan will take into account the study area, as well as monitoring measures for the surrounding areas.

1.19.2 Commercially Sensitive Information

For all instances of project activity, whether already verified or in the process of verification, any contract signed between the Project Proponent (PP) and landowners, as well as commercial, financial, scientific, technical, or other information whose disclosure could reasonably result in material financial loss or gain, compromising contractual terms, agreements, or other negotiations declared by the project proponent, must be considered commercially sensitive information.

Also considered sensitive is any information related to decisions regarding internal policies, financial, commercial, scientific, or technical matters, where public disclosure could reasonably be expected to harm or negatively affect the development and/or implementation of any project activity.

Information related to the project's social activity, the determination of the baseline scenario, the demonstration of additionality, and the estimation and monitoring of GHG emission reductions (including operational and capital expenditures) is not considered commercially sensitive and is provided in the public versions of the project documents.

1.19.3 Further Information

Not applicable. No additional information to be shared.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.1.1 Stakeholder Identification

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.1.2 Stakeholder Consultation and Ongoing Communication

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.1.3 Free Prior and Informed Consent

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.1.4 Grievance Redress Procedure

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.1.5 Public Comments

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.2 Risks to Stakeholders and the Environment

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.2.1 Management Experience

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.2.2 Risk Assessment

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.3 Respect for Human Rights and Equity

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.3.1 Labor and Work

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.3.2 Human Rights

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.3.3 Indigenous Peoples and Cultural Heritage

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.3.4 Property Rights

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.3.5 Benefit Sharing

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.4 Ecosystem Health

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.4.1 Rare, Threatened, and Endangered Species

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

2.4.2 Introduction of Species

Esta seção ainda está em desenvolvimento e será apresentada na próxima versão do Relatório de Descrição/ Monitoramento do Projeto.

2.4.3 Ecosystem Conversion

Esta seção ainda está em desenvolvimento e será apresentada na próxima versão do Relatório de Descrição/ Monitoramento do Projeto.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The list below refers to the methodology, modules, and tools that have been and will be used within the scope of the project.

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0007	VM0007 REDD+ Methodology Framework (REDD+MF)	1.6
Module	VMD0001	VMD0001 Estimation of carbon stocks in the above- and below ground biomass in live tree and non-tree pools (CP-AB)	1.2

Module	VMD0002	VMD0002 Estimation of carbon stocks in the dead wood pool (CP-D)	1.1
Module	VMD0003	VMD0003 Estimation of carbon stocks in the litter pool (CP-L)	1.1
Module	VMD0004	VMD0004 Estimation of stocks in the soil organic carbon pool (CP-S)	1.1
Module	VMD0005	VMD0005 (CP-W) Estimation of carbon stocks in the long-term wood products pool	1.2
Module	VMD0011	VMD0011 Estimation of emissions from market-effects (LK-ME)	1.2
Module	VMD0013	VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E–BPB),	1.3
Module	VMD0015	Methods for monitoring of greenhouse gas emissions and removals (M-REDD)	2.3
Module	VMD0016	VMD0016 Methods for Stratification of the Project Area (X-STR)	1.3
Tool	VT0001	VT0001 Tool for the demonstration and assessment of additionality in VCS agriculture, forestry,	3.0

		and other land use (AFOLU) project activities (T-ADD)	
Tool	AFOLU NonPermanence Risk Tool	AFOLU Non-Permanence Risk Tool	4.2
Tool	CDM Tool for testing significance of GHG emissions in A/R CDM project activities (T-SIG)	CDM Tool for testing significance of GHG emissions in A/R CDM project activities (TSIG)	1.0

3.2 Applicability of Methodology

The list below refers to the methodology, modules, and tools that have been and will be used within the scope of the project, in addition to providing information on applicability conditions and the respective justifications of compliance for each module, tool, or methodology.

Methodology ID	Applicability condition	Justification of compliance
VM0007	Land classified as forest for at least 20 years; and baseline deforestation/degradation classified as APD (planned deforestation).	Land classification by DETER/PRODES; baseline falls under APD.
VMD0001, VMD0002	Applicable to all forest types and age classes.	The project aims to reduce non-tree aboveground biomass through degradation reduction.
VMD0003	Applicable to all forest types and age classes on lands classified as forest for more than 20 years.	Applied according to prior forest qualification.
VMD0004	Applicable to non-organic soils under all forest types and age classes.	The project area is mainly composed of Latosols and Argisols.

VMD0005	Applicable where timber is harvested for commercial markets.	Activities comply with conditions for commercial timber harvest.
VMD0011	Applicable for calculating market leakage from projects expected to substantially and permanently reduce timber harvest.	Reduction in timber harvest may shift production to other areas, including forested peatlands.
VMD0013	Applicable to REDD projects with emissions from biomass and/or peat burning.	The GPD qualifies; emissions from fires in the baseline and project scenario will be calculated using this module.
VMD0015	Applicable to forest management and selective timber harvesting operations, following best practices, FSC certification requirements, and minimal impact conditions.	Assessment conducted according to the Forest Code (Law 12.651/2012) and technical criteria; continuous monitoring by the BRC team.
VMD0016	Applicable for delineating and stratifying peatland areas in rewetting and conservation projects.	Used for aboveground biomass stratification, as required by the module.

3.3 Project Boundary

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

3.4 Baseline Scenario

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

3.5 Additionality

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

3.5.1 Regulatory Surplus

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

3.5.2 Additionality Methods

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

3.6 Methodology Deviations

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

4.2 Project Emissions

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

4.3 Leakage Emissions

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

5 MONITORING

5.1 Data and Parameters Available at Validation

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

5.2 Data and Parameters Monitored

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.

5.3 Monitoring Plan

This section is still under development and will be presented in the next version of the Project Description/Monitoring Report.